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Water Transport, Retention, and Survival of *Escherichia coli* in Unsaturated Porous Media: A Comprehensive Review of Processes, Models, and Factors

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*The vadose zone can function as both a filter and a passage for bacteria. This review evaluates when and why either effect will apply based on available literature. It summarizes theories and experimental research that address the related, underlying bacterial attenuation processes, the applied macro-scale modeling approaches, and the influencing factors - including the cell, soil, solution and system characteristics. Results point to that the relative importance of each removal mechanism depends on the moisture content and the solution ionic strength. The limitations of available modeling approaches are discussed. It remains unclear in which contexts these are reliable for predictions. The temporal first-order kinetic *Escherichia coli* (*E. coli*) removal coefficient ranges three orders of magnitude, from 10^{-4} to 10^{-1} /min. Results suggest that this rate depends on the pore-water velocity. Spatial filtration of *E. coli* increases with slower flow and higher collector surface heterogeneity. It could be insignificant in the case of heavy and sudden infiltration and subsequent transport in preferential flow paths, induced, for example, by plant roots or cracks in clayey soils. Future research*

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thus needs to address transport as an effect of extreme weather events such as droughts and subsequent floods.

KEY WORDS: attachment, *Escherichia coli*, preferential transport, straining, subsurface modeling, survival, unsaturated zone

1. INTRODUCTION

In the United States the estimated total cases of food-related illness due to different strains of *Escherichia coli* (*E. coli*) is nearly 270,000 per year: *E. coli* O157:H7: 73,480 cases; *E. coli*, non-O157 Shiga toxin-producing: 36,740 cases; *E. coli*, enterotoxigenic: 79,420 cases; and *E. coli*, other diarrheogenic: 79,420 cases (Mead et al., 1999). The infectious dose has been estimated to be as low as 10 cells for some *E. coli* strains (Pachepsky et al., 2006). A potential source of microbial contamination is irrigation water on ready-to-eat crops, such as carrots and radish (Forslund et al., 2011; Wood et al., 2010). Forslund et al. (2011) reported that subsurface drip irrigation with water containing *E. coli* O157:H7 had contaminated potato tubers in sandy loam and coarse sand, which are soils typical for Danish agriculture. Irrigation water is often provided by groundwater; for example, in the Central Coast area of California it relies heavily on groundwater resources (Gelting et al., 2011). Investigating a multistate outbreak of *E. coli* O157:H7 infections in the United States, Gelting et al. (2011) argued that the most probable irrigation water-related factor that pose a risk to ready-to-eat crops is groundwater-surface water interactions. It was further concluded that the vertical, spatial dimension needs to be considered in future analyses (Gelting et al., 2011). Pathogenic coliforms generally enter the groundwater through the unsaturated zone, since contaminant sources are often located close to the ground surface (Wan and Tokunaga, 1997). For example, contamination could occur due to manure used as fertilizer in agriculture, septic tank effluents, disposal of excreta using pit-latrines, leakage of sewer lines, or irrigation of wastewater (Downs et al., 1999; Pachepsky et al., 2006; Pang et al., 2004; Scandura and Sobsey, 1997). The latter practice has both agronomic and economic benefits, and is likely to become more widespread as the demand for fresh water intensifies (Hamilton et al., 2007; Westcot, 1997).

The health impact of pathogenic bacteria in groundwater is particularly severe in developing regions. In sub-Saharan Africa and South Asia shallow groundwater is a common drinking water source for low-income communities (Howard et al., 2003; Knappett et al., 2011b). Groundwater-fed springs in periurban Kampala, Uganda, are frequently, particularly in the rainy season, infected by fecal coliforms (Howard et al., 2003; Kulabako et al., 2007), which are often considered to represent *E. coli* (Unc and Goss, 2003). The use of water from springs and wells in this region has been linked to the

incidence of acute diarrhea and cholera in this region (Howard et al., 2003; Nasinyama et al., 2000). An important contaminant source is local recharge from stagnant surface water, e.g., in solid waste dumps, ponds, or drainage channels (Howard et al., 2003). In Bangladesh, most households use water from ponds for hygiene, such as bathing and brushing teeth (Knappett et al., 2011a). These could receive latrine effluents and cattle manure, hence contaminating nearby aquifers, which are tapped by tube-wells that provide drinking water (Knappett et al., 2011b). Knappett et al. (2011a) detected cultured *E. coli* in 42 of 43 surveyed ponds in rural Bangladesh.

1.1 Focus of *E. coli*

Here we consider *E. coli* in general, for several reasons. Primarily, water quality guidelines have been developed for the ensemble of *E. coli* strains (World Health Organization, 2011). Considering water treatment and disinfection procedures, there is no indication that enteropathogenic *E. coli* strains differ from other *E. coli*, according to World Health Organization (2011). Further, despite that only some *E. coli* strains are pathogenic, all of them have common characteristics: they are enteric, facultatively anaerobic, Gram-negative, and rod-shaped (Bergey and Breed, 1957); moreover, they have a similar size, in the range of $2.0\text{--}6.0\text{ }\mu\text{m} \times 1.1\text{--}1.5\text{ }\mu\text{m}$, and they are considered to be hydrophilic (Foppen and Schijven, 2006). This suggests that the key transport and removal mechanisms in soil are similar for most *E. coli* strains, even though the attenuation rates might vary. Moreover, there are very few studies that consider transport of pathogenic *E. coli* in unsaturated porous media, and the current understanding of which *E. coli* strains provide suitable, conservative representatives of the pathogenic ones considering transport in soil is still very limited. Pathogenic *E. coli* strains can be both motile, such as some strains of the enterohemorrhagic *E. coli* (EHEC), of which O157:H7 most often cause illness, and nonmotile, such as the enteroinvasive *E. coli* (EIEC; Todar, 2012). Thus, both motile and nonmotile *E. coli* strains are considered in this review. Foppen and Schijven (2006) also addressed *E. coli* in general, however, focusing on saturated conditions.

Moreover, since *E. coli* are negatively charged they generally attach unfavorably in soil, and could therefore potentially be suitable for predictive modeling, for example, related to protection area delimitation and placement of wastewater reclamation sites (Foppen and Schijven, 2006; Pang, 2009; Pang et al., 2004). *E. coli* have also been considered a representative of the microorganisms that are used for soil bioremediation (Chen et al., 2010b). On the other hand, the use of *E. coli* as an indicator bacteria in soil has been questioned, which will be discussed below. Lastly, relatively many studies have addressed *E. coli* as compared to other microbes under unsaturated, dynamic conditions; nevertheless, findings are limited, and this review also

includes recent important results relating to *E. coli* in the saturated zone, as well as colloids with similar characteristics. The latter approach is common; however, the difference in removal behavior between abiotic colloids and bacteria has not yet been fully clarified. The difference in rheology could for example affect the attachment behavior (Ohshima, 1995; Ohshima and Kondo, 1991).

1.2 Aim of This Review

Previous reviews have typically addressed larger groups of particles in the subsurface as a whole. Pang (2009) assessed the transport of microbes in field studies that included saturated and unsaturated filter media. Pachepsky et al. (2006) reviewed modeling of manure-borne pathogens in both surface and subsurface aquatic environments. Early literature studies, such as McDowell-Boyer et al. (1986) and Yates et al. (1988), examined colloid transport processes in porous media in general. Since then, the importance of the presence of air has been highlighted (Wan and Wilson, 1994a, 1994b; Wan et al., 1994). More recent reviews have qualitatively discussed the physical, chemical and biological factors that influence microbial attenuation in soil (Jamieson et al., 2002; Stevik et al., 2004); however, they have not evaluated predictive models. In the saturated zone findings on microbial deposition and fate have been assessed by Ginn et al. (2002), Murphy and Ginn (2000), and Tufenkji (2007), as well as Foppen and Schijven (2006). The latter focused particularly on *E. coli*. Relating to transport in porous media in general, Shein and Devin (2007) and Sen (2011) addressed colloids and biocolloids, respectively. The extensive review by Schijven and Hassanizadeh (2000) focused on virus removal under saturated conditions. Research relating to unsaturated conditions is, however, relatively rare: a search for “microbes saturated soil” and “microbes unsaturated soil” results in 178 versus 86 hits, respectively, in the Web of Knowledge, as well as 51,300 versus 26,000 hits, respectively, in Google Scholar (December 6, 2012). Even so, studies by Keller and Auset (2007), and Bradford and Torkzaban (2008) addressed the unsaturated zone, in particular concentrating on pore-scale mechanisms. DeNovio et al. (2004) analyzed transport and retention of colloids under unsaturated conditions, with particular focus on fluctuations in pore-water flow rates. Rockhold et al. (2004) reviewed the coupling of microbial transport and other constituent processes in the vadose zone. In this article we focus on unsaturated porous media and one particular bacterial species. Our aims are the following:

- to provide an in-depth review of the theories and empirical studies that relate to the processes that are involved in *E. coli* transport and fate in unsaturated filter media;

- to critically compare the models that have been applied to predict these processes; and
- to evaluate the findings that relate to the physical, chemical, and biological factors that affect them.

The emphasis of the study is placed on common environmental conditions and sandy soils i.e., negatively charged filter media, a transporting solution of low ionic strength and approximately neutral pH.

1.3 Background of Flow and Transport Modeling

Water flow in the unsaturated zone is comparatively slow, as it is limited by capillary forces and generally takes place in small pores (Bradford and Torkzaban, 2008). It is often assumed to be vertical and uniform, and it is, as such, typically modeled using a one-dimensional version of Richards's equation (Richards, 1931; Schijven and Simunek, 2002):

$$\frac{\partial \theta(h)}{\partial t} = \frac{\partial}{\partial x} \left[K_s K_r(h) \left(\frac{\partial h}{\partial x} + 1 \right) \right] - S \quad (1)$$

where h is the soil water pressure head [L]; t is time [T]; θh [–] is the volumetric water content, as described by, for example, the van Genuchten (1980) and Brooks and Corey (1964) equations; x is the vertical distance (positive upward) [L]; S is a source/sink term (e.g., representing the root water uptake) [T^{-1}] = ; K_s is the saturated hydraulic conductivity [LT^{-1}]; and $K_r(h)$ represents the relative hydraulic conductivity [–]. Non-equilibrium descriptions of flow will be addressed in section V.

The most commonly used equation for describing mass transport in aqueous systems at a representative elementary volume scale is the Fickian-based advection-dispersion equation (Reddy et al., 1981; Schwartz and Zhang, 2003). In a general case, it can be extended to account for bacterial fate (Rockhold et al., 2004):

$$\frac{\partial(\theta C)}{\partial t} + \Lambda_{fate} = \frac{\partial}{\partial x} \left(\theta D \frac{\partial C}{\partial x} \right) - \frac{\partial(\theta v C)}{\partial x} \quad (2)$$

where C [NL^{-3}] is the bacterial concentration in the water and N is the number of particles; D [L^2T^{-1}] is the hydrodynamic dispersion, the sum of the molecular diffusion and the mechanic dispersivity, which is equal to the product of the pore-velocity, v [LT^{-1}], and the dispersivity, λ [L]; and Λ_{fate} is the total bacterial reaction-rate source-sink term. Thus, the Darcy velocity q [LT^{-1}] equals $v\theta$, and the dispersion flux is given as $-D \frac{\partial C}{\partial x}$ in coherence with Fick's first law of diffusion.

The most straightforward attenuation model assumes constant first-order attenuation (Pang, 2009), according to

$$\Lambda_{fate} = \theta k_{remt} C \quad (3)$$

where Λ_{fate} is the fate term in eq. (2); and k_{remt} [T^{-1}] is the temporal removal rate. The latter is typically empirically obtained based on $k_{remt} = k_{remx}v$, in which k_{remx} [L^{-1}] is the spatial removal rate, inferred from experimental breakthrough curves. This equation assumes that the removal parameter is scale-invariant and that the effect of heterogeneities is not very important (Pang, 2009). In better detail, the bacterial attenuation mechanisms involved are decay and growth, as well as deposition and release, which relate to the exchange of colloids between the gas, liquid and solid phases (Vega et al., 2003). However, the models used to describe such processes are yet to be established, and the definitions of the various related mechanisms are still under debate. In the current study the distinction is made between attachment, or physicochemical deposition at a single interface, and straining, i.e., retention due to multiple interfaces (Figure 1). This is in agreement with Bradford and Torkzaban (2008), who found that different torques and forces

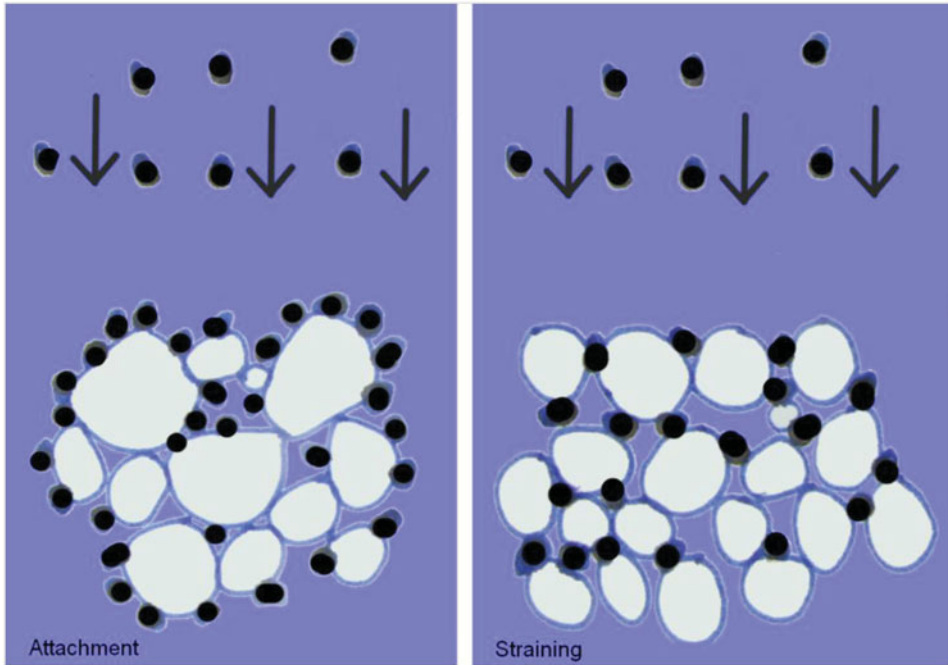


FIGURE 1. A conceptual image of attachment (i.e., retention at one interface), which occurs due to physical-chemical interactions (left), and straining (i.e., retention between multiple interfaces (right), which is related to physical filtration. The figure is redrawn from McDowell-Boyer et al. (1986).

acted on attached versus strained colloids. In the current paper the terms deposition and retention relate to both mechanisms, and attenuation and removal are used interchangeably and refers to the total effects of retention and decay. In eq. (2) Λ_{fate} can thus be specified by

$$\Lambda_{fate} = \Lambda_{att} + \Lambda_{str} + \Lambda_{sur} \quad (4)$$

where Λ_{att} , Λ_{str} , and Λ_{sur} represent attachment (section 2), straining (3), and survival (4), respectively. All of these mechanisms are influenced by the presence of preferential flow paths in the vadose zone, which is addressed in a separate section (5).

2. ATTACHMENT THEORIES AND EXPERIMENTAL FINDINGS

First, this section presents the most common approaches to subsurface transport predictions. Subsequent subsections address experimental findings that are inconsistent with these models and outline alternative methodologies. Deviations from traditional theories relate to both the whole subsurface and to unsaturated filter media in particular, as well as to the fact bacteria are biological particles. Additionally, a subsection is dedicated to the importance of transient flow for deposition and release processes. The last subsection accounts for the factors that influence attachment.

2.1 Modeling Attachment in Saturated Filter Media Using CFT and DLVO

The most important theory for particle deposition in saturated soil is the classic Colloid Filtration Theory (CFT) (Tufenkji and Elimelech, 2004b; Yao et al., 1971). In this model the only attenuation process is attachment to the solid-water interface, i.e., $\Lambda_{sur} = \Lambda_{str} = 0$ in eq. (4), and the concentrations of attached and suspended particles are reduced exponentially with distance:

$$\Lambda_{fate} = \Lambda_{att} = \rho_b \frac{\partial C_{Satt}}{\partial t} = \theta_s k_{Satt} C \quad (5)$$

where Λ_{fate} and Λ_{att} are the fate and attachment terms in eq. (4), respectively; ρ_b [ML⁻³] is the soil bulk density; C_{Satt} [NM⁻¹] is the solid phase concentration of attached bacteria; θ_s [-] is the saturated volumetric water content; and k_{Satt} [T⁻¹] is the solid phase attachment rate, which is modeled according to (Logan et al., 1995)

$$k_{Satt} = \frac{3(1 - \theta_s)v}{2d_{50}} \eta_0 \alpha \quad (6)$$

where d_{50} [L] is the median grain diameter; η_0 [—] is the single-collector contact efficiency, which can be calculated theoretically and is dependent on sedimentation, interception and Brownian diffusion (Rajagopalan and Tien, 1976; Tufenkji and Elimelech, 2004a; Yao et al., 1971); and α [—] is the total sticking efficiency. The latter is often considered to be an empirical parameter, typically derived from fits of eq. (2) and eq. (5) to a breakthrough curve and calculations of η_0 , and the applications of eq. (6). It can theoretically be calculated using the classic DLVO theory (Derjaguin and Landau, 1941; Verwey and Overbeek, 1948), which determines the forces between charged particles in a fluid, based on surface thermodynamic properties. The total Gibbs energy is calculated as the sum of electrostatic and van der Waals forces (Figure 2), as dependent on the distance between the particles and the interface, typically effective at distances < 50 nm (Vigean et al., 2002). Primarily, repulsive electrostatic interactions are encountered as a particle approaches a collector, and attractive interactions increase and begin to dominate as the distance decreases, e.g., due to hydrodynamic forces (Chen et al., 2008). The magnitude of the energy barrier could be used to theoretically evaluate the probability of attachment in the primary minima, α , using the Interaction Force Boundary Layer model (Spielman and Friedlander, 1974), which thus links CFT to DLVO. Alternatively, it could be assumed that only second energy minima attachment occurs, and that release takes place for sufficiently high thermal energy of the attached particles (Hahn and O'Melia, 2004; Syngouna and Chrysikopoulos, 2011). Such attachment is weaker and occurs at a greater distance, and, therefore, the fluid drag force has a larger effect (Torkzaban et al., 2009).

Electrostatic forces depend on both the particle and the filter media surface charge, typically considered to be equal to the surface ζ -potential (Haznedaroglu et al., 2009). This has traditionally been calculated from the particle electrophoretic mobility using the Smoluchowski equation (Ohshima and Kondo, 1991). Soil, air, and *E. coli* generally carry negative charges; typically, the ζ -potential of *E. coli* is larger than soil grains, which is larger than air (most negative). The calculated sticking efficiency is thus different for the cell-air and cell-grain interfaces. Further, organic matter is also typically negatively charged (Pang, 2009). This implies that the electrostatic interactions between *E. coli* and the collectors are generally repulsive (Bradford and Torkzaban, 2008; Haznedaroglu et al., 2009; Kim et al., 2009a). CFT and eq. (6) only apply to saturated conditions. However, DLVO interactions have been considered to be the most important factors for retention in the literature, also considering unsaturated conditions, as stated by Steenhuis et al. (2006). CFT and DLVO thus provide foundations for particle transport theory in the subsurface; nevertheless, their relevance for the predictions of bacterial transport remains limited, as addressed subsequently.

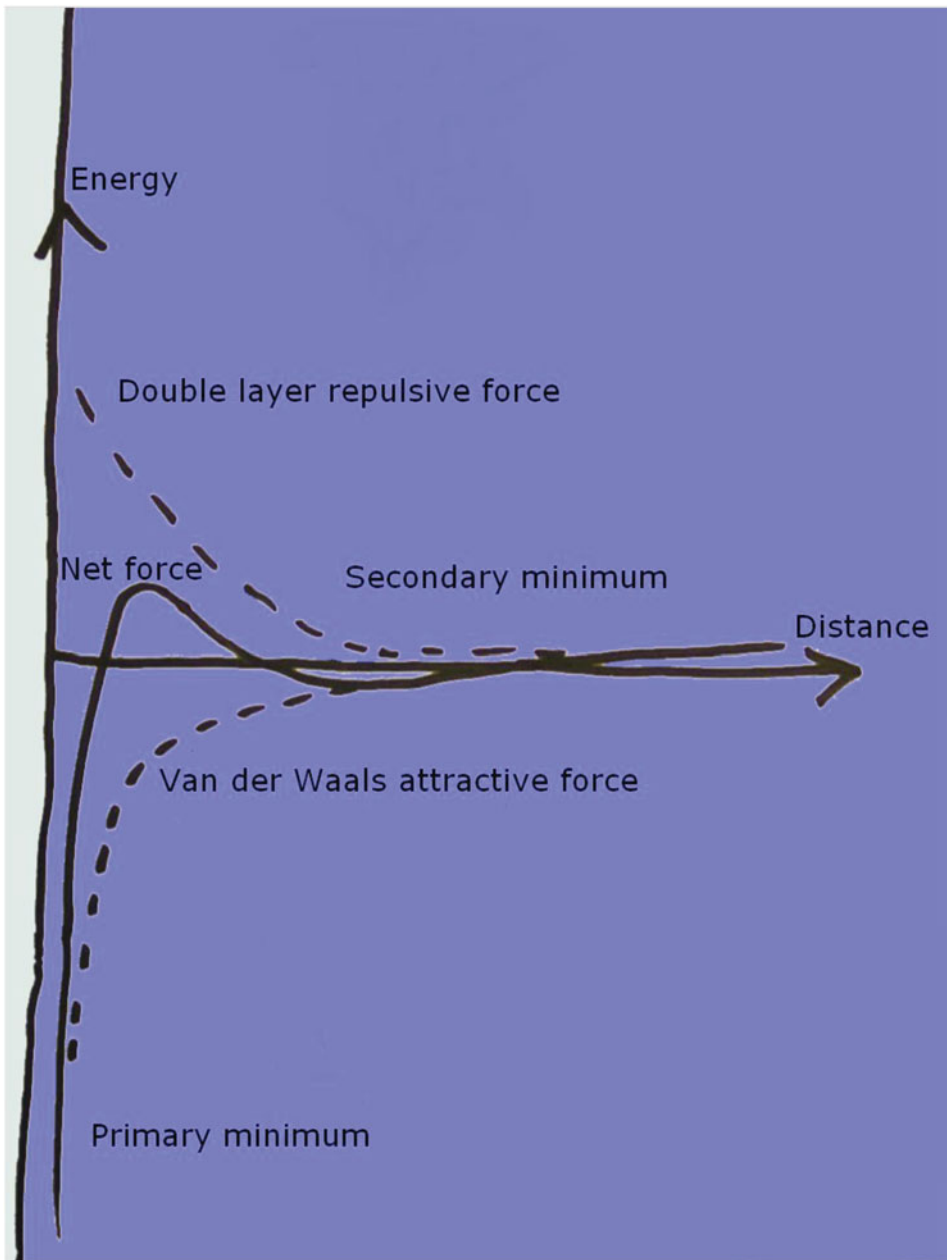


FIGURE 2. Conceptual representation of interaction forces according to the classic DLVO theory (not to scale): the total Gibb's energy is calculated as the sum of electrostatic interactions, double layer repulsive forces, which are generally repulsive, and van der Waals forces, which are attractive. The primary and secondary energy minima are found at the distance where the net force is negative. The figure is redrawn from DeNovio et al. (2004).

2.2 Inadequacies of CFT and DLVO in Both Saturated and Unsaturated Media

This section addresses traditional theories in relation to the subsurface in general.

2.2.1 ATTACHMENT TO GRAINS UNDER UNFAVORABLE CONDITIONS AND THE IMPORTANCE OF HETEROGENEITY

Originally, CFT was developed for favorable attachment conditions, in which case the sticking efficiency, α , in eq. (6) equals one (Bradford and Torkzaban, 2008). It was assumed that deposition was irreversible and only occurred in the primary energy minimum of the DLVO energy distribution (Bradford and Torkzaban, 2008; Ryan and Elimelech, 1996). Thus, the sticking efficiency theoretically depends on the amount of favorable attachment sites, which could occur when natural sand contains patches of Al- and Fe-oxides that carry positive charges (Zhang et al., 2010). However, favorable attachment conditions are rare for most natural subsurface environments, since the filter media is generally negatively charged (e.g., Bradford et al., 2009). For unfavorable attachment environments, α , and hence k_{Satt} in eq. (6), is thus theoretically equal to zero (Bradford and Torkzaban, 2008); nevertheless, *E. coli* have been shown to indeed deposit in such conditions (e.g., Jiang et al., 2007).

Recent research indicates that this type of attachment could be attributed to microscopic heterogeneities on the collector surfaces, which could induce attachment despite that the average surface properties are unfavorable (Figure 3, mechanism H) (Redman et al., 2004; Santore and Kozlova, 2007; Torkzaban et al., 2010). Bradford and Kim (2012) reported that the classic DLVO theory did not explain *E. coli* D21g transport behavior in saturated Ottawa sand (360 μm), as it does not account for physical (surface roughness) or chemical (iron oxides) surface microscopic heterogeneities. They investigated the reason for the previously observed colloid retention-release hysteresis as a result of varying solution ionic strength (at pH 10) in Torkzaban et al. (2010). Bradford and Kim (2012) argued that the dominant cause was the presence of microscopic heterogeneities, based on the observation that the hysteresis effect increased for more heterogeneous filter media (only salt treated sand instead of salt and acid treated sand), and that release was dependent on the relative size of the particle and the heterogeneity. The impact of these micro-heterogeneities increased with higher ionic strength and lower particle size, which was attributed to the smaller size of the electrostatic zone of influence that deepened the local energy minima, and thus increased the local deviation from the average conditions (Figure 2) (Bradford and Kim, 2012). Via mechanistic modeling of attachment in saturated porous media Ma et al. (2011) found that the incorporation of

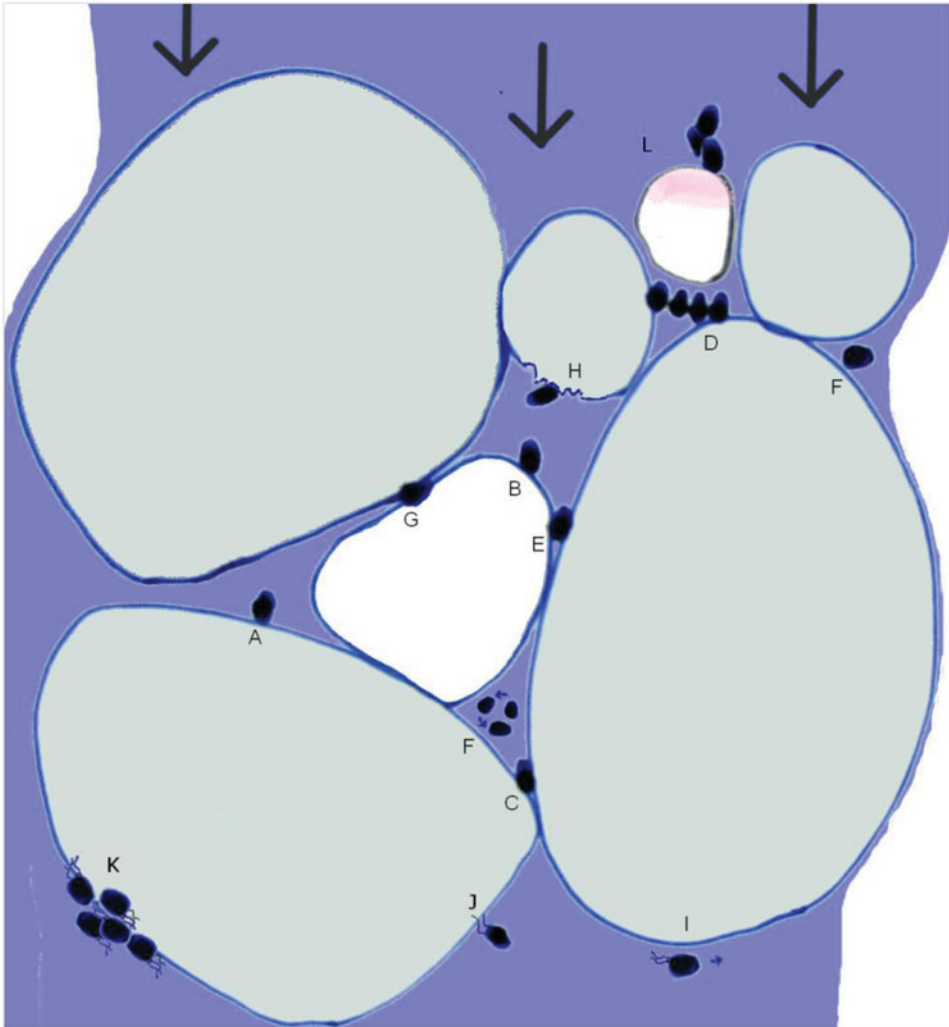


FIGURE 3. Conceptual drawing of various proposed retention mechanisms in unsaturated media (not to scale): (A) solid-water-interface attachment; (B) air-water-interface attachment; (C) wedging at the grain-grain contact point; (D) bridging, clogging, or straining near already deposited bacteria; (E) straining in the air-water meniscus-solid interface, where capillary forces and opposing friction forces induce pinning; (F) retention due to bacterial rotation or immobilization in low-velocity regions (e.g., as a result of weak solid-water-interface attachment and subsequent translation; (G) film straining induced by fluctuations in the water film around a grain; (H) attachment due to heterogeneities that cause a secondary minima locally (i.e., in the zone of influence of the cell); (I) a motile bacterium translating, or swimming, along a grain surface; (J) a bacterium with flagella tethered to a grain; (K) attachment induced by a sticky biofilm surface; (L) floc unit transport and bridging flocculation due to the retention of dissolved organic matter (humic acid) on the particles and the air-water interfaces. Parts of the figure are redrawn from Bradford and Torkzaban (2008).

surface heterogeneities in the hemisphere-in-cell model can explain a variety of observed attachment processes under unfavorable conditions. This model predicts that larger hetero-domains cause greater retention. Accordingly, Shen et al. (2011) reported that heterogeneities in the form of valleys on sand grains can host colloids in second energy minima, shielding them from hydrodynamic shear. Hence, *E. coli* might be attached in soil that is in average unfavorable for attachment if microscopic heterogeneities are present on the collector surfaces.

2.2.2. HYPER-EXPONENTIAL AND NON-MONOTONIC DEPOSITION PROFILES

In CFT the removal rate is constant; however in several studies of *E. coli* the deposition rates have been reported to decrease with the distance (Foppen et al., 2007; Jiang et al., 2007; Kim et al., 2010; Lutterodt et al., 2009a; Torkzaban et al., 2008b). The underlying mechanisms have been debated. One explanation is that the sticking efficiencies vary within the cell population; however, depth-dependency has also been reported for abiotic colloids (Bradford et al., 2002; Li et al., 2004; Torkzaban et al., 2008a), which indicates that the phenomenon is not related to variations in particle characteristics. On the other hand, Li et al. (2004) argued that interaction potentials could indeed vary for abiotic particles as well. In models accounting for collector or bacteria heterogeneities, attachment sites have been divided into fractions with distinct attachment rates (Simoni et al., 1998).

Then again, Torkzaban et al. (2008a) observed depth-dependency even though batch results indicated that colloids neither attached to air nor to soil; hence, colloidal surface charge heterogeneity was an unlikely explanation. Instead, the authors argued that straining in low flow velocity regions was the underlying mechanism (Figure 3, mechanism F). Coherently, depth-dependency has been reported to increase with increasing colloid size and decreasing filter media size (Bradford et al., 2003; Bradford et al., 2006b; Tufenkji and Elimelech, 2005), and when attachment conditions are more unfavorable (Li et al., 2004). The phenomenon has thus been attributed to straining (Bales et al., 1991; Bradford et al., 2002; Bradford et al., 2003; Bradford et al., 2006a; Bradford et al., 2006b; Foppen et al., 2004; Gargiulo et al., 2008; Haznedaroglu et al., 2009; Torkzaban et al., 2008a).

However, addressing microbial transport in general Pang (2009) evaluated 87 field studies in soil, vadose zones, and aquifers and found that the assumption of a constant kinetic attenuation rate fitted well with data in 70% of the cases, and that deviation from this approach applied particularly to organically contaminated, fine filters. In field-scale natural environments with little organic matter this supports the assumption of a kinetic microbial attenuation in accordance with eqs. (3) and (5). In coherence with Pang (2009), reduced colloidal attenuation rates with distance was observed as organic waste, manure and sewage sludge was applied to 4 cm long mini-columns ($k_{remx} \approx 20\text{--}80/\text{m}$), 50 cm long undisturbed columns

($k_{remx} \approx 20\text{--}40/\text{m}$), and in a field scenario of 450 cm ($k_{remx} \approx 2/\text{m}$) (Unc et al., 2012). One hypothesis is that organically contaminated filters increase aggregation, which cause deposition rate time- and depth-dependency, since the presence of humic acid in the transporting solution has been found to induce colloidal flocculation (Morales et al., 2011). Tong et al. (2010) reported that the presence of *E. coli* extracellular polymeric substances, which has been associated with cell-cell interactions and bioclogging (Kim et al., 2010; Or et al., 2007b), enhances the hyper-exponential shapes of the deposition profiles. Another possible explanation to hyper-exponential deposition in the field is that fast flow paths, such as fingers (section 5.1.1), are developed below a water distribution layer in the top soil (Morales et al., 2010); coherently, high colloidal removal in the top soil and small attenuation below this zone has been reported in the field (e.g., Cey et al., 2009).

In addition, non-monotonic profiles have been reported under unfavorable attachment conditions. In intact cores Bolster et al. (1999) found that the amount of deposited bacteria increases with distance near to the inlet and then decreases. Coherently, Li and Johnson (2005) reported that the removal increases within the first 5 cm and then decreases for abiotic colloids under unfavorable attachment conditions in sand, considering different ionic strengths and flow rates in the absence of organic matter. The non-monotonic behavior increased with decreasing ionic strength and disappeared for favorable attachment conditions (Li and Johnson, 2005). These results indicate that neither biological particle properties nor organic matter induces the non-monotonic behavior. Li and Johnson (2005) argued against aggregation as a possible explanation. Downward sloping breakthrough curve plateaus were only manifest for some cases and it was apparent that deposition was non-monotonic already after 1.3 pore volumes; the underlying reasons for the shape of the deposition profiles were not identified in Li and Johnson (2005). Bradford et al. (2006b) however reported that non-monotonic deposition of *E. coli* O157:H7 increased in importance for finer grains and at lower flow rates in saturated sand, and the result was attributed to straining in the form of cell-aggregation and subsequent release of the aggregated cells due to hydrodynamic shearing forces in the event that a critical concentration was reached.

In summation, several hypotheses have been put forward to explain the reasons for the observed non-exponential deposition profiles and the underlying processes remain unknown. Bioclogging, or cell-aggregation on the collector surfaces, appears to be a possible explanation. Findings relating to unsaturated filter media are, however, limited. Further studies are recommended, preferably including micrometer-scale imaging in different parts of a flow system, or imaging of transport in two-dimensional flow cells at the centimeter scale.

2.2.3 INSTANTANEOUS ATTACHMENT AND RELATED MULTI-SITE APPROACHES

An equilibrium isotherm could be applied if deposition is assumed to occur instantaneously (e.g., Toride et al., 1999). A linear isotherm implies that $C_{Satt} = K_{Seqatt}C$, where K_{Seqatt} [L_3M^{-1}] is the equilibrium distribution coefficient:

$$\rho_b \frac{\partial C_{Satt}}{\partial t} = \rho_b K_{Seqatt} \frac{\partial C}{\partial t} \quad (7)$$

Other applied equilibrium models include Freundlich or Langmuir isotherms (Sen et al., 2004). Chen (2008) applied a kinetic rather than an equilibrium equation to describe *E. coli* deposition in unsaturated soil, observing long tails of the breakthrough curve, a sign of kinetic behavior. Rockhold et al. (2004) found that most models in porous media have used kinetic modeling approaches in relation to microbial transport in variably saturated porous media. Considering saturated, dynamic microbial deposition Tufenkji (2007) underlined that the equilibrium assumption implies instantaneous deposition, retarded breakthrough, and negligible irreversible removal, which have rarely been observed; thus, it was concluded that this type of deposition generally does not apply. The same conclusion has been drawn in relation to virus transport in saturated soil (Schijven and Hassanizadeh, 2000), and to bacterial transport in saturated sand columns (Chen and Zhu, 2004).

Equilibrium sorption effects have, nevertheless, often been included in models with several retention mechanisms. Morales et al. (2009) found that colloidal fate according to eq. (7) in an addition to eq. (5) (kinetic deposition) fairly well reproduces breakthrough curves of colloidal transport in fine sand; however, under unsaturated conditions the retardation factor, $1 + \rho_b K_{Seqatt}/\theta$, was equal to 1 and 0.84 for rough surfaces and smooth surfaces, respectively, which indicates that bacterial transport was similar and accelerated as compared to the fluid, respectively, and the latter value lacks a physical foundation as it implies negative concentrations.

Moreover, equilibrium and kinetic solid-water interface attachment are accounted for in the deterministic two-region non-equilibrium transport model (Toride et al., 1999; van Genuchten and Wagenet, 1989), described in section 5.2. With various adjustments this model has been applied to column studies of *E. coli* in unsaturated sand (e.g., Chen et al., 2010b; Jiang et al., 2007). Such models have often showed good breakthrough curve fits; however, they include many fitting parameters and therefore inference relating to the underlying processes is limited. This issue is further addressed in section 6.3.3.

2.3 Alternative Approaches in Unsaturated Systems

This subsection is concentrated on deviations from the DLVO theory in unsaturated porous media.

2.3.1 STERIC EFFECTS AND CAPILLARY FORCES

DLVO theory has further not been able to entirely explain retention behavior in unsaturated media (Lazoukaya et al., 2006; Schäfer et al., 1998a; Sirivithayapakorn and Keller, 2003b; Steenhuis et al., 2006). Attachment to air has been attributed to non-classic DLVO forces, such as steric, capillary, and hydrophobic forces (Bradford and Torkzaban, 2008; Grasso et al., 2002; Lazoukaya and Jin, 2008). Steric interactions occur as two polymer-coated surfaces approach each other, and can be both repulsive and attractive (Grasso et al., 2002). The latter occurs when surface polymers extend far from their surface and bind to another surface, causing a bridging force, typically effective at large distances; however, the most important type of steric interactions are weakly repulsive and arise due to the restricted thermal movement of the polymer chains as the surfaces approach each other (Grasso et al., 2002). Kim et al. (2010) found that steric interactions were important to understand *E. coli* attachment in saturated systems; however, no results that relate to the importance of steric interactions for *E. coli* attachment to air have been found.

Capillary interactions depend on the surface tension, the size and shape of the colloids as well as the contact angles of the soil and the colloids (Shang et al., 2009). They could be calculated using the Young-Laplace equation for spherical particles; however, larger capillary forces are experienced for irregularly shaped particles, such as bacteria, in which case there are no theoretical approaches (Shang et al., 2009). Capillary forces particularly affect detachment in the presence of a mobile air phase, as well as straining near the air-water interface. The related mechanisms are further addressed in subsections 2.5 and 3.2.1.

2.3.2 AIR-WATER-INTERFACE ATTACHMENT AS RELATED TO HYDROPHOBIC INTERACTIONS

Hydrophobic forces occur principally due to Lewis acid-base interactions, which account for hydrogen-bonding and structural effects (Bellon-Fontaine et al., 1996; Van Oss et al., 1986; Van Oss et al., 1988). These can be stronger than van der Waals and electrostatic interactions (Chen and Strevett, 2002; Schäfer et al., 1998a). It is considered that sand and *E. coli* are relatively hydrophilic, whereas air is hydrophobic (Crist et al., 2005; Schäfer et al., 1998a). It has further been argued that it is sufficient if only one of the materials is hydrophobic for attraction to arise (van Oss, 1993); hence, hydrophobic interactions could induce *E. coli* attachment to air. It has been argued that hydrophobicity is a determinant factor for air-water-interface attachment of colloids (Schäfer et al., 1998b). Hydrophobic interactions can be calculated using the particle and collector contact angles (Schäfer et al., 1998a); however, the understanding of their origin is still limited (Bradford

and Torkzaban, 2008). Bradford and Torkzaban (2008) calculated that particles with contact angles $<30^\circ$ had a positive total interaction energy near the air-water-interface (solution ionic strength 10 mM). This indicates that *E. coli*, with a contact angle of 13° reported in Chen (2008), would not attach to air. However, Chen (2008) calculated that the Lifshitz–van der Waals and Lewis acid-base interactions were both attractive and that the sum of all energies was negative at the *E. coli*–air interface (de-ionized water); hence, it was argued that air-water interface attachment had occurred. The importance of hydrophobic interactions for attachment to air of hydrophilic colloids ($<30^\circ$ contact angle), such as *E. coli* remains unresolved (Lazouskaya and Jin, 2008).

2.3.3 MODELING AIR-WATER-INTERFACE ATTACHMENT

Air-water interface attachment could be modeled according to Saiers and Lenhart (2003b) and Kim et al. (2008a):

$$\Lambda_{att} = \rho_b \frac{\partial C_{Satt}}{\partial t} + \theta_A \frac{\partial C_{Aatt}}{\partial t} \quad (8)$$

where Λ_{att} is the attachment term in eq. (4); θ_A [–] is the volumetric air content; and C_{Aatt} [NL $^{-3}$] is the air-phase concentration of attached bacteria. Air-water interface deposition has generally been modeled using a kinetic equation (e.g., Wan and Tokunaga (1997)). In addition, Lenhart and Saiers (2002) included a factor accounting for a maximum retention capacity of the air, in accordance with

$$\theta_A \frac{\partial C_{Aatt}}{\partial t} = \theta k_{Aatt} \left(\frac{C_{Aattmax} - C_{Aatt}}{C_{Aattmax}} \right) C \quad (9)$$

where $C_{Aattmax}$ [NL $^{-3}$] is the maximum air-water interface retention capacity; and k_{Aatt} [T $^{-1}$] is the air-phase attachment coefficient. Under steady-state conditions, attachment to air has generally been considered to be irreversible (Chen et al., 2008; Schäfer et al., 1998a; Schäfer et al., 1998b; Shang et al., 2009; Sirivithayapakorn and Keller, 2003b).

2.3.4 AIR-WATER VERSUS AIR-WATER MENISCUS-SOLID DEPOSITION

The understanding of colloidal attachment to air is, as yet, evolving. In early pore-scale imaging studies Wan and Wilson (1994a) argued that it applied to both negatively and positively charged colloids, as well as hydrophobic and hydrophilic ones; however, Wan and Tokunaga (2002) later reported that air-water interface attachment only had occurred for positively charged (clay) colloids. More recent pore-scale imaging in partially saturated translucent silica sand, cleaned through rinsing and combustion, has revealed that retention of hydrophilic microspheres mainly occurs at the air-water meniscus-solid interfaces (Zevi et al., 2005), and not at the air-water interfaces (Steenhuis

et al., 2006) (Figure 3, mechanism E), and Steenhuis et al. (2006) reported that the retention rates of abiotic colloids and unaltered *E. coli* are similar. Crist et al. (2004) re-investigated the images of Wan and Wilson (1994a) and argued that the colloids had, in fact, been retained at the air-water meniscus-solid interface. Nevertheless, Auset et al (2005) did observe that *E. coli* attachment to air occurred in variably saturated micro-models; however, only two dimensions were assessed, and it is possible that the top or bottom surface had influenced the bacterial retention (Keller and Auset, 2007). Lazouskaya and Jin (2008) argued that more retained colloids are likely to be associated with the three-phase contact line than with the air-water interface in natural filter media. Based on direct visualization neither Zevi et al. (2005), nor Chen and Flury (2005), reported significant attachment of hydrophilic colloids to the air-water interface.

2.4 Importance of Biological Properties

This subsection addresses the limitations of CFT and DLVO theory with regard to the effects of the characteristics of *E. coli*.

2.4.1 EFFECTS OF SURFACE MACROMOLECULES

According to the classic DLVO theory the attaching particle is a rigid, smooth spherical, abiotic colloid. Such differ in rheology from *E. coli*, which have soft and heterogeneous surfaces, with biomolecules, such as proteins, lipopolysaccharides, and extracellular polysaccharides (Chen and Walker, 2012; McClaine and Ford, 2002b; Tufenkji, 2007). The length of these surface macromolecules is generally greater than the interaction distances accounted for in DLVO theory (Ginn et al., 2002). It has thus been proposed that soft-particle theory, which accounts for particles coated by ionpenetrable polymeric layers (Kim et al., 2009b), should be used to predict the surface potential instead of the Smoluchowski equation, which links the electrophoretic mobility with the surface potential (Morisaki et al., 1999; Ohshima and Kondo, 1991; Ohshima, 1995). Based on soft-particle theory McClaine and Ford (2002b) reported that wild-type *E. coli* had a surface potential of -24 mV, as compared to -29.5 mV considering the Smoluchowski equation (20 mM). With this approach the height of the repulsive energy barrier is thus reduced. However, for lower solution ionic strengths, the difference between the two approaches was small (McClaine and Ford, 2002b).

Moreover, the presence extracellular macromolecules implies that the steric effects are more important for bacteria than for abiotic colloids. In experimental studies of transport in saturated porous media Kim et al. (2010) inferred that retention behavior of *E. coli* O157:H7 could not be predicted using classic DLVO theory that neglects steric interactions; it was argued that the presence of macromolecules had hindered irreversible attachment (repulsive steric interactions) under favorable attachment conditions (ionic

strengths ≥ 1 mM), and had caused increased attachment and ripening (attractive steric interactions) under unfavorable conditions (ionic strength ≤ 0.1 mM) (Kim et al., 2010).

2.4.2 MOTILITY AND ITS EFFECT ON ATTACHMENT

The limitations of DLVO theory have also been attributed to the motility of bacteria. For low ionic strengths (20 mM) attachment was higher for motile (wild type) than paralyzed cells (McClaine and Ford, 2002). McClaine and Ford (2002) hypothesized that motile cells can get closer to the collector surface, since their swimming velocity is less reduced near the collector as compared to the settling velocity of nonmotile particles. In accordance with findings for wild type *E. coli* in McClaine and Ford (2002). Vigeant et al. (2002) found that the attachment of motile cells to glass plates was not affected by electrostatic interactions. The authors argued that motile bacteria swim parallel to the collector surfaces at distances that are too great for DLVO forces to have an effect (Figure 3, mechanism I), whereas nonmotile, irreversibly immobilized cells are indeed attached due to DLVO forces. Vigeant et al. (2002) further proposed that the cells occupy one of three compartments: bulk fluid, at distances $> 10 \mu\text{m}$ from the surface, where they experience no effect of the collector surface; near-surface bulk, at about $20 \text{ nm} - 10 \mu\text{m}$ from the surface, where the hydrodynamic effect of the surface is experienced; and near-surface constrained, at $< 20 \text{ nm}$ from the surface, where both hydrodynamic and physicochemical influences of the surface are experienced. In unsaturated conditions mobility is, however, likely to be reduced. In regions of locally high saturation, where water films are much thicker than the width of the *E. coli*, near surface bulk retention could cause increased deposition rates for *E. coli* as compared to colloids; however, Or et al. (2007b) and Wang and Or (2010) argued that mobility is restrained due to capillary forces, and that free swimming microbes are not probable in most unsaturated systems.

2.5 Bacterial Remobilization and Its Relation to Transients in Water Content

Re-mobilization of colloids could be a passive bacterial process, induced by diffusive and hydrodynamic forces, which could cause interfacial shear stress (Corapcioglu and Choi, 1996; Wan and Wilson, 1994a), or an active one, for example, in response to changes in nutrient flux (Tufenkji, 2007; Wang et al., 2011). Release from soil is usually modeled with the addition of a sink term to eq. (5) (Bradford et al., 2004):

$$\Lambda_{att} = \rho_b \frac{\partial C_{Satt}}{\partial t} = \theta k_{Satt} C - \rho_b k_{Sdet} C_{Satt} \quad (10)$$

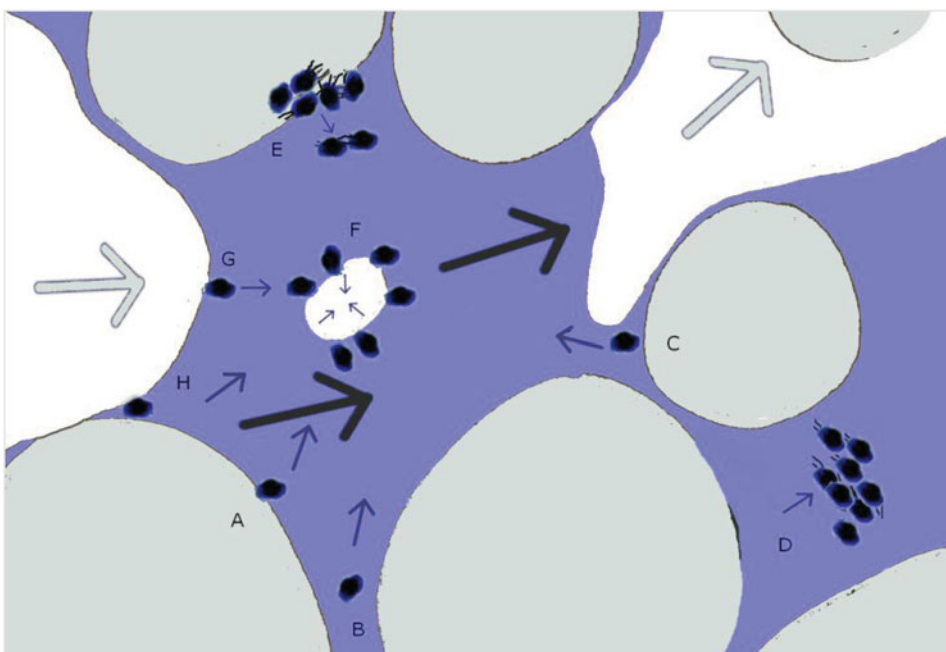


FIGURE 4. Conceptual drawing of various proposed re-mobilization mechanisms under transient and steady-state conditions (not to scale). The large black and light blue arrows indicate the direction of movement of the water and air interface, respectively, and the small blue arrows indicate the direction of the re-mobilizing cells. (A) detachment from the solid-water interface due to hydrodynamic shear stress; (B) re-mobilization from a low-velocity region between grains due to increased water flux; (C) release from the air-water meniscus-solid interface straining induced by increased water content; (D) sloughing of a biofilm in a low flow velocity region, due to reduced diffusion of nutrients to cells deep in the film; (E) detachment of the outer layer of a biofilm due to shear forces in a high flow velocity region; (F) detachment from the air-water-interface due to dissolution of the gas phase and air bubble disappearance; (G) cell transport with the moving air-water-interface toward the outlet; (H) sloughing of a weakly attached cell due to capillary forces induced by the approaching air-water-interface.

where k_{sdet} [T^{-1}] is the detachment rate. In breakthrough curves, detachment can be inferred from significant tailing (Tufenkji, 2007). Figure 4 depicts various proposed detachment processes in variably saturated soil, in which mechanism A shows detachment from the solid-water interface due to shear forces, and mechanism B shows release from a previously stagnant region due to increased flux, and the other mechanisms will be addressed subsequently.

2.5.1 RELEASE DURING STEADY-STATE FLUX

In unsaturated systems during steady-state flow (i.e., without chemical or hydrodynamic perturbations), colloid mobilization is generally considered to be very small (DeNovio et al., 2004). In this case the release is dependent

on the slow exchange of water between the immobile and the mobile zones (Gao et al., 2006). In a review relating to microbial subsurface transport Ginn et al. (2002) stated that detachment rates are generally low and do not significantly contribute to concentrations during the main breakthrough pulse. The authors, however, added that slow release over long time-periods could increase long-distance transport, which could have severe consequences when small amounts of bacteria cause great damage, such as in drinking water; on the other hand, this type of transport has rarely been monitored in the field, as discussed by Tufenkji (2007). In undisturbed, unsaturated lysimeters under effluent irrigation Pang et al. (2008) found that the detachment rate of fecal coliforms and *Salmonella* bacteriophages was on average 1% of the attachment rate, and argued that detachment could be considered negligible. Steady-state conditions with relatively stable moisture levels typically occur in the lower part of the unsaturated zone, where infiltration changes are damped by both time and matric forces, whereas variably saturated conditions apply mainly in the upper part of the vadose zone, where the moisture content varies with the time of day and rainfall rates (Holden and Fierer, 2005; Powelson and Mills, 2001).

2.5.2 RE-MOBILIZATION AS A RESULT OF DRAINAGE

Imbibitions could cause microbial mobilization due to faster flow, which increases shear forces on the attached microbes (Bradford and Torkzaban, 2008). However, under transient flow recent research suggests that drainage induces detachment to a higher degree than imbibitions. Shang et al. (2009) found that moving air-water interfaces provide an effective mobilization mechanism for particles attached to the solid-water interface (Figure 4, mechanism H). The associated repulsive capillary forces are especially important for irregularly shaped particles with diameters $< 2 \mu\text{m}$, such as *E. coli* (Shang et al., 2009). In this case the capillary forces exceed gravity forces and DLVO interactions, according to Shang et al. (2009). Cheng and Saiers (2009) evaluated colloidal mobilization from sand during drainage and imbibitions and found that the relative mass recovered was significant (1.5–24.3%) and that the release of colloids due to drainage was always higher than due to imbibitions; the effluent concentration peaked shortly after the passage of the drying front and declined soon afterward. Accordingly, Powelson and Mills (2001) found a mismatch between the pulses of water and *E. coli*. Studying sand columns with intermittent flow Auset et al. (2005), coherently, reported a delay of 20 min between the hydraulic and *E. coli* peaks; most of the released bacteria came after the second water pulse, and small but significant amounts of *E. coli* ($10^{-7} < C/C_0 < 10^{-4}$) were present in the effluent 72 h after the initial pulse with bacteria. In micro-models Auset et al. (2005) argued that detachment from the air-water interface occurred only when the air bubbles had been completely dissolved (Figure 4, mechanism F). The passing air front swept up loosely attached colloids and pushed

them together. A hypothesis relating to the results in Auset et al. (2005) is that, at first, the water front re-mobilized loosely deposited cells due to shear and repulsive capillary forces resulting from the moving air-water interface; subsequently, with the drainage front, the approaching air-water interface captured the free cells and pushed them towards the outlet or nearby grains. The effect would be that the effluent cells in the early pulses originated from the lower parts of the column, and that the effluent cells arriving later had once been weakly deposited higher in the column and then been released, re-deposited and re-released with the subsequent cycles of drainage and imbibitions (Fig. 4, mechanism G). Possibly, some cells were pushed by the air-water interface towards the primary minima of the solid-water interfaces. With the imbibitions and drainage cycles, the consequence would thus be that the cells were consistently, but to a decreasing degree, transported towards the outlet, which is coherent with breakthrough data in Auset et al. (2005).

2.5.3 RELEASE AS RELATED TO DISTANCE AND FREQUENCY OF PULSES

The frequency of pulses can affect the total re-mobilization rate. Cheng and Saiers (2009) argued that colloidal transport was more sensitive to rainfall frequency than total rainfall amount, based on the observation that short infiltration events could mobilize nearly as many colloids as few and long-lasting ones in columns with sand. However, in undisturbed lysimeters Jiang et al. (2010) argued that more frequent irrigation events reduced the formation of preferential flow paths (section 5), which can induce very fast colloidal transport in the case of sudden and heavy irrigation (Cey et al., 2009; Jiang et al., 2010). In this way more frequent and less intense irrigation events, as compared to few and heavy ones, could reduce transport. These findings underline the difficulty to generalize results from controlled environments in the laboratory to complex conditions in the field.

Further, research indicates that the importance of release could decrease with increased travel distances. Bridge et al. (2009) reported that the mobilization of carboxylate-latex microspheres decreases with the distance travelled by the drainage front in variably saturated quartz sand ($d_{50} = 713 \mu\text{m}$). The mobilization rates varies directly with the drainage rate (0.2–1.0 mL/min), and inversely with the ionic strength (2–50 mM) in Bridge et al. (2009), who inferred that re-deposition of the colloids had occurred some times after the release.

2.5.4 TRANSIENT FLOW AND RE-MOBILIZATION MODELING

Saiers and Lenhart (2003a) developed a model that accounts for the effects of transient water conditions on deposition and release. It includes different compartments of immobile colloids that each entails a particular relationship between the mobilization rate and a particular critical moisture content. It

has the following structure:

$$\Lambda_{att} + \Lambda_{str} = \rho_b \frac{\partial C_{SAdep}}{\partial t} = \theta k_{SAdep} C - \rho_b \sum_{i=1}^{N_C} k_{SArel_i} C_{SAdep_i} \quad (11)$$

where Λ_{att} and Λ_{str} is the attachment and straining term in eq. (4), respectively; C_{SAdep} [NM⁻¹] is the total number of deposited cells per bulk mass of filter media; k_{SAdep} [T⁻¹] is the total solid- and air-water interface deposition coefficient; N_C is the number of compartments; k_{SArel_i} [T⁻¹] is the release rate from compartment i ; and C_{SAdep_i} is the concentration immobilized colloids in the i th compartment. The deposition rate is dependent on the saturation and the pore-water velocity according to (Saiers and Lenhart, 2003a):

$$k_{SAdep} = \left[1 - \left[\frac{\theta - \theta_r}{\theta_s - \theta_r} \right]^\mu \right] N v^{0.5} \quad (12)$$

and the release rate is equal to:

$$\begin{aligned} k_{SArel_i} &= N_2 v^\kappa, \quad \text{for } \theta \geq \theta_{crit_i} \\ k_{SArel_i} &= 0, \quad \text{for } \theta < \theta_{crit_i} \end{aligned} \quad (13)$$

where $\mu[-]$, $N[(LT)^{-0.5}]$, $\kappa[-]$ and $N_2[L^{-\kappa}T^{\kappa-1}]$ are empirical parameters; and θ_{crit_i} [-] is the critical moisture content in compartment i , which reflects the water content needed for the water films around grains to connect and induce release. This model was found to fit *E. coli* breakthrough curve data fairly well for columns fed intermittently with wastewater (Auset et al., 2005). Due to hysteresis in hydraulic properties the retention processes could also depend on the wetting history of the porous medium (Lenhart and Saiers, 2002). Cheng and Saiers (2009) developed the model above, assuming that k_{SAdep} is constant with velocity, similar for all compartments, and can vary between drainage and imbibitions; further, release varies with each compartment and occurs only when there is a change in flow during drainage and imbibitions. The advantage of the models discussed in this subsection is that retention and release kinetics are linked to physical properties of the porous medium; however, the amount of fitting parameters remains high, especially if several compartments are assumed considering that the critical moisture content of each compartment needs to be specified empirically. Parameter identification issues are discussed in 6.3.3.

2.6 Factors That Affect Attachment Behavior

Attachment is dependent on the physicochemical interactions between the bacteria, the collectors and the solution (Elimelech and O'Melia, 1990; McDowell-Boyer et al., 1986). An evaluation of which factors affect attachment is, however, complicated by the fact that a limited amount of studies distinguish between attachment and straining.

2.6.1 COLLECTOR CHARACTERISTICS

According to DLVO theory attachment rates increase with increasing difference in surface charge between the colloid and filter media interfaces. *E. coli* are negatively charged (Foppen and Schijven, 2006) and allophanic soils, derived from volcanic ash, have a net positive charge (Pang et al., 2008); coherently, it has been reported that the fecal coliform attenuation rates are higher in volcanic soils than in greywacke-derived silt loams, and lowest in granular young sandy soils (Pang et al., 2008). Correspondingly, in positively charged filter media, Kim et al. (2008b) found that the *E. coli* mass recoveries decreased as the content of Al- and Fe-coated sands increased. However, Dong et al. (2002) found that the attenuation was primarily controlled by grain size rather than by the abundance of clay as well as iron and aluminum hydroxide coatings in the case of the bacteria *Comamonas* sp. DA001, which are similar to nonmotile *E. coli* with regard to size, surface charge, and hydrophily.

Further, an increased filter media surface area is expected to increase attachment due to the enlarged particle-collector contact area. Capillary forces are increased in finer grained media, which affects hydraulics and decreases the flow velocity. A lower flow rate has been reported to increase the *E. coli* removal rate (Stevik et al., 1999a; Stevik et al., 1999b). Microbial attenuation has, accordingly, been found to be high in pumice sand, fine sand, and highly weathered aquifer rocks, and low in structured clayey soils, stony soils, coarse gravel aquifers, and fractured rocks (Pang, 2009). In addition, increased solid-water interface attachment of colloids due to increased surface roughness has been reported. Chen et al. (2010a) studied deposition of negatively charged, relatively hydrophobic colloids to stainless steel and aluminum alloy surfaces that were bare (positively charged in 1 and 10 mM ionic strength) and zeolite-coated (negatively and positively charged in 1 mM and 10 mM ionic strength, respectively). The general trend, considering the four surface types and three ionic strengths (1–100 mM), is that deposition increases with a higher surface roughness—also when the physical heterogeneities are two orders of magnitude smaller than the colloids (Chen et al., 2010a). Grain surface roughness, which ranges from a few to hundreds of nm, could reduce interaction energies due to reduced asperity radius, and locally increased colloid-collector distances (Bhattacharjee et al., 1998; Chen

et al., 2010a). This reduces the repulsive forces in negatively charged filter media.

Recent findings indicate that surface heterogeneity is essential for solid-water interface attachment and grain-grain retention to take place. In saturated column studies Syngouna and Chrysikopoulos (2011) reported that *E. coli* CN-13 mass recoveries are almost complete (94.5–100%) in sand, considering grain sizes of 1180–1700 μm , 425–600 μm , and 150–212 μm and Darcy velocities of 0.16, 0.31, and 0.51 cm/min and a solution ionic strength of 2 mM (nine studies in total). Further, they did not observe any important relationships between removal and the grain size or the pore-water velocity (Syngouna and Chrysikopoulos, 2011). The quartz sand had been cleaned with 70% nitric acid during 3 h to remove physicochemical heterogeneities, such as organic coatings and iron hydroxide. Acid treatment could also reduce surface roughness. Bradford and Kim (2012) exposed salt treated sand to storage in 12 N hydrochloric acid overnight and found that the BET surface area was reduced by 19.6%. It was, moreover, found that hydrophilic, negatively charged microspheres (1.1 μm) are almost completely transported (97%) through a saturated column containing the salt and acid treated sand (360 μm) at an ionic strength of 6 mM (Bradford and Kim, 2012). These results indicate that smooth surfaces preclude retention at the solid-water interfaces or near the grain-grain interfaces.

2.6.2 SOLUTION IONIC STRENGTH AND pH

Increased solution ionic strength is expected to increase attachment; in this case the diffuse double layers surrounding the cells are compressed, thus reducing the repulsive electrostatic double-layer forces, increasing the ζ -potential (making it less negative), and increasing the depth of the second energy minimum (Keller and Auset, 2007). Accordingly, a higher ionic strength has been shown to increase deposition of two non-flagellated K-12 strains (*E. coli* D21g and XL1-Blue) in the range 10–100 mM at a pH of 5.6–5.8 (Kim and Walker, 2009). This trend was also reported for *E. coli* O157:H7 at pH 5.8 (Kim et al., 2009a); however, for the same strain deposition decreased with higher ionic strength at high pH of 8.4 and 9.2 (Kim et al., 2009a). Haznedaroglu et al. (2009) reported that *E. coli* O157:H7 deposition was independent of the ionic strength in the range 1–100 mM.

Typically, a higher solution pH reduces deposition, which has been reported for *E. coli* O157:H7 at a given ionic strength (Kim et al., 2009a). This has been attributed to increased double layer thickness and decreased ζ -potential (Morales et al., 2011; Wan and Tokunaga, 2002), as well as macromolecule conformational changes that increase the cell separation, called pH-associated electrosteric stabilization (Kim et al., 2009a). Morales et al. (2011) found that the ionic strength had higher effect on retention than pH in the absence of humic acid (dissolved organic matter); however, pH had large importance in its presence. Nevertheless, the pH of groundwater and manure

slurry is approximately neutral (Schijven and Hassanizadeh, 2000; Unc et al., 2012) and groundwater and sewage water generally have low ionic strengths that are less than 10 mM (Powelson and Mills, 2001; Wang and Zhou, 2010; Yang et al., 2007). Hence, at least in the case of the pathogenic *E. coli* strain O157:H7 the results above indicate that ionic strength variations are not very important under typical environmental conditions; nevertheless, differences in behavior between various *E. coli* strains underline the need for further clarification.

For transient chemical conditions reduced ionic strength theoretically implies colloid re-mobilization, since the depth of the second energy minima is reversible with ionic strength. Coherently, Torkzaban et al. (2010) found that detachment of colloids from sand and glass beads occurred as the solution ionic strength was decreased. However, they also reported that the ionic strength history mattered: attachment is higher for an ionic strength decrease than an increase. Colloid release is particularly pronounced as the solution ionic strength decreased to 0 mM; in this case the secondary minimum is almost completely removed (Torkzaban et al., 2010), and deposition was attributed to grain-grain interface straining. This retention hysteresis was later found to increase with more micro-heterogeneities (2.1.1) (Bradford and Kim, 2012). However, Bradford and Kim (2012) reported that reversibility increased and that hysteresis decreased for *E. coli* as compared to smaller particles, due to the higher effect of the hydrodynamic torque and the smaller depth of the local energy minima for the larger particle due the larger local zone of influence.

2.6.3 DISSOLVED ORGANIC MATTER CONTENT AND SURFACTANTS

Moreover, the presence of dissolved organic matter can increase transport, especially in the case of low pH and high ionic strength. Morales et al. (2011) evaluated the effect of humic and fulvic acids (20 mg/L) on retention of carboxylated spheres in unsaturated quartz sand (400–590 μm) in solutions of ionic strengths 0 and 1 mM, at acid, neutral, and basic pH (61% saturation). It was found that the presence of humic acid increases transport significantly as compared to de-ionized water, whereas the fulvic acid only has a marginal effect. Adsorption of humic acid to the colloids and the air-water interface increased electrostatic repulsion and repulsive steric surface structures—the colloids had obtained more pronounced soft particle characteristics. This decreased solid-water interface attachment rates considerably and induced bridging flocculation at the air-water interfaces (Figure 3, mechanism L); taken as a whole, the removal was reduced and the mass recovery was 5–13 times higher than in de-ionized water (Morales et al., 2011). Morales et al. (2011) argued that humic acid had augmented the hydrophobicity of the colloids. Yang et al. (2012) coherently found that the presence of humic acid in the transporting solution decreases *E. coli* BL21 (nonmotile) and *E. coli* C3000 (motile) retention in saturated quarts sand, at different ionic

strengths (5 and 20 mM) in both the presence and absence of extracellular polymeric substances on the cell surfaces, even at concentrations as low as 1 mg/L; this was attributed to the competition of retention sites by humic acid and enhanced electrostatic and steric repulsions due to the presence of suspended humic acid around bacteria. Natural organic matter is ubiquitous in the natural environment and in wastewater (Morales *et al.*, 2011; Yang *et al.*, 2012) and concentrations can range from orders of few hundred $\mu\text{g/L}$ to a few hundred mg/L (Yang *et al.*, 2012). Thus, reduced retention due to the presence of humic acid is likely to be important in unsaturated, organically contaminated superficial soil.

Lastly, the presence of surfactants in the transporting solution, such as alcohols and aromatics, reduces the surface tension, thus decreasing retention (Zhuang *et al.*, 2010). Zhuang *et al.* (2010) investigated colloid transport with wetting fronts and reported that a lower surface tension resulted in lower attachment to soil: a decrease in surface tension from 73 to 30 mN/m increases the recovery by 104% (ionic strength 2 mM). Lowering the surface tension both reduces the capillary forces, thus increasing drainage, and increases the electrostatic repulsion forces, consequently decreasing deposition (Zhuang *et al.*, 2010).

2.6.4 EFFECTS OF SATURATION AND FLOW RATE UNDER STEADY-STATE CONDITIONS

Reduced saturation has often been reported to increase colloidal and microbial retention (Chen and Flury, 2005; Keller and Sirivithayapakorn, 2004; Lenhart and Saiers, 2002; Morales *et al.*, 2009; Powelson and Mills, 1998; Schäfer *et al.*, 1998b; Wan and Wilson, 1994b). A multitude of processes have been proposed to account for this, such as increased attachment to the solid-water interface due to reduced diffusive length (Torkzaban *et al.*, 2006), increased attachment to the air-water interface due to its enlarged area (Schäfer *et al.*, 1998b; Wan and Wilson, 1994b), and increased straining, such as trapping in thin water films (section 3.1.1) (Lenhart and Saiers, 2002; Wan and Tokunaga, 1997). The flow rate and the water content are related, as the flow rate depends on the hydraulic conductivity, which varies with the saturation (Brooks and Corey, 1964; Richards, 1931; Schijven and Simunek, 2002; van Genuchten, 1980). Slower flow results in an increased bacterial-collector contact time, which increases attachment (Pang, 2009). It also decreases the hydrodynamic shear. Torkzaban *et al.* (2007) quantified deposition as related to hydrodynamic forces, which depend on the pore-water flow velocity and act near collector surfaces; according to this theory attachment occurs if the adhesive torque, calculated as the DLVO adhesion force multiplied by the radius of the colloid-surface contact area, is larger than the applied torque, calculated based on the hydrodynamic shear force and the colloid radius. With this model attachment is predicted to occur near stagnation points, where the adhesion torque is higher than the applied one;

if the ionic strength is reduced, deposition decreases nonlinearly and takes place in regions of lower velocity (Torkzaban et al., 2010).

2.6.5 BACTERIAL SURFACE CHARGE, HYDROPHOBICITY AND GROWTH STATE

Recent research suggests that the effect of bacterial surface charge is more important than hydrophobicity: Kim and Walker (2009) compared the transport of two non-flagellated strains, *E. coli* D21g and *E. coli* XL1-Blue, in packed bed columns (ionic strengths of 10–100 mM). It was reported that deposition is consistently greater for the former strain, which was attributed to its lower negative surface charge (higher ζ -potential). The bacterial diameters and viabilities were similar; however, the hydrophobicity of *E. coli* XL1-Blue was considerably higher, indicating that this factor is not as important as surface charge for deposition (Kim and Walker, 2009). Coherently, Haznedaroglu et al. (2009) reported that the retention of *E. coli* O157:H7 was higher than retention of *Salmonella pullorum* SA1685, despite that their hydrophobicities are similar; the result was linked to the difference in surface charge. Considering transport of eight *E. coli* isolates in saturated quartz sand Bolster et al. (2010) reported that the attachment efficiencies in quartz sand is strongly inversely correlated to the bacterial log ζ -potential (lower negative surface charge). On the other hand, when investigating 54 different *E. coli* strains Foppen et al. (2010) reported that the correlation between *E. coli* ζ -potential and attachment efficiency is weak ($R^2 = 0.01$ and $p = 0.53$). This indicates that the effects of surface charge and hydrophobicity are not key factors determining the retention rates of *E. coli*.

The bacterial surface thermodynamics can be affected by the cell's growth state: for example, attachment of *E. coli* D21g has been observed to be lower during the mid-exponential growth phase, as compared to the subsequent, stationary, one (Walker et al., 2005). This was attributed to that mid-exponential cells have a uniform charge distribution, whereas stationary-phase cells have higher surface heterogeneity, implying a decreased electrostatic repulsion between the cells and the quartz surface; further, it was found that the number of hydrophilic (acidic) proteins on the bacterial surface decreases with culture age, which results in higher bacterial hydrophobicity. Moreover, when bacteria are dying, their deposition mechanisms change (e.g., due to size reduction) (Foppen and Schijven, 2006). In saturated filter media Haznedaroglu et al. (2008) found that starved *E. coli* cells were less retained than non-starved cells. This was attributed to the higher hydrophobicity of the latter due to their higher presence of extracellular polymeric substances. Coherently, Han et al. (2013) found that starvation increased transport, which was attributed to changes in bacterial size, ζ -potentials, and extracellular polymeric substances. However, Bolster et al. (2010) reported that transport depends more on the depositional environment (Fe-coated vs. uncoated quartz sand) than the bacterial growth stage

(stationary vs. exponential growth phase), when comparing eight different porcine *E. coli* isolates in saturated filter media; in uncoated sand, growth had minimal impact in seven of the investigated isolates.

2.6.6 INTERACTIONS DUE TO BACTERIAL SURFACE BIOMOLECULES

Other bacterial characteristics, such as motility, size and density, have further been shown affect the attachment rates (Pang, 2009; Schijven and Hasanzadeh, 2000). Surface properties, such as texture and presence of surface biomolecules, have been considered to be particularly important (Kim et al., 2009b; Schäfer et al., 1998b; Stevik et al., 2004; Tufenkji, 2007; Walker et al., 2004). Gargiulo et al. (2007) found that enzyme-treated cells with removed surface proteins and lipopolysaccharides are attached to a lesser extent (17%) than untreated ones (33%). Kim et al. (2010), additionally, found that extracellular macromolecule-related interactions contribute to attachment of *E. coli* O157:H7. Tong et al. (2010) and Long et al. (2009) reported that the presence of extracellular polymeric substances on *E. coli* BL21 increases deposition to quartz sand and silica surfaces. Lutterodt et al. (2009b) studied the impact of cell-surface properties of different *E. coli* strains, and found that, particularly, bacterial motility and expression of Ag43 increase the sticking efficiency. McClaine and Ford (2002b) found that the attachment of motile cells was 10 and four times as efficient as that of paralyzed bacteria, at flow rates of 2.64 cm/min and 0.26 cm/min, respectively (ionic strength 20 mM); however, at an ionic strength of 200 mM the attachment of motile and non-motile cells were not statistically different. Increased motility could increase attachment due to flagellar rotation, which could enable swimming to the collector, as well as tumbling that could cause flagella to attach to a collector by acting as an anchor (Figure 3, mechanism J) (McClaine and Ford, 2002b). In unsaturated sand Steenhuis et al. (2006) coherently reported that the sticking efficiency of *E. coli* PHL 628 GFP-1 with removed curli is smaller than that of unaltered bacteria, and larger than that of flic mutant cells without either curli or flagella. At low flow rates (0.26 cm/min) it has further been argued that motile *E. coli* detach from glass more easily than nonmotile cells, which can only be detached due to fluid shear or Brownian motion (McClaine and Ford, 2002a, 2002b).

However, knowledge on which biotic mechanisms affect removal is still limited, and often based on assumptions and empirical observations (García et al., 2010; Tufenkji, 2007). In identical saturated sand columns Foppen et al. (2010) investigated the impact of sphericity, motility, ζ -potential, cell aggregation, lipopolysaccharide composition, and the presence/absence of 22 genes that coded for different surface proteins on the sticking efficiency of 54 different *E. coli* strains, and found that no single factor could statistically significantly explain the observed transport variations within the strain.

3. STRAINING PROCESSES, MODELS, AND FACTORS

Straining has typically been considered to occur when cells sample pathways that are smaller than their own diameter, thus accumulating at the surrounding interfaces (Corapcioglu and Haridas, 1984). Other mechanisms have additionally been proposed, such as straining in grain-grain contact points (Figure 3, mechanism C) (Bradford et al., 2006b; Herzig et al., 1970); retention in low flow-velocity regions (Figure 3, F) (Torkzaban et al., 2008b); retention through adhesion to already deposited bacteria (Figure 3, D) (Bradford and Torkzaban, 2008); film straining, or retention due to encapsulation in the thin water films that surrounds the grains (Figure 3, G) (Wan and Tokunaga, 1997); and straining at the air-water meniscus-solid interfaces (Figure 3, E) (Steenhuis et al., 2006).

3.1 Straining in Water Films, Grain-Grain Contacts and Due to Aggregation

This subsection and the subsequent one explain various straining mechanisms.

3.1.1 FILM STRAINING AND RETENTION BETWEEN GRAINS

Under unsaturated conditions water flow takes place in narrow, water-filled pores and in pendular rings, or water films retained due to capillarity in the grain-grain interfaces (Keller and Auset, 2007; Lenhart and Saiers, 2002; Wan and Tokunaga, 1997). Film straining occurs where these films are too narrow to allow for particle passage. Wan and Tokunaga (1997) formulated a quantitative model for predicting the efficiency of film-straining, in which the straining rate is a function of the capillary pressure head, the probability of pendular ring discontinuity, and the film thickness. According to this theory, however, the water film has a width of about 1/20 of the *E. coli* diameter (Keller and Auset, 2007; Lenhart and Saiers, 2002; Wan and Tokunaga, 1997); thus, film straining is only likely to be important in the case of fluctuations in water content. In steady-state systems direct visualization has revealed that film straining is insignificant for synthetic hydrophilic microspheres of sizes larger than $2.6\ \mu\text{m}$, whereas some film straining occurs for colloids that are $0.8\ \mu\text{m}$; however, the dominant retention mechanism is air-water meniscus-solid interface straining (flow $0.015\ \text{cm/min}$; Zevi et al., 2005). This indicates that some but limited film straining could occur for *E. coli*, since their size is in the range of $2.0\text{--}6.0\ \mu\text{m}$ by $1.1\text{--}1.5\ \mu\text{m}$ (Foppen and Schijven, 2006).

Another proposed mechanism is retention in locally saturated regions in grain-grain interfaces, due to wedging, rotation or immobilization. Cells could be translated to these hydrodynamically disconnect zones (Bradford et al., 2007). Direct imaging studies of saturated systems have shown that particles are often retained in such points under unfavorable attachment conditions in

systems with colloid to grain size ratios, a_p/d_{50} , larger than 0.004 (Johnson et al., 2010; Torkzaban et al., 2010; Yoon et al., 2006). Unsaturated systems could locally resemble saturated ones when the water is unevenly distributed (Chen et al., 2008). Under such conditions it is thus possible that bacteria are wedged or retained in immobile zones between grains. However, retention at the grain-grain interfaces has been found to be minor, based on pore-scale imaging of hydrophilic, negatively charged microspheres (2.6 μm) in unsaturated sand, at approximately 50% saturation and ionic strengths of 0–1 mM (Zhang et al., 2010).

3.1.2 FLOC FORMATION AND BRIDGING

Colloids can aggregate in the fluid or when they are retained on a surface if the particle surfaces interact (Kim et al., 2010; Schinner et al., 2009; Sirivithayapakorn and Keller, 2003b; Stevik et al., 1999a). Aggregations in the fluid, or flocs, affect straining due to the larger size of the transported unit (Schinner et al., 2009; Stevik et al., 1999a). Higher bacterial hydrophobicity implies a higher inclination to form aggregates (Gargiulo et al., 2008). It has been calculated, using extended DLVO theory, that the energy barriers for hydrophilic colloids are lower for colloid-colloid interactions than for colloid-solid or colloid-air interfaces (Zevi et al., 2005). Further, pore-scale imaging has revealed that hydrophilic particles often adhere to already deposited microspheres (Tong et al., 2008; Torkzaban et al., 2010; Zevi et al., 2005; Zhang et al., 2010). Steenhuis et al. (2006) reported that hydrophilic colloids frequently are retained to other colloids forming bridges between sand grains on the air-water meniscus-solid interfaces. Coherently, imaging in two-dimensional variably saturated micro-models has revealed that colloids were often deposited near the colloids that had already been retained, and that they were deposited in a similar way as *E. coli* (Auset et al., 2005). Local clustering on grains could enable ripening of the system. Microscope imaging of untreated cells revealed that ripening occurs for *E. coli* O157:H7 in saturated sand (Kim et al., 2009a; Kim et al., 2010). It has, however, been argued that aggregation of cells is not very important under natural conditions, as in this case concentrations are relatively low as compared to the number of retention sites, which reduces the probability of collision of two bacteria (Chen and Zhu, 2005; Jiang et al., 2007; Schäfer et al., 1998b). On the other hand, local coverage of colloids (e.g., near the grain-grain contact regions), can be high even if the overall coverage is low, as revealed by direct imaging of colloids in a saturated flow chamber (Tong et al., 2008). Tong et al. (2008) argued that secondary minima often merged in these regions, causing collision of particles and subsequent attachment and ripening. In saturated filter media Kim et al. (2009a) coherently observed that the bacteria only are attached at a small fraction (2–23%) of the quartz surfaces, which was associated with retention in low-velocity regions, such as the grain-grain interfaces

or in small pore spaces. Nevertheless, Haznedaroglu et al. (2009) found that the effect of bacterial concentration in the form of increased transport is only obvious for influent concentrations $\geq 10^{10}$ CFU/100 mL (ionic strength 10 mM) in saturated filter media. This is very high considering that the occurrence of fecal coliforms is typically 10^5 to 10^9 /100 mL in untreated wastewater (World Health Organization, 2011), and liquid swine and solid cow manure have been found to contain *E. coli* concentrations in the order of 10^6 /100mL and 10^8 /100 mL, respectively (Unc and Goss, 2003). The effect of concentration might, nevertheless, be more important under unsaturated conditions. Zhang et al. (2010) found that the influent concentration of colloids increased retention in unsaturated systems (ionic strength $0.1 \geq$ mM), and that results in unsaturated and saturated systems were in agreement; however, the study only addressed a range of very high input concentrations (10–1000 mg/L), which approximately corresponds to *E. coli* concentrations of 10^9 to 10^{11} cells/100 mL, considering a wet weight of 10^{-9} mg (Milo et al., 2010). Thus, the effect of the influent concentration needs further studies of unsaturated filter media and more realistic *E. coli* concentrations in sewage water and manure.

3.2 Air-Water Meniscus-Solid Interface Straining

Many studies have reported that retention at or near the air-water meniscus-solid interface is an important retention process for *E. coli* (Chen, 2008) and colloids of similar surface charge, size and hydrophily (Chen and Flury, 2005; Chen et al., 2007; Crist et al., 2004; Crist et al., 2005; Gao et al., 2006; Gao et al., 2008; Lazouskaya and Jin, 2008; Morales et al., 2009; Saiers and Lenhart, 2003b; Steenhuis et al., 2006; Zevi et al., 2005; Zevi et al., 2006; Zevi et al., 2009), of which the majority include pore-scale visualizations. For example, video films of hydrophilic, negatively charged latex microspheres revealed that retention primarily occurred in the thin film of water at the edge of the menisci of pendular rings (Crist et al., 2005). In studies using confocal laser and bright field microscopes Steenhuis et al. (2006) and Zevi et al. (2005) reported that retention of hydrophilic (contact angle $<10^\circ$) and negatively charged colloids (ζ -potential -31.1 mV in distilled water) primarily occurred in the air-water meniscus-solid interface. It is probable that this finding is relevant for *E. coli*, which has similar characteristics. The *E. coli* contact angle was 13.3° and the ζ -potential was -10.6 mV in de-ionized water in Chen (2008), and the *E. coli* surface charge ranges between -17.9 and -32.1 mV (ionic strength 4.7 mM) in Foppen et al. (2010). Generally, transport to this interface has been attributed to hydrodynamic forces, since, for laminar flow, the colloidal velocity decreases when it approaches the grain (Crist et al., 2005). Proposed retention mechanisms near the air-water meniscus-solid interfaces are addressed below.

3.2.1 PINNING DUE TO CAPILLARY AND FRICTION FORCES AND COUNTER FLOW

Gao et al. (2008) argued that the capillary, or immersion, energy potentials for colloids at this interface were many orders of magnitude larger than the classic DLVO interactions (section 2.1). Accordingly, Steenhuis et al. (2006) and Zevi et al. (2005) stated that air-water meniscus-solid interface straining had likely taken place due to capillary forces. The water meniscus has been compared to an elastic membrane that strives to achieve the smallest area; it pushes a colloid that attempts to protrude it due to its kinetic energy back into the pendular ring, and a frictional force prevents the particle from moving, resulting in pinning (Gao et al., 2008; Steenhuis et al., 2006). Zevi et al. (2012) investigated transport to the air-water meniscus-solid interface in an open triangular micro-channel using a confocal microscope. It was found that retention at the interface is important for meniscus-wall contact angles smaller than 16° and minimal for contact angles larger than 20° . This was attributed to the formation of counter flow near the air-water meniscus-solid interface for the smaller contact angles but not for the larger ones, which enables the transport of colloids to the surface just below the air-water meniscus-solid interface and secondary minima attachment for low drag forces. It was inferred that capillary forces push colloids to the primary energy minima for small contact angles when fluctuations in the flow rate occur (Zevi et al., 2012). It was further found that the friction forces needed for retention near the three-phase contact line increases with higher contact angles and ionic strengths, and that air-water meniscus-solid interface retention decreases with an increase in ionic strength, which was explained by the decreasing depth of the primary minima (Zevi et al., 2012). This indicates that air-water meniscus-solid interface straining is primarily important for lower ionic strengths (1 mM rather than 50 or 100 mM) and lower moisture contents; however, it is not clear for which saturations in soil these findings apply.

3.2.2 ENTRAPMENT IN LOW FLOW-VELOCITY REGIONS

Zhang et al. (2010) used DLVO theory that included Lifshitz–van der Waals, electric double layer, and Born repulsive (steric) interactions and argued that the studied colloids could not penetrate the air-water interfaces, due to the significant energy barrier and the menisci stability under steady-state conditions; thus, the importance of capillary forces, as put forward in Gao et al. (2008), was questioned. It has instead been proposed that particles can be funneled to and retained in low-velocity regions near air-water meniscus-solid interfaces due to hydrodynamic forces (Lazouskaya and Jin, 2008; Torkzaban et al., 2008a; Zevi et al., 2009; Zhang et al., 2010). Crist et al. (2005) observed that a centrifugal motion in pendular rings transports hydrophilic microspheres toward the three-phase interface, where they are

retained due to low laminar flow velocities, or strained between the grain and the air-water interfaces. Zhang et al. (2010) observed that retained colloids often act as new deposition sites for suspended colloids, and that the deposition rate increased with higher input concentrations at solution ionic strengths of 0.5 and 1 mM but not of 0 mM; the result was attributed to weak attachment, translation along grains, and subsequent straining in the air-water meniscus-solid interface wedges. Their computational hydrodynamic simulations, in which the air-water interface was assumed to be a slip boundary, showed that the wedges can form a hydrodynamic trap, in which retention occurs due to low-velocity vortices. Lazouskaya et al. (2006) found that flow through a micro-channel is analogous to the traditional Poiseuille flow through a closed conduit, indicating that the air-water interface is unlikely to be a no-slip boundary, and that low-velocity regions can occur near this interface as well.

3.2.3 LIFSHITZ–VAN DER WAALS, ELECTROSTATIC, AND LEWIS ACID-BASE FORCES

Chen (2008) argued that *E. coli* are, primarily, transported to the three-phase interface due to hydrodynamic forces, and, subsequently, attached to the nearest air-water interface, because of the attractive total effective forces at this interface. Chen (2008) calculated the forces at the equilibrium distance using extended DLVO theory, including Lifshitz–van der Waals, Lewis acid-base, and electrostatic interactions. It was found that Lifshitz–van der Waals forces are attractive toward both the grain and the air, and that the Lewis acid-base interactions are repulsive toward the grains and attractive toward the air-water interface. Chen (2008) therefore concluded that the *E. coli* had likely been pushed toward the air-water interface, due to the stronger repulsive forces toward the solid-water interface. The total deposition coefficient was linearly correlated to the total of the effective forces exerted on the bacteria, i.e., the sum of repulsive and attractive interactions (Chen 2008). In summation, it is clearly not yet fully understood why colloids are transported to and retained near the three-phase contact line. Different theories have been proposed, depending on which interactions are accounted for and at which distance they are calculated.

3.3 Straining Modeling

A model for colloidal straining in saturated systems was proposed by Bradford et al. (2003). It has also been applied to describe transport of bacteria in unsaturated columns (e.g., Gargiulo et al., 2008) and has the structure:

$$\Lambda_{str} = \rho_b \frac{\partial C_{str}}{\partial t} = \theta k_{str} C \psi \quad (14)$$

where Λ_{str} is the straining term in eq. (4); k_{str} [T⁻¹] is the straining coefficient; C_{str} [NM⁻¹] is the concentration of the strained particles; and $\psi[-]$ can be defined as a function of distance or time, e.g., describing the blocking of favorable sites (Gargiulo et al., 2007; Lenhart and Saiers, 2002). Blocking is typically inferred to occur if the breakthrough curve plateau slopes upward despite that the injection of bacteria is constant (e.g., in Gargiulo et al. (2008)). This implies that the amount of transported particles increases with time and this can be modeled according to (Gargiulo et al., 2007):

$$\psi = \left(1 - \frac{C_{str}}{C_{strmax}}\right) \quad (15)$$

where ψ is from eq. (14) and C_{strmax} [NM⁻¹] is the maximum of the strained concentration. In this case the deposition curve becomes more uniform with time as the retention sites are successively occupied in the top of the column. Blocking has also been applied to describe air-water interface attachment (eq. (9) (Corapcioglu and Choi, 1996; Lenhart and Saiers, 2002). The opposite process, ripening, is assumed if the height of the breakthrough curve plateau decreases with time. Ripening can thus cause a successively accentuated hyper-exponential shape of the deposition curve. This is modeled such that ψ instead increases with C_{str} in eq. (15). The following model can be applied to represent a hyper-exponential deposition profile (Bradford et al., 2003; Gargiulo et al., 2008):

$$\psi = \left(\frac{d_{50} + x}{d_{50}}\right)^{-\beta} \quad (16)$$

where ψ is from eq. (14); d_{50} [L] is the median grain size; and β [-] is a fitting parameter that increases with increasing depth-dependency of the deposition profile (Bradford et al., 2003).

3.4 Key Factors That Influence Straining Under Steady-State Conditions

This section considers the factors that affect straining. The effects of transient flux have been addressed previously (section 2.5).

3.4.1 GRAIN SIZES AND THEIR HETEROGENEITY

Straining is affected by physical constraints and its importance is naturally related to the colloid: collector size ratio (a_p/d_{50}) (Herzig et al., 1970; McDowell-Boyer et al., 1986). The bacterial sticking efficiency, α , has often been found to increase with reduced matrix grain size (Gargiulo et al., 2007). This has frequently been attributed to straining rather than attachment since α is theoretically independent of the grain size in saturated media (section

2.1). Straining in saturated media has traditionally been considered to contribute to deposition if the critical ratio $a_p/d_{50} > 0.05$ – 0.2 (Herzig et al., 1970; Wan and Tokunaga, 1997); however, recent studies suggest that straining is of considerably greater importance, especially under unsaturated conditions, and smaller ratios have been proposed, such as $a_p/d_{50} > 0.002$ – 0.005 (Bradford et al., 2002; Bradford et al., 2004; Bradford et al., 2007; Johnson et al., 2010; Li et al., 2004). In the case of *E. coli* transport this implies a grain size lower than $500\ \mu\text{m}$ in average. In saturated columns Jiang et al. (2007) found that *E. coli* retention was larger in fine sand (75 – $500\ \mu\text{m}$) than in coarse sand (500 – $2360\ \mu\text{m}$): the recovery was 17% and 81% considering the former and the latter, respectively. This result was hypothesized to occur because the adsorption rates were higher in the finer sand, in which the flow was slower due to smaller pore throats. Another possible explanation is that straining occurred in the finer media, as its median grain size was approximately $300\ \mu\text{m}$, thus $a_p/d_{50} \approx 0.003$. Straining could also have occurred in other experiments that originally assessed attachment, e.g., in Foppen et al. (2010), in which $a_p/d_c \approx 0.004$. According to Morales et al. (2009) straining at the air-water meniscus-solid interface implies that air-water meniscus-solid retention is higher for colloids with rougher surfaces, since in this case the amount of cavities and friction increase, whereas the meniscus shape remains approximately the same.

3.4.2 EFFECTS OF SOLUTION IONIC STRENGTH

Straining has traditionally been considered to be independent of the solution ionic strength (e.g., Saiers and Lenhart, 2003b). The capillary force, which affects air-water meniscus-solid interface straining (3.2), is primarily dependent on the surface tension and the deformation of the meniscus (Kralchevsky and Nagayama, 2000), and it remains approximately the same for various ionic strengths in the range of 0 – $100\ \text{mM}$, according to Zevi et al. (2009). However, a higher ionic strength has both been argued to decrease and increase straining. An increase in the ionic strength from 1 to $100\ \text{mM}$ increases colloidal attachment to grains and decreases air-water meniscus-solid interface straining, due to the higher attraction to solid-water interfaces and a subsequent competition of deposition sites (Zevi et al., 2009). It has, on the other hand, been proposed that that an increase in ionic strength increases weak attachment and funneling to, and retention in, low-velocity regions—a straining mechanism (Bradford et al., 2007; Bradford and Torkzaban, 2008; Bradford et al., 2009; Torkzaban et al., 2008a; Torkzaban et al., 2008b). Accordingly, at higher ionic strengths the critical straining ratio has been found to be considerably reduced (Shen et al., 2008). Solution chemistry could, moreover, affect aggregation in the presence of dissolved organic matter (humic acid): in unsaturated sand Morales et al. (2011) observed that colloid movement often occurs in flocs, and that air-water interface retention due to bridging flocculation and subsequent ripening is the most important

deposition mechanism in solutions that are more favorable for attachment, with higher ionic strengths or lower pH.

3.4.3 SURFACE ROUGHNESS AND SATURATION EFFECTS

Surface roughness has additionally been shown to affect colloidal straining at the air-water meniscus-solid interfaces and in zones of immobile water under unsaturated conditions. Morales et al. (2009) reported that the relative mass recovery was consistently larger for smooth than rough surfaces: 0.21 higher ($p = 0.047$) for fine sands (300–420 μm), and 0.05 higher ($p = 0.036$) for coarse sands (800–1200 μm). Hence, the importance of surface roughness was reduced with grain size. The higher surface roughness was hypothesized to cause higher friction forces, increasing the pinning of colloids near the air-water meniscus-solid interface. It was observed that colloids initially are retained at the air-water meniscus-solid interface, and then oscillated about the air-water meniscus-solid interface or air-water interface, and subsequently are able to slide into the pendular ring, from where they cannot be re-mobilized (Morales et al., 2009).

Moreover, air-water meniscus-solid interface straining is clearly influenced by the saturation. A reduced moisture content entails that capillary forces cause flow to occur in more narrow regions, decreasing the size of the wedges and increasing the size of the three-phase contact line (Gargiulo et al., 2008; Torkzaban et al., 2008a). Gargiulo et al. (2008) found that depth-dependency increases with a decreasing water content, which was attributed to straining, considering that more flow occurs in narrow pores as a result of less moisture. A reduced flow rate further increases the number of low velocity compartments that are often found near the air-water meniscus-solid interface. Counter flow, enabling secondary minima attachment near the air-water meniscus-solid interface, is more likely to occur for lower water contents (Zevi et al., 2012). Thus, reduced water content is expected to increase retention under steady-state conditions. Under transient flow, a saturation increase can result in expanded pendular rings and increased flow connection to zones of stagnant water and subsequent release of strained particles (Cheng and Saiers, 2009; Gao et al., 2006; Saiers and Lenhart, 2003a; Sirivithayapakorn and Keller, 2003a).

4. SURVIVAL AND THE EFFECTS OF BIOFILMS

Microbial subsistence processes include growth and decay, for example through natural die-off or predation. Biofilms are surface-associated multicellular communities that are embedded in a matrix of extracellular polymeric substances (Branda et al., 2005; Strathmann et al., 2007). They can serve as reservoirs and protect microbes from the environmental stresses that occur

due to temporal variations in temperature as well as in moisture and nutrient content (Hall-Stoodley et al., 2004; Morales et al., 2010). The presence of various types of biofilm could potentially affect both *E. coli* decay and persistence. After the initial deposition, the presence of biofilm is crucial in order to enable *E. coli* survival and growth (Liu and Li, 2008). The understanding of survival processes of indicator bacteria is however limited, due to high field complexities and a small amount of data (Taylor et al., 2004). In column studies die-off and growth are further often intentionally limited by means of low temperature, darkness and lack of nutrients (e.g., Jiang et al., 2007). Other reviews have covered various aspects of microbial survival; for example, Rockhold et al. (2004) addressed the coupling of microbial and substrate transport and fate processes, and Foppen and Schijven (2006) reviewed *E. coli* survival mechanisms and the influencing factors under saturated conditions. *E. coli* survival processes in unsaturated soil are addressed subsequently.

4.1 *E. coli* Removal and Growth Processes in the Presence of Biofilm

This subsection discusses retention and decay processes an effect of biofilm, as well as biofilm formation and colony expansion of *E. coli*.

4.1.1 ATTENUATION IN THE FORM OF PREDATION, RETENTION, AND RELEASE

In the field *E. coli* often originate from septic tank systems or manure application, and it is therefore probable that other bacteria or protozoa will be present, which could enable predation (Chabaud et al., 2006), and the development of biofilms (Wang et al., 2011). It has been argued that removal in biofilms are responsible for the major part of the microbial processing that occurs in constructed wetlands (García et al., 2010). Regarding *E. coli*, specifically, Wang et al. (2011) reported that attenuation in sand prior and subsequent to the promotion of growth of biofilm by groundwater microbes was approximately 9% and 47%, respectively. Moreover, Bauman et al. (2009) reported that *E. coli* O157:H7 was removed to a higher degree in porous media colonized with biofilms than in clean porous media: retention was 9.37% and 0.22% in reactors with a three-week biofilm, versus those without biofilm, respectively. Zhang et al. (2011) evaluated the attenuation of *E. coli* O157:H7 in sandy loam, silica sand, and mulch loaded intermittently with synthetic urban storm-water runoff during 18 months and reported that initially the removal efficiency was 72%, and after six months it increased to 97%. The improvement was attributed to decreased porosity and increased dispersion with time, facilitating retention, as well as growth of protozoa, enabling predation—it was concluded that bioretention systems are viable (Zhang et al., 2011). Removal of fecal coliforms was further at least 5.6 log₁₀ as septic tank effluents were applied intermittently at rates of 2–8 cm/day in 12 filters containing sorted ($d_{50} = 650 \mu\text{m}$) and unsorted

sand ($d_{50} = 840 \mu\text{m}$), as well as crushed light-weight aggregates ($d_{50} = 700 \mu\text{m}$) in Ausland et al. (2002). However, the natural unsaturated subsurface habitat is typically fragmented, and knowledge regarding the distribution and attenuating properties of microbial biofilms in the vadose zone is, as yet, insufficient (Morales et al., 2010; Or et al., 2007b; Wang and Or, 2010).

It has been argued that biofilms can increase the attachment rate, since they have sticky properties and could provide attachment sites at the extracellular polymeric substances, the cell walls, the lipid membranes, as well as the cytoplasm (Figure 3, mechanism K) (Strathmann et al., 2007). Bioclogging can moreover influence retention due to increased surface roughness, reduced porosity and permeability, or preferential channeling that could locally increase the conductivity (Bauman et al., 2009; Liu and Li, 2008). The type of biofilm also influences the retention rate. Liu and Li (2008) reported that a significantly higher amount of *E. coli* JM109 cells were initially captured in the presence of *Pseudomonas aeruginosa* wild type biofilm than the corresponding mucoid biofilm, even though the former biofilm has a lower thickness and roughness. The biofilm surface charges were both more negative than the surrounding media, crushed glass beads, and the result was attributed to the differences in the biofilm hydrophobicity and polymeric interactions between the two biofilms (Liu and Li, 2008).

Research further indicates that the effect of predation is more important than retention in the presence of biofilm. The fecal coliform removal was approximately 0.15 log10, 0.20 log10, and 0.35 log10 in configurations without biofilm, with biofilm and inactivated *protozoa*, and with biofilm and active *protozoa*, respectively (Chabaud et al., 2006). This indicates that the presence of a biofilm only has a small effect on fecal coliform removal if there are no predators present.

Additionally, the presence of biofilms can influence the re-mobilization of cells through sloughing (Figure 4, mechanism D) or detachment (Figure 4, E) (Trulear and Characklis, 1982). The latter occurs due to shear stress, and is mainly affected by hydrodynamics, whereas sloughing occurs due to the re-entrainment of large portions of the biofilm due to nutrient depletion deep in the film (Trulear and Characklis, 1982). The presence of extracellular polymeric substances can limit nutrient diffusion to cells (Holden and Fierer, 2005), and sloughing typically occurs for thick, low density films, developed under low shear stress conditions (Trulear and Characklis, 1982). Wang et al. (2011) found that biofilm sloughing and erosion had likely contributed to the release of *E. coli*. Liu and Li (2008) reported that biofilm detachment was the most important *E. coli* release process that occurred several days after retention. These processes could be particularly pertinent in a transient scenario with sudden changes in the flow rate (section 2.5).

4.1.2 *E. COLI* SURVIVAL AND GROWTH IN THE NATURAL ENVIRONMENT

E. coli are enteric bacteria and are therefore generally considered to die gradually when they are transported in aquatic environments (Foppen and Schijven, 2006). The assumption that *E. coli* are not soil borne is a prerequisite for their use as indicator bacteria; recent findings, however, points to that *E. coli* indeed can persist lengthily in soil (Nautiyal et al., 2010). In non-sterile soils in the laboratory it has been shown that environmental *E. coli* are able to grow well when incubated at 30 or 37°C, and that they can survive more than one month at temperatures less than 25°C (Ishii et al., 2006). Further, faster growth of environmental than laboratory strains has been reported by Brennan et al. (2010a). Recent research has shown that *E. coli* strains can become residents in many different types of soils, in both temperate and tropical environments (Brennan et al., 2010a; Brennan et al., 2010b; Goto and Yan, 2011; Ishii et al., 2006; Ishii et al., 2010; Nautiyal et al., 2010). Brennan et al. (2010b), for example, reported that *E. coli* had persisted in temperate maritime soils for more than nine years. Furthermore, bacterial populations could be dynamic, as seasonal variations in *E. coli* pollutions have been observed in Hawaii (Goto and Yan, 2011) and Lake Superior watersheds (Ishii et al., 2006). Thus, *E. coli* growth in the natural environment could be substantial.

4.1.3 BIOFILM FORMATION AS RELATED TO MOTILITY AND SURFACE PROPERTIES

Bacterial colonies subsist in biofilm (Morales et al., 2010), of which the development is affected by both the environment and the bacterial genome (Hall-Stoodley et al., 2004). Research indicates that motility is important for *E. coli* biofilm formation; however, the findings are not always incoherent. Pratt and Kolter (1998) reported that motility but not chemotaxis, is important for the initial *E. coli* biofilm formation; it was argued that motility had enabled initial adhesion and migration along the collector surface. Moreover, studying eight *E. coli* strains, Wood et al. (2006) reported a strong relationship between motility, flagella, and biofilm architecture: the best biofilm-formers were the most motile, whereas the worst biofilm-former were motility-impaired. Yang et al. (2006) further compared retention behavior of 14 environmental *E. coli* feedlot isolates and concluded that motility was determinant for biofilm formation: all four nonmotile isolates were found to be thin-biofilm formers. On the other hand, studying glucose-minimal medium Reisner et al. (2003) argued that *E. coli* K-12 biofilm development was attributed to cell-cell adhesion rather than motility: flagella, type 1 fimbriae, curli, Ag43, or cell-cell signaling were not essential for biofilm maturation in that study. Yang et al. (2004) evaluated approximately 280 *E. coli* isolates from a bovine feedlot and found that motility varied widely and was positively correlated with biofilm formation in glucose-minimal medium ($p < 10^{-4}$) but not in rich medium

(Luria broth). This indicates that motility is particularly important for biofilm formation in environments with limited access to nutrients. In addition, Yang (2005) reported that enhanced *E. coli* biofilm formation is positively correlated with cell surface hydrophobicity ($R^2 = 0.66$), and that the isolates that exhibited the highest surface hydrophobicity formed visible clumps (section 3.1.2). Moreover, Liu and Li (2008) reported that *E. coli* JM109 can survive and grow in the presence of a biofilm of another bacterium, *Pseudomonas aeruginosa*. Findings on biofilm formation are, however, currently mainly based on controlled and saturated conditions. In unsaturated systems the interdependence of growth, motility and the physicochemical characteristics of the soil matrix are more complex and dependent on the moisture content (Wang and Or, 2010).

4.1.4 CHEMOTAXIS AND COLONY EXPANSION IN UNSATURATED CONDITIONS

Chemotaxis is the ability to move in response to a chemical gradient. The cell moves in a chaotic (Brownian) motion, with at least one flagellum rotating clockwise, until a nutrient gradient is found, and then the bacterium is pushed forward in the direction of the gradient, all flagella rotating counterclockwise (McClaine and Ford, 2002b; Parkinson, 2009). A study of *E. coli* HCB1 in a saturated system determined that an increase of 18% of the total population, as compared to the control, had exited through a more conductive core due to transverse chemotaxis in response to an attractant gradient in the core, at an average fluid velocity of 0.35 cm/min (Wang and Ford, 2009b). Chemotaxis increases growth of *Pseudomonas putida* F1 in saturated porous media, according to Wang et al. (2008). Chemotaxis could cause faster transport and colony expansion; for example, swarming motility, or movement in groups, can enable colony expansion at velocities of 0.012–0.060 cm/min (Harshey, 2003).

However, in unsaturated soil the cell mobility is constrained by viscous and capillary forces, and, thus, it has been argued that the configuration of pore-water has a dominant influence on the development of bacterial habitats (Berg, 2005; Dechesne et al., 2010; Harshey, 2003; Or et al., 2007b; Wang and Or, 2010). In these environments bacteria subsist in water-films of micrometer thickness and habitats are fragmented and connected by thin water-films (Dechesne et al., 2008; Dechesne et al., 2010). According to Long and Or (2009) abundance of water is critical for bacterial swimming to take place, and Dechesne et al. (2010) argued that flagellar motility only occurs in very wet conditions. Direct observations of these processes in soil are difficult to obtain; alternatively, microbial growth has been evaluated on partially saturated rough surfaces, such as ceramic porous plates, where the water film resembles the one in unsaturated soil (Dechesne et al., 2008; Long and Or, 2009). Dechesne et al. (2008) and Wang and Or (2010) reported that an increase in the size of the water films on such surfaces increases the mean cell velocity and the substrate transport, as well as the connectivity among the

pathways between colonies: the colony expansion decreases rapidly with a decreasing matric potential and can only occur in the range 0–2 kPa (film thickness larger than 1.5 μm). Dechesne et al. (2010) studied the impact of motility and moisture content of *Pseudomonas putida* KT2440 and its non-flagellated isogenic mutant on colony expansion on ceramic plates. At low water contents motile and nonmotile strains develop small colonies of similar size, whereas at high water contents populations of motile cells increase greatly in size, while the growth of nonmotile cells is negligible. When the matric potential increased from -3.6 to -0.5 kPa the colony expansion rate of the motile cells increased from 1.5×10^{-5} to 8.7×10^{-4} cm/min (Dechesne et al., 2010). Dechesne et al. (2010) modeled colony expansion using parameters for *E. coli* and found that predictions are reasonably matching the experimental results on the motile *Pseudomonas putida*. Survival is also affected by transient flow regimes. Dechesne et al. (2010) found that even brief and infrequent wet events enable *Pseudomonas putida* to disperse by flagellar motility, yielding larger colonies. The translation of these results to unsaturated soil is, however, complicated by the difference in pores sizes; for example, the maximum pore size was 1.7 μm in Dechesne et al. (2010) and Jiang et al. (2007) reported that the pore sizes mainly ranged between 30 and 60 μm in fine sand (75–500 μm grain size). Nevertheless, the findings point to that *E. coli* colony expansion is likely to be considerable only if the cells are motile and the moisture content is high.

4.2 Modeling Survival

In unsaturated soil there is still a knowledge gap in relation to the evolution of microbial populations as related to depth, temperature, nutrient content, and habitat architecture: “we do not yet know enough about microbial processes in the vadose zone to build predictive models” (Holden and Fierer, 2005). Key concepts are nevertheless addressed subsequently.

4.2.1 BACTERIAL GROWTH MODELING

The most basic bacterial survival processes are first-order growth and decay models. Rockhold et al. (2004) described a simple model for microbial growth:

$$\Lambda_{sur} = -\theta k_{gro} l C \quad (17)$$

where Λ_{sur} is the survival term in the water in eq. (4); $k_{gro}[\text{T}^{-1}]$ is the growth rate, which depends on the nutrient level; and l represents the metabolic time lag. The growth rate can be different in the aqueous versus the solid phase (Rockhold et al., 2004); however, only the liquid phase growth is considered above. This model further assumes that cell division in the solid phase does not release bacteria to the aqueous one (Murphy and Ginn, 2000).

The growth of biomass as a function of substrate concentration is generally modeled using a Monod-type kinetic equation (Monod, 1949; Rockhold et al., 2004; Soleimani et al., 2009):

$$k_{gro} = k_{gromax} \frac{C_{sub}}{(K_{half} + C_{sub})} \quad (18)$$

where C_{sub} [ML⁻³] is the water concentration of substrate (organic matter); k_{gromax} [T⁻¹] is the maximum specific utilization rate, and K_{half} [ML⁻³] is a half-velocity coefficient. This equation is multiplicative in the case of several substrate types, i.e., electron donors and acceptors, such as glucose and oxygen, respectively (Rockhold et al., 2004). Alternatively, k_{gro} could also depend on Y , the synthesis yield coefficient that relates growth to substrate utilization (Soleimani et al., 2009).

There are a wide range of models that account for other microbial survival processes in the subsurface, such as competitive inhibition that takes place as several bacterial populations compete for the same nutrients (Murphy and Ginn, 2000). In saturated porous media motility and chemotaxis have been accounted for in the advection-dispersion equation (eq. (2)) by adding a chemotactic velocity, directed toward the chemical gradient, to the pore-velocity in the convective term, and a bacterial random motility term to the dispersion coefficient (Wang and Ford, 2009a). Moreover, models could keep track of biomass with regard to space, time and exposure time, thus enabling descriptions of microbial residence time on a surface, as well as exposure time to nutrients (Ginn et al., 2002; Murphy and Ginn, 2000). The tendency of a cell to be irreversibly attached could also depend on the residence time near a surface and the implication is that the bacterial concentration has the form $C_{Satt} = C_{Satt}(x, t, w)$, where w [T] is the exposure time or residence time, and the attachment and detachment rates are: $k_{Satt} = k_{Satt}(w)$ and $k_{Sdet} = k_{Sdet}(w)$ in eq. (10). These models are described in detail by Ginn et al. (2002). Their limitations for predictions are often related to the difficulty in reliably determining the many model parameters for use in predictions.

4.2.2 MODELING THE INTERDEPENDENCE OF GROWTH AND PERMEABILITY

Other modeling approaches account for the impact of biofilm on porosity and permeability, bioclogging, and its affect on transport and removal (Cunningham et al., 1991; Rosenzweig et al., 2009; Taylor et al., 1990; Yarwood et al., 2006). For example, Model no. 1 in Mostafa and Van Geel (2007) formulated a dependence of the relative permeability, K_r [LT⁻¹], on microbial growth, based on the equation by Mualem (1976) and using the van Gencuhten (1980) capillary pressure-saturation relationship:

$$K_r(S_w) = S_T^{0.5} \left[\left(1 - S_M^{1/m} \right)^m - \left(1 - S_T^{1/m} \right)^m \right]^2 \quad (19)$$

where S_M [–] is the effective microbial saturation or the volume of pore-space occupied by biomass; S_W [–] is the effective water saturation; $S_T = S_W + S_M$; and m [–] is the van Genuchten parameter. In this model the microbial growth initially occurs in small pores, and the increasing biomass displaces water to successively larger pores to maintain a constant water flux. Mostafa and Van Geel (2007) reported that this model does not significantly affect the relatively permeability until the biomass is so large that the flow path is significantly altered. Soleimani et al. (2009) applied this model to simulate bioclogging in an unsaturated biofilter, and the Monod equation (Monod, 1949) was used to calculate microbial and solute mass as related to time, corresponding to eq. (17) and eq. (18). As expected, it was found that clogging is slower in soils with higher porosities and saturated hydraulic conductivities, and additionally, clogging results in a biomat that is displaced downward with time (Soleimani et al., 2009). Rosenzweig et al. (2009) compared the impact of biofilm presence in small pores first, uniform biofilm thickness in all pores, and a constant pore-fraction covered with biofilm, and concluded that the type of spatial distribution had a significant effect on the soil hydraulic properties. Pore-scale observations are needed for improved understanding of the underlying processes. However, nutrient-dependent processes and the effects of biomass growth on hydraulic properties are most often overlooked in microbial models of growth (Rockhold et al., 2004).

4.2.3 BACTERIAL DECAY MODELING

As addressed in section 4.2.1 *E. coli* growth could be considerable in the environment; however, survival is generally assumed to be dominated by die-off or predation, since the natural environments are not expected to be the natural habitat of enteric bacteria (Gerba et al., 1991). Typically, only inactivation in the water is included in macro-scale models, which has the following basic structure (Mubiru et al., 2000; Reddy et al., 1981):

$$\Lambda_{sur} = \theta k_{ina} C \quad (20)$$

where Λ_{sur} is the survival term in the water in eq. (4); and k_{ina} [T⁻¹] is the first-order inactivation rate in the solution. If deposited cells are assumed to be irreversibly retained the inactivation and growth of the deposited cells are generally ignored. There is a significant variation in reported *E. coli* inactivation values. Westcot (1997) reported that fecal coliforms can survive up to 70 days in field soils but usually survive less than 20 days (temperatures of 20–30°C). Decay parameters, as related to temperature and predation, are discussed in subsection 4.3.2.

4.3 Additional Factors and Processes Influencing Growth and Decay

This subsection outlines factors that have been found to influence bacterial persistence.

4.3.1 LINKS BETWEEN SURVIVAL, DEPTH, NUTRIENTS, AND MOISTURE CONTENT

Bacterial survival is clearly related to the access to nutrients, and in the vadose zone the availability of organic carbon and other nutrients is highly dependent on hydrology (Holden and Fierer, 2005; Konopka and Turco, 1991). Coherently, the microbial die-off rate has generally been found to increase with reduced saturation (Fine and Hass, 2007; Jamieson et al., 2002; Reddy et al., 1981). Mubiru et al. (2000), for example, found that the inactivation rate is $2.2 \times 10^{-4}/\text{min}$ and $2.7 \times 10^{-4}/\text{min}$ for nonpathogenic *E. coli* and *E. coli* O157:H7, respectively, in Pope silt loam, and only $1.4 \times 10^{-4}/\text{min}$ and $1.8 \times 10^{-4}/\text{min}$ for the nonpathogenic *E. coli* and *E. coli* O157:H7, respectively, in Zanesville silt loam; the higher inactivation rate in the former case was attributed to the lower soil moisture. Coherently, the *E. coli* inactivation rate, k_{ina} , is $2.0 \times 10^{-4}/\text{min}$ and $3.1 \times 10^{-4}/\text{min}$ for 50% and 10% soil moisture content, respectively, in Weld fine sandy loam (Boyd et al., 1969; Reddy et al., 1981). Microbial populations decline sharply with depth in the vadose zone, as do concentrations of oxygen, nitrogen, phosphorous, and organic matter (Holden and Fierer, 2005; Konopka and Turco, 1991). Organic matter, which controls access to organic nutrients, can further retain moisture (Jamieson et al., 2002). Autocorrelation studies of bacterial distribution in soil at the microscale has showed that colonies are preferentially situated in patches that are likely to be situated near local deposits of substrate (Nunan et al., 2003). These are randomly distributed in the top soil and can be found near pores further down in the subsurface (Nunan et al., 2003). Environmental scanning electron microscopy in unsaturated sand has revealed that sand grain colonization of *Pseudomonas aeruginosa* is sparse and that most cells are found in small, monolayer clusters rather than in thicker biofilms (Holden et al., 2002).

4.3.2 INACTIVATION INCREASES WITH PREDATION, TEMPERATURE, UV LIGHT, AND pH

It has been argued that predation is the dominant bacterial attenuation process in both slow sand filters and constructed wetlands (Ellis and Wood, 1985; García et al., 2010). This takes place in the Schmutzdecke (dirt layer) that develops in the top soil as a result of successive deposition of organic waste, plankton and microbes, such as *protozoa*, bacteria, rotifers, bacteriophages, and fungi; particularly, the bacterivores *protozoa ciliate* have large effect on decay (Ellis and Wood, 1985; Puigagut et al., 2007; Stott et al., 2003). The Schmutzdecke can improve in attenuation capacity with time

(Schijven et al., 2013), and the effect of predation increases with higher temperatures (Ellis and Wood, 1985). Moreover, in subsurface-flow constructed wetlands the removal of *E. coli* can be enhanced in the presence of vegetation, since plants compete with bacteria for nutrients, and some plants produce secondary metabolites with antibacterial properties (García et al., 2010). In addition, the filter media size could affect the biofilm structure: a higher production of exopolysaccharides has been reported on a gravel of 5 mm size than of 20 mm (Larsen and Greenway, 2004).

However, the presence of a Schmutzdecke is dependent on the access to nutrient and water, which varies with time and space in the natural environment. Grazing can also be important in the absence of thick biofilm. Chabaud et al. (2006) found that the decay rates of total coliforms were 10 times higher in untreated septic effluent containing *protozoa* (1100 CFU/mL per day) than in effluent treated with an inhibitor of *protozoa* (96 CFU/mL per day). Foppen and Schijven (2006) concluded that the *E. coli* die-off rate increased by on average one log-unit due to other organisms, such as predation and competition. Coherently, Foppen et al. (2008) reported that the *E. coli* O2:K1 decay rate coefficient was 2×10^{-5} /min for transport in sterile water and 1×10^{-4} /min in non-sterile water in a laboratory microcosmos, and the result was attributed to the presence of other organisms.

Moreover, an increase in temperature increases the die-off rate (Reddy et al., 1981; Westcot, 1997). Reddy et al. (1981) reported that the inactivation rates increase by a factor of approximately two for a 10°C rise within the interval 5–30°C, considering microbial pathogens and indicator organisms. Foppen and Schijven (2006) focused on *E. coli* in the saturated zone and reported that the average die-off rate coefficient is 1.0×10^{-4} /min at 10°C and 3.5×10^{-4} /min at 20°C. Sjogren (1994) found that the die-off rate of an antibiotic-resistant bacteria in an unsaturated (15% saturation) Richmond sandy loam is 2×10^{-4} /min at 37°C, whereas the corresponding rate is 4×10^{-5} /min at 5°C. The variation in temperature is high in the top soil but more constant deep in the soil (Holden and Fierer, 2005). Moreover, UV-radiation reduces bacterial survival (Crane and Moore, 1986; Fine and Hass, 2007), and *E. coli* survive longer in a neutral pH and prefer an alkaline pH to acidic soils (Jamieson et al., 2002; Reddy et al., 1981; Sjogren, 1994; Westcot, 1997). The decay rate was 4×10^{-3} /min for the extreme pH levels of 12 and 2.5 (McFeters and Stuart, 1972; Reddy et al., 1981). For approximately neutral pH the *E. coli* inactivation rate, however, remains in the order of 10^{-4} /min in soil, as listed in Reddy et al. (1981).

5. PREFERENTIAL FLOW AND ITS EFFECT ON TRANSPORT AND SURVIVAL

Processes in homogeneous soil systems with uniform flow provides a valuable starting point for the understanding of *E. coli* water transport in porous

media; however, in the vadose zone bacterial transport rates are strongly dependent on flow processes, and non-equilibrium, or preferential flow, occurs in most types of soil as an effect of heterogeneities at various scales (Jarvis and Dubois, 2006; Martins et al., 2013). Such flow is considered particularly important in the unsaturated zone, where the transport pathways are dependent on the varying moisture content (Cey et al., 2009). The presence of preferential flow paths generally reduces bacterial subsurface removal due to faster transport, as well as reduced deposition in the matrix (Coppola et al., 2009). Preferential transport of *E. coli* has been reported in the field (e.g., Brennan et al., (2012); Smith et al., (1985)). For example, Brennan et al. (2012) evaluated in situ transport of *E. coli* IND1 as related to flow patterns visualized by two tracers (brilliant blue and bromide), in three grassland soil types in the wet spring and the dry summer. The correlations between *E. coli* and bromide as well as brilliant blue were significant and equal to 0.59 and 0.56, respectively; thus, it was concluded that vertical bacterial transport mainly occurs in preferential flow paths.

5.1 Flow Processes

Three types of preferential flow were distinguished by Jarvis and Dubois (2006): finger and heterogeneous (funnel) flow, as well as macropore flux (Figure 5). These will be described below. The type of preferential flow typically depends on the soil characteristics and the moisture content.

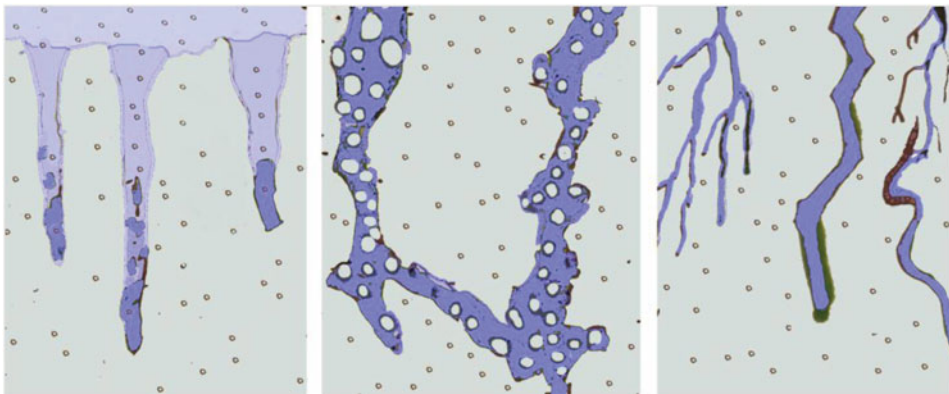


FIGURE 5. Conceptual drawing of three types of preferential flow (not to scale). The small circles represent the grains in the matrix and the degree of saturation of the blue color indicates the water saturation. Left: finger flow, which typically occurs in narrowly graded porous media, induced by instability in the wetting front. Middle: heterogeneous flow in graded soil, where fast flow takes place in the regions of the soil with connected pores of similar size. Right: macropore flow, which can be induced by heavy infiltration, shrinkage and crack formation in clayey soils, or in the presence of earth worm burrows and plant roots.

5.1.1 FINGER AND HETEROGENEOUS FLOW

Finger and heterogeneous flow typically occur in the matrix in sandy soils (Jarvis and Dubois, 2006). Finger flow is caused by wetting front instability and saturation overshoot at the bottom of the fingers: the moisture content is high and uniform immediately above the finger tips and low and unstable higher above in the finger tails (Hillel, 1987; Morales et al., 2010). Saturation overshoot does not occur for flow rates that are very low or very high, i.e., near the saturated hydraulic conductivity (Morales et al., 2010). In the top soil uniform flow that is driven by capillarity typically applies; however, this distribution zone could be as thin as a few cm and preferential flow patterns that are driven by gravity typically develop further down in the soil (Darnault et al., 2004; Morales et al., 2010). Thus, removal is likely to be highest in the top soil where flow is more uniform. Cey et al. (2009) found that colloidal removal was more than two orders of magnitude in the top 5–10 cm of the soil, whereas it was very small below this depth due to the presence of preferential flow paths. Heterogeneous flow occurs in coarse-textured and graded soils, with local presence of materials with different grain sizes than the rest of the matrix (Jarvis and Dubois, 2006). The pathways depend on the saturation, as flow occurs in small and connected pores, in fine grained soil, at low saturation, and in larger pores, in coarser-textured soil, for high saturation (Jarvis and Dubois, 2006). Darnault et al. (2004) evaluated the transport of *Cryptosporidium parvum* Oocysts applied in calve feces to columns containing disturbed sand and undisturbed silt loam as a result of irrigation. It was found that finger flow occurs in the former case, whereas macropore flow takes place in the latter one.

5.1.2 FLOW IN MACROPORES

Macropores are generated from cracks and fissures, due to wetting and drying, and in burrows and roots, as well as in voids found between dense soil in cultivated topsoil (Jarvis and Dubois, 2006). Jarvis and Dubois (2006) found that literature evidence supported the functional definition of macropores to be pores with diameters larger than 0.3–0.5 mm, which correspond to water-entry pressures that are equal to –10 to –6 cm. This indicates that the soil must be nearly saturated in order for macropore flow to occur (Jarvis and Dubois, 2006). Considering field sites and silt loam soils Cey et al. (2009) reported that macropore flow had occurred when the maximum pressure heads was larger than –3 cm, corresponding to infiltration rates larger than 0.033 cm/min.

Pang et al. (2008) stated that the microbial attenuation is more affected by soil lithology than macroporosity. However, the presence of macropores has been found to almost completely preclude colloidal attenuation (e.g., Natsch et al., (1996)). Cey et al. (2009) reported that the transport times of colloids are to a higher degree determined by the flow systems than

the colloid deposition rates. In studies of disturbed and undisturbed soil Smith et al. (1985) found that the share of water transported in macropores determines the water flow rate and the fraction of transported *E. coli* K-12. It was argued that flow can be divided in two components: water flow in the soil matrix, in which bacteria will be almost completely retained, and flux through the macropores, where the interaction with the soil is small, retention is negligible and *E. coli* are transported at the same rate as a tracer. In one intact core with silt loam 96% of the influent cells were found in the effluent (28 cm long columns) (Smith et al., 1985).

Jarvis and Dubois (2006) argued that macropore flow is the most important of the preferential flow types, based on the ubiquity of macropores and the assumption that the transport volume of the soil is generally smaller in this case. Travel times in macropores can be up to two orders of magnitude faster than in the matrix (Beven and Germann, 1982). Morales et al. (2010), however, argued that the flow patterns are similar for the various preferential flow types. Nevertheless, it has been put forward that modeling needs to be different as dependent on the preferential flow type (Jarvis and Dubois, 2006; Köhne et al., 2009b), as addressed subsequently. Further, fast preferential transport due to heterogeneous flow and macropore flow is particularly critical in systems with high saturation (Jarvis and Dubois, 2006).

5.2 Modeling Preferential Flow and Transport

Traditional flow and transport models need to be adjusted to account for preferential flow. Richards's equation (eq. (1)) and the advection-dispersion equation (eq. (2)) assume that unique values of pressure and concentration can be defined for representative elementary volumes, which is not possible in the case of non-equilibrium flow, due to lateral differences in the moisture content and the flow rate (Beven and Germann, 1982; Jarvis and Dubois, 2006). There are, however, no generally accepted models for describing the related microbial flow and transport processes. Several complex models have been developed to account for preferential flow and transport, as discussed by Simunek et al. (2003), Gerke (2006), Jarvis and Dubois (2006), and Köhne et al. (2009a; 2009b).

5.2.1 COMMON FLOW EQUATIONS

Preferential flow can be accounted for in a network approach or as multiple overlaying regions with various flow rates (Köhne et al., 2009a; 2009b). The dual-porosity model assumes two continua (Simunek et al., 2003), in which water flow is fast in one domain (preferential pathways) and occurs through diffusion in the other (soil matrix). The dual-permeability model adds the possibility of advection in the matrix to this conceptualization (Gerke and van Genuchten, 1993; Simunek et al., 2003). Bradford et al. (2009), e.g., used such a model to account for colloid retention in low-velocity regions. Flow in

the preferential domain can further be assumed to be driven only by gravity (Jarvis, 1994), or additionally account for capillary forces, in accordance with Richards's equation (Simunek et al., 2005). Köhne et al. (2009a) argued that capillary models are suitable for graded soils such as structured loamy soils, whereas gravity driven flow models are more appropriate for soils with large macropores, such as clayey soils with cracks.

Flow and fecal coliform transport in undisturbed soil lysimeters with structured sandy loam soils were modeled using the dual-porosity model, accounting for capillarity in the preferential domain in Jiang et al. (2010). In the mobile region this was modeled according to (Simunek et al., 2003)

$$\frac{\partial \theta_m(h)}{\partial t} = \frac{\partial}{\partial x} \left[K_s K_r(h) \left(\frac{\partial h}{\partial x} + 1 \right) \right] - \Gamma_w - S \quad (21)$$

and in the matrix (immobile water region)

$$\frac{\partial \theta_{im}(h)}{\partial t} = \Gamma_w \quad (22)$$

where the subscripts m and im refer to the mobile and immobile water region, respectively; $\theta_m + \theta_{im} = \theta[-]$; and Γ_w represents the rate of transfer of water between the mobile and the immobile regions and is proportional to their difference in pressure head. In Jiang et al. (2010) the upper boundary was time-variable and accounted for irrigation and rainfall data, and the bottom boundary was a seepage face (atmospheric pressure).

5.2.2 TRANSPORT EQUATIONS

Fingering and heterogeneous transport modeling approaches are not well established. Jarvis and Dubois (2006) argued that transport due to heterogeneous and finger flow can be accounted for by increasing the dispersion coefficient in the advection-dispersion equation (eq. (2)). Alternatively, Morales et al. (2011) modeled colloidal pore-exclusion by replacing the water content θ in the advection-dispersion equation with the colloid accessible water content: $\theta_C = \theta E$, where $E[-]$ is the pore exclusion factor, which obtained values of 0.7–1.0 after breakthrough curve fitting. Moreover, the active region model assumes that preferential flow and transport only takes place in a part of the subsurface, the active region, whose fraction increases with the saturation (Liu et al., 2005; Sheng et al., 2011).

Considering flow and pesticide transport modeling in structured soil Köhne et al. (2009a; 2009b) concluded that dual-continua approaches are the most widespread. These vary with regard to, e.g., what types of attenuation processes are accounted for and whether deposition from the mobile flow region occurs. Jiang et al. (2010) applied the two-region mobile-immobile model (dual-continua) to describe bacterial transport (Toride et al., 1999; van Genuchten and Wagenet, 1989), in which the concentration in the liquid

phase was divided into one mobile and one immobile region. In the mobile zone this was modeled according to:

$$\frac{\partial(\theta_m C_m)}{\partial t} = \frac{\partial}{\partial x} \left(\theta_m D_m \frac{\partial C_m}{\partial x} \right) - \frac{\partial(q C_m)}{\partial x} - \Gamma_{mim} - k_{remt} \theta_m C_m \quad (23)$$

and, in the immobile zone

$$\theta_{im} \frac{\partial C_{im}}{\partial t} = \Gamma_{mim} - k_{remt} \theta_{im} C_{im} \quad (24)$$

where C_m [NL⁻³] and C_{im} [NL⁻³] are the concentrations in the mobile and immobile region, respectively (measured in terms of mass in Jiang et al., (2010)); θ_m and θ_{im} are the mobile and immobile water fractions; and Γ_{mim} is the transfer rate between the regions [NL⁻³ T⁻¹] (Jiang et al., 2010). With the assumption of single-porosity flow (eq. (1)) the mass exchange was diffusional in Jiang et al. (2010): $\Gamma_{mim} = \alpha_{mim}(C_m - C_{im})$, where α_{mim} [T⁻¹] governs the rate of bacterial exchange between the mobile and immobile liquid regions. For dual-porosity flow (eq. (21) and eq. (22)) both diffusive and convective fluxes were accounted for, according to (Simunek and van Genuchten, 2008)

$$\begin{aligned} \Gamma_{mim} &= \alpha_{mim}(C_m - C_{im}) + C_m \Gamma_w, \quad \text{for } \Gamma_w > 0 \\ \Gamma_{mim} &= \alpha_{mim}(C_m - C_{im}) + C_{im} \Gamma_w, \quad \text{for } \Gamma_w < 0 \end{aligned} \quad (25)$$

It has been argued that the relevance of accounting for preferential flow is dependent on the infiltration rate (Chen et al., 2002; Jiang et al., 2010). Jiang et al. (2010) compared the dual-porosity flow model (eq. (21) and eq. (22)) to a single-porosity flow model (eq. (1)). Based on breakthrough curve fitting it was reported that water flow can be well described by the single-porosity flow model before artificial irrigation, i.e., with only natural precipitation, and for spray irrigation (5 cm for 2 h fortnightly); however, the coefficients that had been calibrated to fit natural conditions were not relevant after spray irrigation, since the soil hydraulic properties had been altered (Jiang et al., 2010). It was reported that the dual-domain model for flow was more adequate to describe water flow in the case of flood irrigation (10 cm for 0.5 h fortnightly) (Jiang et al., 2010).

5.3 Factors That Induce Preferential Flow

This subsection outlines important results relating to the factors that affect the tendency for preferential flow.

5.3.1 SOIL CHARACTERISTICS

Loamy soils generally contain more macropores that are caused by biological activity such as plant roots and worm burrows. Furthermore, structured soils that contain clay enhance preferential flow as compared to non-structured, sandy soils (Flury et al., 1994). Clay soils are prone to shrinking and cracking (Carlander et al., 2000). The tendency for preferential flow also increases with higher soil heterogeneity (Pang et al., 2008), which is related to the wider range of pore sizes in this case. Pang et al. (2008) found that mobile water contents and dispersivities are lower for fecal coliforms than tracers in silt loam, silt loam over gravels, and deep silt loam, which was attributed to the exclusion of microbes from small pores. Size exclusion of *E. coli* has been reported in the literature (Foppen et al., 2008; Jiang et al., 2007; Unc and Goss, 2003). Furthermore, vertical differences in hydraulic conductivity, as well as entrapment of air and soil water repellency can induce wetting front instability and fingering (Morales et al., 2010).

Findings by Pang et al. (2008) indicate that the tendency order for preferential flow in undisturbed lysimeters is, in declining order of importance: clay soil, silt loam, sandy loam, dune sand soil, pumice soil, allophanic soil, considering a steady application rate of 0.008 cm/min. Coherently, using computed tomography Luo et al. (2010) reported that sandy soil has fewer macropores than silt loam; this result was linked to the higher amount of rock fragments and the weaker structure of the sand, enabling less biological activity. Further, the difference in *E. coli* K-12 removal between intact and repacked, mixed soil columns was 37%, 22%, and 74% in silt loam (Crider), silt loam (Maury), and sandy loam soil (Bruno, Kentucky river), respectively, considering an irrigation rate of 0.033 cm/min (Smith et al., 1985). This indicates that sandy loam results in more preferential flow than silt loam.

5.3.2 LAND USE IMPACTS BIOLOGICAL ACTIVITY WHICH AFFECTS FLOW

Land use influences the type and occurrence of preferential flow paths in the field. Luo et al. (2010) found that columns with pasture soil had a higher macroporosity and macropore network structure as compared to row crop soil, which was attributed to the higher content of organic matter and biological activity in the former case. Natsch et al. (1996) reported that macropores were to a higher degree caused by earthworms in a grassland plot and more often caused by roots channels in a wheat field. It was argued that successively more pronounced preferential flow paths are the effects of longtime growth of ley without tillage (Natsch et al., 1996). Cey et al. (2009) reported that worm burrows with diameters of 2–10 mm, which often contained plant roots, provided the most important source of macropores in variably saturated silt loam soils. Pryor Williams et al. (2006) observed survival and transport of *E. coli* O157:H7 under 21 days in contaminated soil and farm-yard manure, as dependent on earthworm digestion. The results showed

that earthworms significantly induce both lateral and vertical transport; for example, in soil the vertical travel distance was up to 0.25 m versus up to 0.4 m, in the absence and presence of earthworms, respectively. However, long-term survival was not affected by their presence (Pryor Williams et al., 2006).

5.3.3 INITIAL MOISTURE CONTENT AND INFILTRATION RATES

The initial moisture content affects preferential flow. The risk of microbial leaching has been found to be higher in the summer, when the initial water content is lower and the risk of crack formation in the clayey topsoil is higher (Jiang et al., 2008). The leached coliform concentration was, almost immediately after the time of application, similar to the applied one in the summer in a lysimeter with high clay content—it was inferred that preferential flow paths had enabled fast flow and precluded bacterial retention in the soil (Jiang et al., 2008). However, the co-occurrence of *E. coli* IND1 and two different tracers was generally higher in the wetter spring season in Johnstown (Luvic Gleysol), Grange (Haplic Luvisol), and Moorepark (Haplic Cambisol) soils in Brennan et al. (2012), suggesting that the transport of *E. coli* in preferential flow paths is more important under more moist conditions. The discrepancy in results could be related to the type of preferential flow: a higher initial moisture content could lead to more pronounced finger and heterogeneous flow in sandy soils, whereas a lower initial moisture content could cause more shrinkage and crack formation, and subsequently increased macropore flow in clayey soils.

Repeated infiltration can perpetuate finger flow patterns due to hysteresis in the soil water retention curve (Morales et al., 2010). After the formation of fingers, these will remain wetter and have a higher conductivity than the matrix, since for a similar pressure a drying curve applies to the fingers that diffuse water to the matrix, whereas a wetting curve applies in the matrix and the consequence is that finger flow is replaced by heterogeneity induced flow (Morales et al., 2010). Coherently, Jiang (2010) found that the soil saturation decreased and that the hydraulic conductivity increased after cycles of imbibitions and drainage, indicating that preferential flow paths had been formed as a result of the irrigation. In undisturbed cores Martins et al. (2013) reported that *E. coli* K-12 breakthrough curves became bi-modal with time as a result of repeated injection/infiltration and drainage cycles. The flow regime had thus gone from uniform flow to preferential flow dominated, and porosimetry showed that this was related to local soil compaction and the formation of macropores; further, it was argued that the most important influencing factor of bacterial transport is the soil structure and its hydraulic properties (Martins et al., 2013).

Further, heavy infiltration has been found to enable macropore flow (Cey et al., 2009; Hruday et al., 2002). Jiang et al. (2008) reported that higher fecal coliform transport occurred in the case of flood irrigation as compared

to spray irrigation in undisturbed field lysimeters with sandy loam. Moreover, Natsch et al. (1996) found that heavy infiltration (sprinkling of 4 cm over 8 h) had induced fast transport of *Pseudomonas fluorescens* CHA0 in structured field soil. The reduction was approximately one order of magnitude in the top 10–15 cm of the soil and negligible in the subsoil at 20–150 cm depth.

5.3.4 INTERDEPENDENCE OF FAST FLOW PATHS AND BIOLOGICAL ACTIVITY

Preferential flow patterns and microbial habitats can influence each other (Morales et al., 2010). For larger and more interconnected pores research indicates that microbes can maintain their activity better (Fredrickson et al., 1997). Concentrations of oxygen, moisture, and organic matter have been found to be higher in the preferential flow paths than in the matrix, and this increases the microbial activity in these regions as compared to the soil matrix (Morales et al., 2010). Microbial populations have been found to be less dependent on depth in the presence of preferential flow paths (Balkwill et al., 1998; Holden and Fierer, 2005). In addition to passive vertical transport with the water, bacteria could migrate to these advantageous zones due to chemotaxis, quorum sensing or co-transportation with other organic colloids (Morales et al., 2010; Pang, 2009). In unsaturated soil it has been reported that *E. coli* can migrate laterally, where the water content is locally high. Considerable concentrations of *E. coli* IND1 have been found in soil matrix samples in the vicinity of macropores (Brennan et al., 2012). However, the bacteria that subside in the vicinity of preferential flow paths need to be resilient to handle the great variations in substrate and moisture content, and the related processes are currently insufficiently understood (Morales et al., 2010).

Further, microbial activity could influence flow paths. The presence of extracellular polymeric substances can decrease hydraulic conductivity and reduce porosity, inducing a layer of lower conductivity above a layer of higher conductivity, thus increasing the wetting instability and the tendency for preferential flow (Morales et al., 2010). Moreover, soil water repellency, which increases preferential flow, can be an effect of the accumulation of hydrophobic organic compounds that are exuded by bacteria (Morales et al., 2010). Sudden water fluxes in preferential flow paths can also cause detachment of biofilm due to the shear stress (Morales et al., 2010). Nevertheless, few studies assess the link between preferential flow and biological activity and the related processes need further clarification.

6. DISCUSSION AND COMPARATIVE ANALYSIS

This section identifies the relative importance of the various factors that have been reported to influence *E. coli* transport in unsaturated porous media, as

well as the key bacterial attenuation processes in various settings. Additionally, it includes simulations of attenuation in a base-case scenario, and provides a discussion around the challenges related to predictive modeling. Lastly, it addresses the use of various types of laboratory results for the prediction of pathogen transport in the field.

6.1 Key Attenuation Factors

The factors that have been proposed to affect bacterial attenuation in unsaturated soil are numerous and include (but are not limited to): filter media properties, such as grain size and size distribution, clay content, surface area and charge, as well as physical and chemical heterogeneities (sections 2.6.1, 3.4.1, 3.4.3 and 5.3.1); solution characteristics, such as ionic strength, pH, and content of nutrients, surfactants and dissolved organic matter (sections 2.6.2, 2.6.3, 3.4.2, 4.3.1 and 4.3.2); *E. coli* properties, such as surface charge, hydrophily, growth state, presence of surface polymers and motility (sections 2.6.5, 2.6.6, 4.1.3); as well as system characteristics, such as temperature, saturation, infiltration rate, land use, presence of worm burrows, vegetation and biofilm, as well as transients in solution chemistry and moisture content (sections 2.5.2, 2.5.3, 2.6.4, 3.4.3, 4.3.1, 4.3.2, and 5.3). The relative importance of each of these factors is difficult to assess, due to the large variety of factors and scales evaluated, as well as the differences in experimental configurations and modeling approaches. The consequence is that inferences on influencing factors often only are made in relation to very specific system settings. For more generally applicable conclusions it is, however, clearly desirable to compare results between studies. Therefore, correlation analyses based on the listed literature data in Table 1 were conducted. There are limitations to this approach considering that the listed experimental settings vary in many ways; nevertheless, it enabled the formulation of hypotheses relating to the key influencing factors, which could be evaluated in later studies.

6.1.1 REMOVAL AS AN EFFECT OF THE PRESENCE OF BIOFILM

Based on 30 literature studies the spatial *E. coli* removal rate, k_{remx} in eq. (3), ranged from 0.62 to 21.41/m in columns, except for one case of complete removal (Table 1). Considering only the results in the presence of biofilm it ranged from 5.45 to 21.41/m, thus the filter capacities were consistently very high in these cases. Coherently, removal of fecal coliforms was almost complete ($\geq \log 10/\text{m}$) as septic tank effluents were applied intermittently to the 9 filters containing various types of sand, at different loading rates, in Ausland et al. (2002). It should, however, be noted that the flow velocity was consistently low in the listed studies with biofilm, which limited inference in relation to the importance of biofilm as compared to the flow velocity. Nevertheless, removal in the case of biofilm and flux at 0.003 cm/min is remarkably high ($\geq 8.98/\text{m}$; Stevik et al., 1999a), as compared to removal

TABLE 1. *E. coli* removal rates in unsaturated columns of various characteristics, sorted by the spatial first-order kinetic removal rate (natural log)

<i>E. coli</i> strain	Flow type/Filter media/ Grain size (μm)/ Column length (cm)	Darcy vel. (cm/min)	Ionic strength (mM)	Saturation (%)	Relative effluent conc. ^a (%)	Temporal removal rate ^b (1/min)	Spatial removal rate ^c (1/m)	Reference
DH5 α	Intermittent/Crushed Light-weight aggregates, biofilm/0–3000/80	0.002 ^k	7.25	6 ^m	0	∞	∞	Stevik et al. (1999a)
DH5 α	Intermittent/Light-weight aggregates, biofilm/0–4000/80	0.002 ^k	7.25	9 ^m	3.6×10^{-6e}	0.0015	21.41	Stevik et al. (1999a)
D	Steady-state/Silica sand/75–500/22	0.299 ^b	0	99.91	1.42 ^f	0.1523	19.34	Jiang et al. (2007) (FD)
D	Steady-state/Silica sand/75–500/44	0.158 ^b	0	99.96	0.03 ^f	0.0769	18.44	Jiang et al. (2007) (FB)
DH5 α	Intermittent/Fine sand, biofilm/ $d_{10} = 0.13$, $d_{60}/d_{10} = 2.39$ /80	0.002 ^k	7.25	—	9.1×10^{-5e}	—	17.38	Stevik et al. (1999a)
DH5 α	Intermittent/Medium sand, biofilm/ $d_{10} = 0.47$, $d_{60}/d_{10} = 1.81$ /80	0.002 ^k	7.25	—	5.2×10^{-4e}	—	15.20	Stevik et al. (1999a)
DH5 α	Intermittent/Crushed Light-weight aggregates, biofilm/0–3000/80	0.003 ^k	7.25	12 ^m	5.9×10^{-4e}	0.0015	15.05	Stevik et al. (1999a)
DH5 α	Intermittent/Crushed Light-weight aggregates, biofilm/0–3000/80	0.003 ^k	97	12 ^m	6.3×10^{-4e}	0.0015	14.97	Stevik et al. (1999a)
DH5 α	Intermittent/Large Light-weight aggregates, biofilm/2500–5000/80	0.002 ^k	7.25	—	0.05 ^e	—	9.50	Stevik et al. (1999a)
DH5 α	Intermittent/Light-weight aggregates, biofilm/0–4000/80	0.003 ^k	7.25	15 ^m	0.06 ^e	0.0012	9.27	Stevik et al. (1999a)

(Continued on next page)

TABLE 1. *E. coli* removal rates in unsaturated columns of various characteristics, sorted by the spatial first-order kinetic removal rate (natural log) (Continued)

<i>E. coli</i> strain	Flow type/Filter media/ Grain size (μm)/ Column length (cm)	Darcy vel. (cm/min)	Ionic strength (mM)	Saturation (%)	Relative effluent conc. ^a (%)	Temporal removal rate ^g (1/min)	Spatial removal rate ⁱ (1/m)	Reference
DH5 α	Intermittent/Light-weight aggregates, biofilm/0–4000/80	0.003 ^k	0	15 ^m	0.076 ^e	0.0012	8.98	Stevik et al. (1999a)
NAR ^l	Steady-state/Repacked clay loam/45	0.013	—	75.4 ^d	5 ^f	0.0027	6.66	Safadoust et al. (2011)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.102 ^b	0	20	20 ^c	0.0538	6.44	Chen et al. (2010b)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.153 ^{lb}	0	40	25 ^c	—	5.55	Chen (2008)
From raw wastewater	Intermittent/Graded sand, biofilm/d ₁₀ = 0.36, d ₅₀ = 910/150	0.028 ^k	3	12 ⁿ	0.028	0.0125	5.45	Auset et al. (2005)
NAR ^l	Steady-state/Repacked sandy loam/45	0.018	—	84.3 ^d	10 ^f	0.0033	5.12	Safadoust et al. (2011)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.102 ^b	0	40	30 ^c	0.0247	4.82	Chen et al. (2010b)
D	Steady-state/Silica sand/500–2360/22	0.526 ^b	0	49	44.25 ^f	0.0945	3.71	Jiang et al. (2007) (CD)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.153 ^{lb}	0	60	40 ^c	—	3.67	Chen (2008)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.102 ^b	0	60	55 ^c	0.0088	2.39	Chen et al. (2010b)
ATCC 25922	Steady-state/Sand/ 354–710/24	0.005 ^b	3	11 ^d	60.5 ^f	0.0014	2.09	Powelson and Mills (2001) (A)
JM109	Steady-state/Sterilized sand/1000–2000/30	0.240	1	53.5	58 ^c	0.0153	1.82	Won et al. (2007)
ATCC 33694	Steady-state/Silica sand/200–500/25	0.153 ^{lb}	0	80	65 ^c	—	1.72	Chen (2008)

NAR ^l	Steady-state/Weathered sandy loam/45	0.066	—	86.9 ^d	50 ^f	0.0037	1.54	Safadoust et al. (2011)
NAR ^l	Steady-state/Weathered clay loam/45	0.136	—	88.7 ^d	50 ^f	0.0026	1.54	Safadoust et al. (2011)
D	Steady-state/Silica sand/500–2360/44	0.434 ^b	0	67	54.36 ^f	0.0222	1.39	Jiang et al. (2007) (CB)
ATCC 25922	Intermittent/Sand/354–710/24	0.004 ^k	3	17 ^d	74.3 ^f	0.0005	1.24	Powelson and Mills (2001) (A)
JM109	Steady-state/Baked sand/1000–2000 ^b /30	0.660	1	61.0	80 ^c	0.0170	0.74	Won et al. (2007)
JM109	Steady-state/Baked sand/1000–2000h/30	2.580	1	83.3	83 ^c	0.0544	0.62	Won et al. (2007)
JM109	Steady-state/Sterilized sand/1000–2000/30	3.360	1	79.8	83 ^c	0.0678	0.62	Won et al. (2007)

— = Value not found or estimated. ^aPeak relative effluent concentration, unless the total recovery rate was found. ^bFlow rate over column cross-sectional area. ^cMaximum relative effluent inferred from visual inspection of the breakthrough curve. ^dFrom given θ and θ_s , and assuming θ_r for sandy loam or clay loam in Carsel and Parrish (1988). ^eMean effluent concentration over influent concentration (%). ^fTotal recovery rate (%). ^gCalculated as $k_{rem} = k_{rem} \times \text{pore-velocity}$, in which the pore-velocity was calculated as the Darcy-velocity over the estimated moisture content, based on the saturation and published or estimated values for θ_s , and θ_r . ^hAssumed to be similar to the sterilized sand used in the same study. ⁱSpatial first-order removal assumed, according to $C = C_0 \exp(-k_{rem} \times \text{distance})$ based on the cumulative recovered concentration rate, if available. ^jApplication rate up to this value, likely to be less for lower saturations. ^kAverage application rate. ^lNalidixic acid-resistant. ^mAverage in column, as estimated based on visual interpretation of the water retention curves and graphs of θ versus time in Stevik et al. (1999b). ⁿAverage in column, as estimated based on visual interpretation of the graph of θ versus time, and published values for θ_s , and θ_r .

considering flux at approximately the same rate, 0.004 cm/min, without biofilm (1.24/m) (Powelson and Mills, 2001). This indicates that the presence of biofilm has a dominant effect on removal. Indeed, predators, which typically inhabit biofilms and the Schmutzdecke, have been argued to be the most important factor for bacterial attenuation in slow sand filters and in constructed wetlands (Ellis and Wood, 1985; García et al., 2010). In the field, thick biofilms hosting predators could develop where nutrients and moisture are in high supply, for example, in the top soil below leaking sewer lines. However, in the vadose zone the access to moisture and substrate varies to a great extent with space and time, and microbial habitats are expected to be patchy, and thus, the presence of thick biofilms is generally unlikely (sections 4.1.4 and 4.3.1). Then again, knowledge in relation to the distribution and characteristics of such habitats in the unsaturated subsurface is currently limited.

6.1.2 HIGH IMPORTANCE OF FLOW RATE AND FILTER MEDIA HETEROGENEITY

The correlations between the spatial removal rate (k_{remx} [L⁻¹]) and the pore-velocity (v [LT⁻¹]), the ionic strength, the saturation and the average grain size were evaluated (Figure 6). No significant inverse correlation was found between k_{remx} and the grain size, which was unexpected (sections 2.6.1 and 3.4.1); on the other hand, both an increased grain size and an increased saturation are likely to increase the flow velocity, and $\log k_{remx}$ was significantly and inversely correlated ($p < 0.05$) with $\log v$ and the saturation. The results thus indicated that the effect of saturation was more important than the effect of the average grain size in the listed studies. Further, the significance of the flow rate for spatial removal was, coherently, underlined by Pang (2009), who stated “removal rates are most closely related to pore-water velocity”. In the same study, Pang (2009) further reported that k_{remx} was proportional to $v^{-0.62}$ in an evaluation of 24 studies of phage transport in gravel aquifers. Further, since the temporal removal rate is typically calculated according to $k_{remt} = k_{remxv}$ (eq. (3)) this implies that k_{remt} [T⁻¹] was proportional to $v^{0.38}$ in that case. In the current analysis the fitted slope of -0.33 suggested that k_{remx} was proportional to $v^{-0.33}$, and thus that k_{remt} was proportional to $v^{0.67}$ (Figure 6). The results suggest that k_{remt} increases with a higher pore-velocity, which is reasonable, since the fluid passes through more filter material per unit time if the flow is faster. They additionally indicate that this effect is more important in unsaturated than in saturated systems. Pang (2009), however, argued that k_{remx} [L⁻¹] is expected to be proportional to v^{-1} . The underlying assumption is that k_{remt} [T⁻¹] is independent of the pore-water velocity. The current study, on the other hand, emphasizes that k_{remt} is in fact likely to increase with an increasing v , at least under unsaturated conditions. The characteristics of this relationship, however, need further research. The results additionally indicate that a model similar to eq. (12) (Auset et al., 2005; Saiers and Lenhart, 2003a) could be appropriate to describe *E. coli* transport

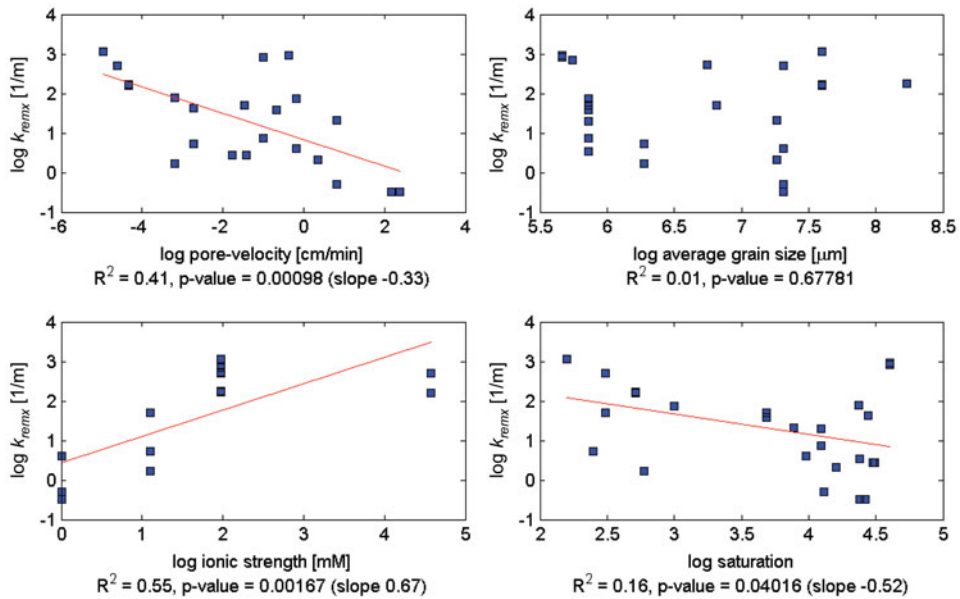


FIGURE 6. Correlations between the *E. coli* spatial first-order total removal rate, k_{remx} , and the pore-velocity, the average grain size, the ionic strength, and the saturation, natural logs (data from Table 1). The middle of the published grain size range was assumed, unless d_{50} had been published. If the pore-velocity had not been published in the cited studies it was set to the Darcy velocity divided by the water content, as estimated based on the saturation. Significant correlations (p values < 0.05) were found between $\log k_{remx}$, and $\log$ pore-velocity, $\log$ ionic strength and $\log$ saturation, for which the slope was calculated.

in unsaturated sandy soils, since the kinetic deposition rate is proportional to $v^{0.5}$ in this approach.

Further, attenuation was higher for crushed light-weight aggregates than spherical ones, even though their grain sizes were similar (Table 1). Coherently, Stevik et al. (1999b) reported that *E. coli* DH5a removal was approximately 5 log10 for crushed light-weight aggregates ($d_{10} = 150 \mu\text{m}$ and $d_{50} = 1500 \mu\text{m}$), versus only approximately 3.5 log10 for spherical ones that were slightly larger in average ($d_{10} = 1100 \mu\text{m}$ and $d_{50} = 1750 \mu\text{m}$) in columns of the same length. These results indicate that the heterogeneities in surface area and grain size are more important than the average grain size. Collector surface roughness is clearly an important factor for retention (sections 2.6.1 and 3.4.1); however, it is rarely quantified, and therefore it was not included in the correlation analyses here. Lastly, the temporal removal rate in the listed columns range from 0.0005/min to 0.1523/min (Table 1). Of these, the average k_{remt} [T^{-1}] was 0.045/min (std 0.055/min) considering only the studies of sandy soils with a d_{50} less than 1 mm in the absence of biofilm.

6.1.3 EFFECTS OF CHEMICAL AND BIOLOGICAL FACTORS

A significant correlation was found between $\log k_{remx}$, and $\log$ ionic strength (Figure 6); however, the $\log$ data points were relatively scarce, and no significant correlation was calculated between k_{remx} , and ionic strength (no $\log$). Further, the high solution ionic strength of 97 mM that applied in some studies is not likely to be very relevant for most groundwater or wastewater (Wang and Zhou, 2010; Yang et al., 2007). Thus, these results show little support for an important impact of ionic strength on attenuation. Contradictory findings have been published (sections 2.6.2 and 3.4.2); on the other hand, considering *E. coli* in particular, many studies have coherently reported a limited effect of ionic strength on deposition (Chen and Walker, 2012; Díaz et al., 2010; Haznedaroglu et al., 2009; Kim et al., 2009a; McClaine and Ford, 2002b). Results further indicate that deposition of motile and nonmotile cells have different susceptibility to solution ionic strength. McClaine and Ford (2002b) found that an increase in the ionic strength from 20 to 200 mM does not significantly affect the deposition of wild type *E. coli* K-12 (motile), due to reversible deposition as a result of bacterial swimming near the collector surfaces that was independent of solution ionic strength. On the other hand, deposition of paralyzed cells was indeed reduced by a higher ionic strength in McClaine and Ford (2002b), suggesting that the effect of ionic strength is more important for nonmotile cells than motile ones. These results, however, applied to saturated conditions, and in unsaturated filter media the effect of solution ionic strength on deposition might be reduced due to capillary forces. In quartz sand ($d_{50} \approx 275 \mu\text{m}$) with 70% saturation Chen and Walker (2012) reported that the effluent concentrations of nonmotile *E. coli* ATCC 12014 were not significantly influenced by an increase in ionic strength from 1 to 100 mM: the relative effluent concentration was $18.1 \pm 2.9\%$ and $15.4 \pm 0.4\%$, for 1 mM and 100 mM, respectively. Moreover, in variably saturated filters, Stevik et al. (1999a) reported that the relative effluent concentration of *E. coli* increased by several orders of magnitude for a doubling of the dosing rate (0.0017/min versus 0.0034/min), and that the change was much less remarkable when the ionic strength was reduced: 0.076% transport rate at 0 mM versus 0.06% at 7.25 mM (Table 1). Stevik et al. (1999a) concluded that physical factors were more important for removal than chemical ones, such as ionic strength and pH.

The importance of the properties of *E. coli* was not assessed in the correlation analyses. The understanding of such factors for attenuation remains limited, particularly in unsaturated media. In saturated porous media it has, nevertheless, been reported that transport of *E. coli* depends more on the depositional environment than the bacterial growth stage (Bolster et al., 2010). In identical saturated columns Foppen et al. (2010) further reported that the sticking efficiency only varied between 0.3 and 1 considering 54 *E. coli* strains, and in unsaturated systems it is likely that this range is narrower

due to the presence of capillary forces that physically hinder the transport of all strains. This indicates that removal rate variations within the strain are limited as compared to those that occur as a consequence of the system settings, such as the flow velocity, considering that the removal rates ranged several orders of magnitude in Table 1.

6.1.4 REDUCTION IN UNDISTURBED LYSIMETERS

The spatial fecal coliform removal rate, k_{remx} , in the lysimeters ranged from 0.35 to 19.74/m (Table 2). This was remarkably similar to the columns, in which the corresponding range was 0.62–21.41/m (Table 1). The temporal removal rate, k_{remt} , in the lysimeters could only be estimated, since the pore-water velocities were not published in Jiang et al. (2008); a rough approximation was that the volumetric moisture content was 0.3 across all values in that study, in which case k_{remt} , calculated based on Table 2, would range from 5×10^{-4} to 0.28/min (average 0.040, std 0.065). Thus, these values were also similar to the ones in the columns, in which the average k_{remt} was 0.045 (std 0.055) for studies with sandy soil and $d_{50} < 1$ mm. This indicated that laboratory studies indeed could be relevant for field analysis; on the other hand, comparisons were limited by the fact that the two data sets were clearly different with regard to flow type and filter media, since the field lysimeters mainly accounted for intermittent flux in loamy soils, whereas the columns mainly included stationary flow in sand. Further, the ranges were wide in both systems. In summation, the results indicated, for the small scales considered (22–150 cm), that the *E. coli* removal rate is generally below 0.1/min in loamy and sandy soils in both columns and lysimeters. Further, the lowest possible removal rates in both types of systems were in the order of magnitude of 10^{-4} /min, which is, as expected, the same as the order of magnitude of the *E. coli* inactivation rate (sections 4.3.1 and 4.3.2). In most of the studies (39 of 46) the temporal removal rate, k_{remt} , was estimated to be in the interval 0.001–0.1/min.

For comparison, Foppen and Schijven (2006) reported that the *E. coli* filter coefficient, corresponding to k_{remx} , ranges from 0.1 to 20/m in saturated soil for a variety of grain sizes and pore-velocities, except for very few, exceptional cases. This is very similar to the results in Tables 1 and 2, which indicates that there is no characteristic of unsaturated media that enables removal rates that cannot be realized under saturated conditions. It also suggests that the flow rate is more important than the saturation for attenuation, since the pore-water velocity can vary in both systems, whereas the saturation cannot. An important difference is, nevertheless, the fact that the pore-water velocity is generally lower in unsaturated soil, which is likely to enable higher spatial removal rates for the same transport distance, considering that the previous results (6.1.2) indicated that k_{remx} increases with a reduced pore-water velocity; further, the underlying attenuation processes

TABLE 2. Fecal coliform removal in lysimeters, sorted by the spatial first-order kinetic removal rate (natural log)

Bacteria	Flow type/Lysimeter length [cm]/Darcy vel. [cm/min]	Filter medium	Cumulative recovered (%)	Removal rate ^c (1/m)	Reference
Fecal coliforms, DSE ^a	Intermittent spray irrigation ^g /70/0.42 ⁱ	Loam	0.0001 ^j	19.74	Jiang et al. (2008) (Trial 2, autumn, C)
Fecal coliforms, DSE ^a	Intermittent spray irrigation ^g /70/0.07 ⁱ	Silt loam	0.001 ^j	16.45	Jiang et al. (2008) (Trial 2, autumn, A)
Fecal coliforms, DSE ^a	Intermittent spray irrigation ^g /70/0.18 ⁱ	Silt loam	0.001 ^j	16.45	Jiang et al. (2008) (Trial 2, autumn, E)
Fecal coliforms, solid cow manur ^e	Drip irrigation, 25 mm for 2 h/75/0.04 ^e	Loam over sandy loam	0.0012 ^f	15.1	Unc and Goss (2003)
Fecal coliforms, DSE ^a	Intermittent spray irrigation, 50 mm for 2 h ^g /70/0.06 ⁱ	Silt loam	0.0054 ^j	14.04	Jiang et al. (2008) (Trial 1, summer, F)
Fecal coliforms, DSE ^a	Intermittent flood irrigation, 100 mm for 0.5 h ^g /70/0.07 ⁱ	Silt loam	0.007 ^j	13.67	Jiang et al. (2008) (Trial 2, autumn, D)
Fecal coliforms, solid cow manur ^e	Drip irrigation, 25 mm for 2 h/100/0.04 ^e	Loam over silt loam	0.0003 ^f	12.6	Unc and Goss (2003)
Fecal coliforms, DSE ^a	Intermittent spray irrigation, 50 mm for 2 h ^g /70/0.42 ⁱ	Loam	0.016 ^j	12.49	Jiang et al. (2008) (Trial 1, summer, C)
Fecal coliforms, DSE ^a	Intermittent spray irrigation, 50 mm for 2 h ^g /70/0.07 ⁱ	Silt loam	0.029 ^j	11.64	Jiang et al. (2008) (Trial 1, summer, D)
Fecal coliforms, DSE ^a	Intermittent spray irrigation, 50 mm for 2 h ^g /70/0.07 ⁱ	Silt loam	0.1 ^j	9.87	Jiang et al. (2008) (Trial 1, summer, A)
Fecal coliforms, DSE ^a	Intermittent spray irrigationh/70/0.21 ⁱ	Sandy loam	0.22 ^j	8.74	Jiang et al. (2008) (Trial 2, autumn, B)

Fecal coliforms, DSE ^a	Intermittent flood irrigation, 100 mm for 0.5 h ^g /70/0.21 ⁱ	Sandy loam	0.48 ^j	7.63	Jiang et al. (2008) (Trial 1, summer, B)
Fecal coliforms, DSE ^a	Intermittent flood irrigation, 100 mm for 0.5 h ^g /70/0.18 ^j	Silt loam	0.54 ^j	7.46	Jiang et al. (2008) (Trial 1, summer, E)
Fecal coliforms, DSE ^a	Intermittent flood irrigation, 100 mm for 0.5 h ^g /70/0.06 ⁱ	Silt loam	0.61 ^j	7.29	Jiang et al. (2008) (Trial 2, autumn, F)
Fecal coliforms, liquid swine manure	Drip irrigation, 25 mm for 2 h/100/0.04 ^e	Loam over silt loam	0.25 ^f	6.0	Unc and Goss (2003)
<i>E. coli</i> , cow manure	Steady-state/40/0.003 ^h	Sand, silt, clay	12	5.24	Mosaddeghi et al. (2010)
Fecal coliforms, liquid swine manure	Drip irrigation, 25 mm for 2 h/75/0.04 ^e	Loam over sandy loam	9.1 ^f	3.2	Unc and Goss (2003)
<i>E. coli</i> , poultry manure	Steady-state/25/0.08	Loamy sand	45 ^d	3.17	Mosaddeghi et al. (2009)
<i>E. coli</i> , poultry manure	Steady-state/25/0.08	Sandy clay loam	55 ^d	2.37	Mosaddeghi et al. (2009)
<i>E. coli</i> , sewage sludge	Steady-state/25/0.08	Sandy clay loam	57 ^d	2.22	Mosaddeghi et al. (2009)
<i>E. coli</i> , sewage sludge	Steady-state/25/0.08	Loamy sand	67 ^d	1.63	Mosaddeghi et al. (2009)
<i>E. coli</i> , cow manure	Steady-state/25/0.08	Loamy sand	77 ^d	1.06	Mosaddeghi et al. (2009)
<i>E. coli</i> , cow manure	Steady-state/25/0.08	Sandy clay loam	92 ^d	0.35	Mosaddeghi et al. (2009)

^aDairy shed effluent. ^bMaximum relative effluent concentration, inferred from breakthrough curves. ^cSpatial first-order removal assumed, according to $C = C_0 \exp(-k_{rem} \times distance)$, based on the cumulative recovered concentration rate, if available. ^dInferred based on published filtration coefficients. ^eApplication rate. ^fInferred from the spatial removal rate. ^g14-day irrigation cycles and natural rainfall. ^hBased on the unsaturated hydraulic conductivity. ⁱWhole-column effective hydraulic conductivity. ^jNot including the day of DSE application.

are likely to vary, as addressed below (section 6.2), which could be particularly important in scenarios with transient flow (section 2.5).

Bacterial attenuation was highest for intermittent spray irrigation on loamy soil in the autumn (Table 2). It was notable that intermittent spray irrigations in the autumn were generally more effective for retention than single drip irrigation events, indicating that intermittent irrigation does not have a negative impact on the filter capacity, as compared to a single irrigation event. Contradictory findings have been reported (section 5.3.3), for example by Martins et al. (2013), who found that preferential transport of *E. coli* K-12 became more pronounced as a result of repeated injection/infiltration and drainage cycles. Further, results suggested that intermittent and drip irrigation was more effective for bacterial attenuation than steady-state flux (Table 2). However, comparisons were complicated by the difference in soil type, and the result could be related to the presence of distinct macropores in Mosaddeghi et al. (2009). Considering the same 12 h total inflow volumes the ratio of transported *E. coli* cells was 60.5% and 74.3% in constant unsaturated and variably saturated sand columns, respectively (Powelson and Mills, 2001) (Table 1). The higher removal in the autumn than in the summer was attributed to cracks and shrinkage in the clayey topsoil, causing preferential flow in the latter, drier season in Jiang et al. (2008). Moreover, flood irrigation in Jiang et al. (2008) generally induced higher transport than spray irrigation, likely due to heterogeneous or macropore flow (section 5.1.1 and 5.1.2). There are several field studies in which heavy irrigation on structured soil has nearly eliminated particle removal (Cey et al., 2009; Natsch et al., 1996; Smith et al., 1985).

6.2 Retention Mechanisms as Dependent on System Characteristics

The previous subsection argued that *E. coli* attenuation in soil essentially depends on the characteristics of the system: it can be practically complete in the presence of biofilms, and practically negligible in the presence of pronounced preferential flow paths and the event of heavy infiltration. This subsection instead addresses literature findings relating to the underlying deposition mechanisms with regard to retention in the soil matrix. Research has shown that the relative importance of each deposition process depends on the filter media and the transporting solution.

6.2.1 SOLID-WATER INTERFACE RETENTION AND THE SIGNIFICANCE OF MICRO-HETEROGENEITIES

Research indicates that the importance of solid-water interface retention processes, such as attachment in second energy minima and straining in grain-grain contact regions are dependent on grain size and surface roughness. Syngouna and Chrysikopoulos (2011) reported that *E. coli* CN-13 deposition is practically negligible in cleaned quartz sand even for small grains

(150–212 μm), and Darcy flow rates as slow as 0.16 cm/min (ionic strength 2 mM). This indicates that deposition near the solid-water interfaces only occurs for grains with micro-heterogeneities. It has further been argued that retention near grains mainly occurs due to grain-grain straining rather than solid-water interface attachment, and that this processes only occurs for colloid to grain ratios (a_p/d_{50}) larger than 0.002 (3.4.1). For example, Wang et al. (2011) found that the transport of *E. coli* O157:H7 was similar to that of a conservative tracer in coarse sand without biofilm, and the low attachment rate was in agreement with results by Bradford et al. (2006b). These results indicate that solid-water interface attachment of *E. coli* is very small for large grains. On the other hand, Auset et al. (2005) reported that *E. coli* ATTC 25922 (motile) attachment to solid-water interfaces indeed occurred in variably saturated two-dimensional micro-models; however, the micro-models might not represent porous media and it is possible that the bottom and top surfaces had influenced this process. Vigeant et al. (2002), nevertheless, observed that *E. coli* HCB437 (motile) indeed attached to clean quartz surfaces in saturated systems, however, the authors noted that most of the attached bacteria were, in fact, swimming along the surface (reversibly adhering). This is a mechanism that is similar to the funneling processes discussed by Torkzaban et al. (2008b). They attributed deposition of *E. coli* D21g (nonmotile) in saturated soil to second energy minima attachment and subsequent funneling along grains until they reached regions of slow flow and small drag forces (solution ionic strengths ≤ 50 mM and $a_p/d_{50} \approx 0.005$)—a straining process (Torkzaban et al., 2008b). It has been found that local second energy minima attachment can be induced by microscopic heterogeneities on grains, and that this effect increases with higher ionic strengths (section 2.2) (Bradford and Kim, 2012; Torkzaban et al., 2010). In summation, these results indicate that, for low ionic strengths, *E. coli* deposition to solid interfaces is only important in porous media with micro-heterogeneities on the solid surfaces, and for grains that are smaller than 500 μm in average ($a_p/d_{50} > 0.002$).

6.2.2 RETENTION NEAR AIR-WATER INTERFACES AS RELATED TO SATURATION AND SOLUTION CHARACTERISTICS

The relative importance of each retention process is further likely to be related to the solution ionic strength and content of dissolved organic matter. It has been argued that straining at the air-water meniscus-solid interface is the main retention process in unsaturated sand considering solutions with low ionic strength; for example, Steenhuis et al. (2006) observed that retention of hydrophilic microspheres mainly occur in the air-water meniscus-solid interface (section 2.3.4 and 3.2).

Morales et al. (2009) found that air-water meniscus-solid interface straining increases for lower saturations. Results based on regression analysis and cm-scale imaging further showed that a reduction in saturation from

100% to 20% had the highest effect on colloidal deposition at ionic strength 1 mM as compared to 10 mM and 100 mM (Bridge et al., 2007). The relatively high importance of saturation indicates that retention induced by the air-water interface, such as air-water meniscus-solid interface straining, was more important in the systems with lower ionic strength. Results by Zevi et al. (2012), coherently, indicate that air-water meniscus-solid interface straining, as an effect of counter flow and capillary forces, is primarily important for lower ionic strengths of 1 mM, rather than 50 or 100 mM, as well as lower moisture contents (section 3.2.1). Direct imaging of negatively charged, hydrophilic colloids revealed that the most important retention locations are likely to be the air-water meniscus-solid interface for solutions of 1 mM ionic strength, whereas attachment mainly occur in clusters on sand grains in solutions of ionic strengths ≥ 100 mM (saturation of 70–90%, sand grain sizes 400–600 μm , and pH 5.7) (Zevi et al., 2009). This was attributed to increased attraction to grains and competition of retention sites for the higher ionic strengths (Zevi et al., 2009). Morales et al. (2011) observed that air-water meniscus-solid interface retention frequently takes place in unsaturated sand (61% saturation) but was not related to solution characteristics. In the same study, mainly solid-water interface attachment and grain-grain interface straining had occurred in conditions that were preferable for attachment in the absence of dissolved organic matter, whereas bridging flocculation at the air-water interface was the most important mechanism in the presence of humic acid for an ionic strength of 1 mM and pH 4 (section 2.6.3). Air-water meniscus-solid interface straining has further been found to increase with more collector heterogeneities, since for rougher surfaces the amount of cavities and friction increases, whereas the shape of the interface remains approximately the same (Morales et al., 2009).

In summation, air-water meniscus-solid interface is likely to be a key retention process in unsaturated sand considering low solution ionic strengths without dissolved organic matter and fine or medium grained sand. For comparison, the spatial attenuation was very low ($k_{remx} < 1/\text{m}$), considering sterilized or baked coarse sand (1–2 mm grains) at saturations and fast flow in solutions of low ionic strength (Won et al., 2007) (Table 1). This indicates that straining near the air-water meniscus-interface is improbable in these systems, likely due to the coarse and smooth filter medium and the high saturations, and that weak attachment to grains is improbable, likely due to the highly repulsive electrostatic forces due to the low solution ionic strength and the smoothened surfaces with reduced micro-heterogeneities.

6.2.3 REMOVAL PROCESSES IN A BASE-CASE SCENARIO

In summation, research indicates that *E. coli* retention occurs as follows, considering sand ($250 \mu\text{m} < d_{50} < 500 \mu\text{m}$) with surface micro-heterogeneities on the grains, in the absence of biofilm, a transporting solution of approximately neutral pH and a solution ionic strength of around 3 mM, as well as

temperatures of around 20°C, and unsaturated, steady-state flow (Darcy velocities 0.001–0.1 cm/min). Assuming high local saturation and the presence of a network of thick and connected pendular rings, the cell is likely to attach to a grain surface in a second energy minimum, found at a locally attractive region (e.g., with high surface roughness; section 2.2.1). Being weakly attached, it could subsequently swim (section 2.4.2) or be funneled along the grain until it approached an air-water meniscus-solid interface (section), or, possibly, a grain-grain contact point. At this location it is expected to become irreversibly retained, due to pinning, induced by capillary and friction forces (section 3.2.1), or spinning and hydrodynamic entrapment (section 3.2.2). The first-order retention coefficient is expected to be somewhere in the range of 10^{-3} to $10^{-1}/\text{min}$ (sections 6.1.2 and 6.1.4). In addition, die-off is like to take place, however at a rate that is at least an order of magnitude inferior to the retention rate, considering the neutral pH, the temperature of 20°C and the absence of biofilm (section 4.3.2). Subsequently arriving cells are likely to be retained in the vicinity of already retained cells, forming bridges (section 3.1.2). One explanation for such cell-aggregation might be that the deposited particles constitute micro-heterogeneities on the grains.

6.3 Steady-State Simulations of Bacterial Transport

This subsection implements macro-scale predictions of particle attenuation in the scenario presented previously (section 6.2.3). Additionally, it includes a discussion relating to bacterial modeling challenges in general.

6.3.1 MODELING OBJECTIVES AND SETTINGS

The model considered one-dimensional transport through sand with properties according Carsel and Parrish (1988), a transport distance of 25 cm, saturations of 25%, 50%, and 75% as well as a pressure head, h , that was constant in space in each scenario. The Darcy velocity in each case, $q = K_s K_r(b)$, was calculated based on the van Genuchten (1980) equations and the specified saturation. The sand bulk density was set to 1.620 kg/L (Jiang et al., 2007). The advection-dispersion equation, eq. (2), with settings for reactions, Λ_{fate} , and dispersivity, λ , according to Table 3 was solved numerically using the COMSOL Multiphysics v4.3a software package (COMSOL AB, 2012). It was assumed that the dispersion overshadowed the molecular diffusion, which was thus set to zero. The model accounted for a steady rate of infiltration and a particle suspension pulse of one pore volume i.e., $C = C_0$ at $x = 0$ cm, with C_0 equal to the mass concentration of 10^9 particles per 100 mL, with the wet weight of one *E. coli* set to 9.5×10^{-16} kg (Milo et al., 2010). The concentration was coherent with Stevik et al. (1999a). The pulse was followed by clean water at the same flow rate (i.e., $C = 0$ at $x = 0$ cm afterward. At the outlet, $x = 25$ cm, it was assumed that mass transport was dominated by convective flux: $\frac{\partial C}{\partial x} = 0$.

TABLE 3. Literature data showing experimental settings and the corresponding models, sorted by saturation

Model Id	Particle/Filter media (grain size [μm])/Saturation (%)/Darcy velocity (cm/min)/Ionic strength (mM)	Attenuation equation, Λ_{rate}	Model parameters	Dispersivity λ^c (cm) (dispersion coefficient [cm/min^2])	Reference
1	<i>E. coli</i> /sand (354–710)/17 ^b /0.005/3	$\rho_b K_d \frac{\partial C}{\partial t} + \theta k_{\text{Sat}} C$	$K_d = 0.0256$ [mL/g] $k_{\text{Sat}} = 0.0009$ [1/min]	1.11 (0.06)	Powelson and Mills (2001), experiment B
2	Spherical silica particles (0.36 μm)/sand (300–355 μm)/22 ^a /0.048 ^b /1	$\theta k_{\text{str}} C + \theta k_{\text{Att}} \left(\frac{C_{\text{Attmax}} - C_{\text{Att}}}{C_{\text{Attmax}}} \right) C$	$C k_{\text{str}} = 0.0018$ [1/min] $k_{\text{Att}} = 0.0058$ [1/min] $C_{\text{Attmax}} = 18.4$ [mg/L]	0.78 (0.32)	Lenhart and Saters (2002), experiment UNSAT2D (wetting history)
3	Spherical silica particles (0.36 μm)/sand (300–355 μm)/36 ^a /0.329 ^b /1	$\theta k_{\text{str}} C + \theta k_{\text{Att}} \left(\frac{C_{\text{Attmax}} - C_{\text{Att}}}{C_{\text{Attmax}}} \right) C$	$C k_{\text{str}} = 0.0026$ [1/min] $k_{\text{Att}} = 0.0071$ [1/min] $C_{\text{Attmax}} = 10.0$ [mg/L]	0.90 (1.79)	Lenhart and Saters (2002), experiment UNSAT1B (drainage history)
4	<i>E. coli</i> /sand (200–500)/40/ ≤ 0.15 /0	$\theta k_{\text{S,Adep}} C + \rho_b k_{\text{S,Arel}} C_{\text{Sat}}$	$k_{\text{S,Arel}} = 1.460$ [1/min] $k_{\text{S,Adep}} = 0.712$ [1/min]	15.58 ^e (11.96)	Chen (2008)
5	Hydrophilic negatively charged microspheres (2.6 μm)/sand ($d_{50} = 400$ μm)/49 ^a /0.077/1	$\theta k_{\text{S,Adep}} C$	$k_{\text{S,Adep}} = 0.065$ [1/min]	0.17 (0.06 ^d)	Zhang et al. (2010), influent concentration 10 mg/L
6	Hydrophilic negatively charged microspheres (2.6 μm)/sand ($d_{50} = 400$ μm)/52a/0.076/0	$\theta k_{\text{S,Adep}} C$	$k_{\text{S,Adep}} = 0.015$ [1/min]	0.18 (0.06 ^d)	Zhang et al. (2010), influent concentration 10 mg/L
7	<i>E. coli</i> /sand (200–500)/60/ ≤ 0.15 /0	$\theta k_{\text{S,Adep}} C + \rho_b k_{\text{S,Arel}} C_{\text{Sat}}$	$k_{\text{S,Adep}} = 1.045$ [1/min] $k_{\text{S,Arel}} = 0.725$ [1/min]	21.60 ^e (11.96)	Chen (2008)
8	<i>E. coli</i> /sand (200–500)/80/ ≤ 0.15 /0	$\theta k_{\text{S,Adep}} C + \rho_b k_{\text{S,Arel}} C_{\text{Sat}}$	$k_{\text{S,Adep}} = 0.378$ [1/min] $k_{\text{S,Arel}} = 0.624$ [1/min]	27.63 ^e (11.96)	chen (2008)

Note. These were used for predictions in the current study. In some cases the referred models had assumed different units for deposited concentration. The main difference between the experimental settings is the saturation and the flow rate: the Darcy velocity varies with two orders of magnitude. The column lengths varied between 10 and 32.8 cm. ^aCalculated from published θ_r and θ_s , assuming θ_r for sand in Carsel and Parrish (1988). ^bCalculated as $q = v\theta$, using published values for v and θ . ^cCalculated as dispersion over pore-water velocity, if not published. ^dThe dispersion term had been ignored when calculating the deposition rate. ^eEstimated based on the published dispersion coefficient (11.96 cm^2/min) divided by the pore-water velocity, estimated based on a flux rate of 3 mL/min and values for θ_s and θ_r for sand in Carsel and Parrish (1988). The pore-water velocity might have been lower in the experiments, in which case the dispersivity would increase.

Clearly, there are no established macro-scale models that accounts for removal in this case. Therefore, the predictions were based on an ensemble of models (Table 3), derived under conditions that were similar to the desired scenarios. All the models had shown good fits to their respective experimental breakthrough curves. The settings that applied when the models had been derived were comparable with regard to: the particle characteristics, which were similar to *E. coli*; the solution characteristics, which had approximately neutral pH and low ionic strength; the filter medium, which was sand in the absence of biofilm; as well as the flow, which was unsaturated and steady-state (Table 3). However, they were different considering the saturation, which varied from 17% to 80%, and the results were therefore expected to vary to some extent. It is reasonable that the context in which the filtration efficiency is to be predicted, e.g., a wastewater reclamation site, will not perfectly match an experimental setting—if it would, the added value of modeling would be very limited. The objectives of these simulations were to define expected ranges of the filtration capacity considering the three scenarios, and to illustrate the effects of the underlying assumptions in the models.

6.3.2 MODELING RESULTS AND DISCUSSION

The ranges of the simulated filtration capacities were very wide (Figure 7); the accumulated mass recoveries for Models 4, 7, and 8 were practically complete for all saturations (Figure 7, bottom). However, in Model 5, for example, the relative recovery was $<10^{-6}$ at 25% and 50% saturation and around 10^{-3} at 75% saturation. Thus, the removal rates varied more than three orders of magnitude, which was remarkable, considering the relatively similar experimental conditions. The effect of Models 4, 7, and 8 was that all of the deposited particles were released when the clean water entered the system; however, insignificant retention is improbable in this steady-state scenario with unsaturated sand with relatively fine grains (Table 3). The results underline the importance of developing models based on deposition curves in addition to breakthrough curves, and suggest that irreversible attenuation needs to be accounted for, which is the case in Model 1, 2, 3, 5, and 6. In these models a higher saturation increased the maximum effluent concentration (Figure 7), which was reasonable, since the coefficients were expressed per unit of time and flow was slower for lower saturations. However, in Model 2, 3, 5, and 6 the relative effluent concentrations were reduced by several orders of magnitude when the saturation decreased from 50% to 25%. This was unexpected, as, for example, the maximum relative effluent concentration of *E. coli* only decreases from approximately 0.55 to 0.2 when the saturation decreases from 60% to 20% in sand, as inferred from Chen et al. (2010) (Table 1). Models 2, 3, 5, and 6 had been developed in studies with pore-velocities that were larger than the ones considered in these simulations, suggesting that the attenuation rate needs to decrease

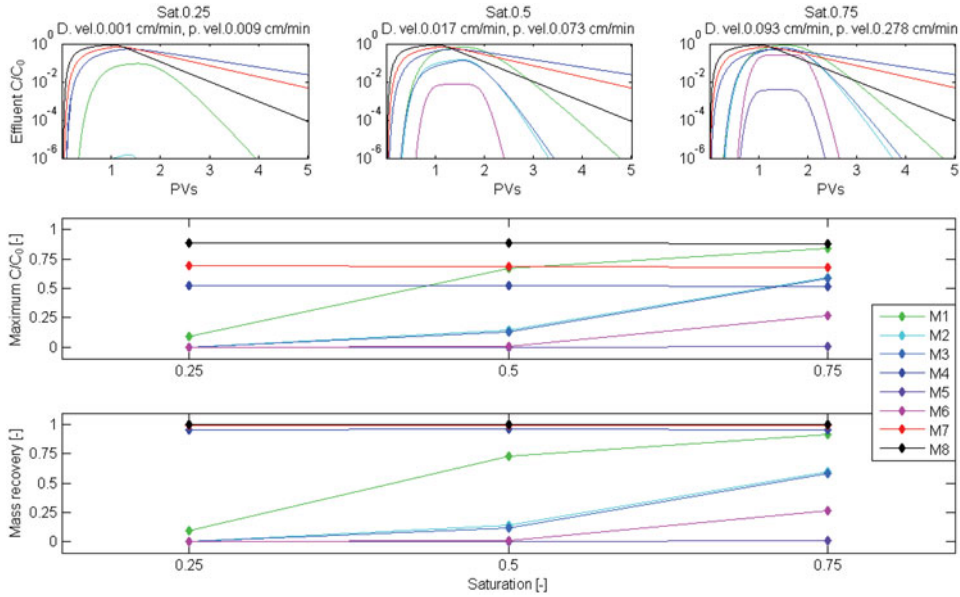


FIGURE 7. Top: relative breakthrough curves, C/C_0 , based on the eight literature models in Table 3, at 25%, 50%, and 75% saturation, respectively. Middle: the maximum relative effluent concentrations versus saturation for the eight models. Bottom: the relative total mass recovery versus saturation. PVs = pore volumes; M: Model. The flow rate and the pressure head were assumed to be constant in each simulation, and the hydraulic conductivity and van Genuchten parameters were those for sand in Carsel and Parrish (1988).

with a decreased flow rate in order to be applicable in scenarios with lower flow rates. This finding is coherent with the results in subsection 6.1.2, in which the correlation analyses pointed to that the temporal first-order kinetic removal rate varied with $v^{0.67}$ (Figure 6).

The equilibrium isotherm in Model 1 apparently did not have a large effect on the accumulated and maximum effluent concentrations, which was reasonable since this term merely delays the breakthrough curve (section 2.2.3). The blocking effect in Models 2 and 3 was evident in the form of upward sloping breakthrough curves for the lower saturations, thus increasing the effluent concentrations in these cases, in which the flow was slower and the concentrations of deposited particles were consequently higher. The result was expected, since the simulated concentration was relatively high (10^9 CFU/100 mL); however, the blocking effect may not be very important in the natural environment, considering that lower concentrations, in the order of 10^6 /100 mL and 10^8 /100 mL, have been reported for liquid swine and solid cow manure, respectively, in Unc and Goss (2003).

It is further remarkable that the dispersivity varied with more than one order of magnitude (Table 3). The larger values resulted in wider breakthrough curves and earlier breakthrough (Figure 7). The lower values were

coherent with results by Bunsri et al. (2008), who found that the dispersivity of tracers varies between approximately 0.6 and 1.2 cm in unsaturated sand (250–500 μm). The dispersion coefficients had typically been derived from tracer breakthrough curves. The difference in dispersion between motile and nonmotile bacteria, colloids and tracers in various soils is as yet unclear. It is possible that the dispersivity of bacteria and tracers differs significantly, considering their dissimilarity in size and motility. Pang et al. (2008) found that the dispersivities of fecal coliforms are lower than tracers in silt loam, silt loam over gravels, and deep silt loam (section 5.3.1). Zhang et al. (2010) neglected the effect of colloidal dispersion in columns, based on the finding that the Péclet number, as calculated from the tracer pore-velocity and dispersion, was larger than 50.

In summation, the wide ranges of the simulated filtration efficiencies, especially for 25% and 50% saturation, when they were more than six orders of magnitude wide, complicated inference in relation to bacterial transport and groundwater vulnerability, even in the simple scenarios considered here—despite that the corresponding experiments were relatively similar. Thus, the predictive power of these models is uncertain. In order to enable the use of a model for predictions it is, therefore, recommended to better evaluate in which systems it is applicable, if ever, by applying it to new experimental data. One possibility is further that the removal capacity varies with the shape of the water retention curve or the soil surface heterogeneity, which calls for quantifying and publishing such data, which is currently rare. Lastly, the results supported the use of a term accounting for temporal, irreversible kinetic attenuation in macro-scale models. This term is likely to vary with the pore-water velocity.

6.3.3 CHALLENGES TO BACTERIAL TRANSPORT MODELING

The relevance of traditional subsurface transport theories is limited for *E. coli* in unsaturated soil (sections 6.2.2, 6.2.3, and 6.2.4). In their place a variety of models have been proposed, for example accounting for: irreversible kinetic attenuation (eq. (3)); equilibrium deposition (eq. (7)); distinct solid-water interface and air-water interface deposition processes (eq. (8)); time-dependent blocking (eq. (9)); particle release (eq. (10)); dependence on saturation and velocity (eq. (11)); distinct straining mechanisms (eq. (14)); hyper-exponential deposition (eq. (15)); preferential flow and transport (eq. (21) and [23]); as well as survival processes, such as growth (eq. (17)), bioclogging (eq. (19)) and inactivation (eq. (20)). Most of these models showed good fits to breakthrough curves. However, oftentimes models are published only after good fits have been found—this was observed by Köhne et al. (2009a) in relation to preferential flow and transport modeling. It clearly limits inference in relation to model relevance for predictions.

Further, a high model fit does not necessarily mean that the relative importance of the involved colloidal fate mechanisms can be inferred from

the fitted coefficients (DeNovio et al., 2004). Breakthrough and deposition curves only reveal results integrated over space and time, respectively. This is related to the concept defined as equifinality in the context of hydrological modeling (Beven and Freer, 2001). It is a problem associated with parameter identification, as, e.g., discussed by van Genuchten and Wagenet (1989) and (Spriet, 1985). Jiang et al. (2007) compared many different attenuation models describing the same *E. coli* breakthrough curves in unsaturated sand, and several of them fitted very well, which illustrates the non-uniqueness problem in this context. Moreover, the models that resulted in the highest fits generally included the most parameters (six), and the use of so many parameters involves the risk of over-fitting to data.

A guiding principle for model selection is that the model structure should relate to the causal mechanisms that act in the system as directly as possible (physicality) (Spriet, 1985). If the coefficients lack a physical foundation, the model might fit to random variations. Theories relating to the underlying mechanisms need to be evaluated using direct observations of retention at the micrometer scale. Important technologies for this application are outlined in the following subsection (section 6.4.1).

The difficulty in using models for predictions was illustrated by the wide attenuation ranges obtained in the simulations in the previous subsection. It appears that the main practical use of modeling is for comparisons of the relative importance of various factors considering a particular laboratory setting. A higher consistency in the use of models would facilitate comparisons between studies; the application of previously derived model structures—and parameters—could develop knowledge in relation to model applicability, and enable the definition of a range of relevant coefficients that could be used in predictions. Models that are found to fit well to many data sets could further be calibrated in worst-case scenarios (Jarvis and Dubois, 2006). Publications would thus sometimes profit from including modeling with bad fits to data. Loewenberg (2013) stated, “it is a basic principle of science that you get things right by analyzing what went wrong.”

It is clear from this review that *E. coli* fate processes in unsaturated subsurface are influenced by a range of factors and processes, which indicates that a number of parameters should be included in models. On the other hand, accounting for all of the discussed mechanisms distinctly would imply a plethora of parameters. Smith et al. (2008) stated, “with any optimized modeling investigation involving several parameters, similarly good fits can be achieved with more than one combination of variables.” A pragmatic approach thus includes only one parameter—parsimony is, coherently, another principle for model selection (Spriet, 1985). Recent studies that have adopted the first-order kinetic approach to particle transport in unsaturated soil include Morales et al. (2011), Zhang et al. (2010), and Unc et al. (2012). Morales et al. (2011) stated that a kinetic first-order removal equation that accounted for all of the contributing sinks acceptably captured the breakthrough curve

shape, and allowed for the interpretation of important relationships between influencing factors and removal rates. Further, as discussed in section 2.2.2, Pang (2009) reported that the assumption of a constant kinetic microbial attenuation rate fitted well with deposition data in 70% of the 87 investigated field cases and that deviations particularly applied to organically contaminated, fine filters.

6.4 Laboratory Studies of Colloids for the Prediction of Field Behavior of Pathogens?

This section provides a discussion in relation to the challenges that arise when laboratory results on colloids and *E. coli* are applied to predict pathogen transport in the field.

6.4.1 TRANSLATION OF RESULTS BETWEEN SCALES

Direct visualization at the pore scale provides crucial knowledge in relation to the underlying particle deposition mechanisms in porous media. It is needed to evaluate hypotheses based on macro-scale results. Imaging could further be used to link pore-scale processes to macro-scale modeling; however, various techniques have different advantages and drawbacks. X-ray computed tomography gives the highest resolution for evaluations of porous media structure and moisture content in three dimensions (Werth et al., 2010). For example, this technique was used in Dann et al. (2011) for the characterization of coastal sand aquifer media. Further, Luo et al. (2010) applied it in undisturbed soil columns for the quantification of macropore networks, in terms of, for example, the volumetric proportion of macropores as well as the number and lengths of macropaths. This methodology is, however, computationally demanding and does not enable observations of distinct colloids and their deposition processes.

Steenhuis et al. (2006) imaged transport of individual, dyed microspheres in unsaturated translucent silica sand with confocal scanning laser and bright field microscopes, and quantified retention at various interfaces using image processing techniques. Morales et al. (2009) also visualized individual colloids using digital bright field microscopy and, in parallel, evaluated effluent particle concentrations using spectrophotometry. However, the link between the processes observed at a few interfaces and the corresponding breakthrough curve is not obvious. At the meso-scale, Bridge et al. (2007) instead applied time-lapse UV-fluorescence imaging of thin-bed flow chambers ($9.8 \times 20 \times 0.67$ cm) and image processing to quantify variably saturated transport of colloids in sand in two dimensions. This enabled time-variable quantification of colloid breakthrough and deposition throughout the filter domain and mass balance was ensured. However, attenuation processes related to individual colloids could not be distinguished in the published

images, and inference relating to the relative importance of attachment to air- and solid-water interfaces was based on regression fits between deposition data and estimated air- and solid-water interface areas for each point in the flow domain. Markov et al (2010) visualized real-time habitat growth and retention of individual bacteria in saturated microfluidic devices using a fluorescence microscope, also at the centimeter scale and in two dimensions. However, the collector surfaces were artificial and it remains to be evaluated to which degree such systems reflect processes in soil.

In addition to direct imaging tools, theoretical approaches to handling the translation of results between scales are under development. The up-scaling from cell- to biofilm-scale has, for example, been developed by Wood and Whitaker (1999), and from biofilm- to Darcy-scale, by Golfier et al. (2009). Common methods involve homogenization and volume averaging that cover means by which closure problems are handled, and imply that sub-pore-scale parameters and problem geometry are incorporated into the definitions of the effective parameters at the macro-scale (Golfier et al., 2009; Wood and Whitaker, 2000; Wood et al., 2002). However, the application of these methods is still limited. Regarding biofilm modeling, Golfier et al. (2009) stated,

Most models have been developed by first hypothesizing the important mass transfer and reaction processes involved, and then heuristically formulating the mathematical model at the Darcy scale by mass balancing. Thus, Darcy-scale parameters are assumed to exist and to be known a priori.

This statement puts light on the subjective nature of current bacterial modeling in the subsurface.

The up-scaling of column-scale findings depends on whether the column can be considered as being a representative element volume (Pang, 2009). It has been argued that laboratory experiments often are not relevant to natural processes, which are very complex due to physical, chemical and biological heterogeneities, and are, as such, difficult to imitate (Tufenkji, 2007). Laboratory studies are nevertheless, clearly, useful for the assessment of individual processes and factors; however, the possibility to compare results between them is still limited, which complicates inference in relation to the field. Further, microbial removal rates can be one to three orders of magnitude times lower in the field than in columns (Pang, 2009). On the other hand, in the current study the ranges of the first order spatial attenuation coefficients in laboratory columns and in undisturbed field lysimeters were remarkably similar (section 6.1); however, the ranges in both cases varied widely and all of the listed studies addressed relatively small scales (≤ 1.5 m) (Tables 1 and 2), which limits inference in relation to deeper vadose zones.

Pang (2009) reviewed and specified ranges of the expected microbial removal rates in various field contexts. Predictions of removal at larger scales are clearly challenging.

6.4.2 RELEVANCE OF ABIOTIC PARTICLES FOR THE UNDERSTANDING OF *E. COLI*

It has been debated as to whether inorganic particles could be used to predict *E. coli*, particularly due to motility, surface macromolecules and subsistence mechanisms. In the presence of biofilms an important difference between bacteria and colloids is the predation of cells (section 6.4.1). Studies of abiotic particles with surface charge and hydrophilicity similar to *E. coli* could thus be useful considering scenarios in which predation is not likely to be very important. Abiotic colloids have been found to well represent *E. coli* in short-term studies, both in the laboratory and in the field. Optical imaging has revealed that the retention mechanisms of *E. coli* and negatively charged, hydrophilic microspheres are similar in artificial micro-models (Auset et al., 2005). In flow chamber tests Steenhuis et al. (2006) further reported that unaltered *E. coli* PHL 628 GFP -1 and hydrophilic microspheres have similar retention characteristics. In the field, Passmore et al. (2010) investigated the relevance of polystyrene latex microspheres ($a_p = 1.1, 3.9, \text{ and } 4.8 \mu\text{m}$) as representatives for *E. coli* RS2g in two unsaturated undisturbed agricultural field sites with silt loam containing macropores. Within 5 cm depth the removal of both *E. coli* and all abiotic colloids were more than 2 log₁₀, and the smallest, carboxylated, microspheres were present in 77% of the samples that contained *E. coli* RS2g. The similarity in transport was attributed to their similar size and surface charge, since the ζ -potentials were -12.1 mV and -18.0 mV, for *E. coli* and the microspheres, respectively (Passmore et al., 2010).

On the other hand, important differences have been reported between *E. coli* and abiotic particles, also in the case of nonmotile *E. coli*. For example, Kim and Walker (2009) reported that the first-order deposition rate at 10°C was approximately 0.09/min versus 0.03/min for *E. coli* D21g (nonmotile) and abiotic colloids under saturated conditions, respectively, whereas the inverse trend applied at 25°C, during which the deposition rate was approximately 0.03/min versus 0.09/min for the cells and the colloids, respectively. The reason could not be identified, however, it was argued that biological factors, such as membrane fluidity, surface hydrophobicity and polymer conformation, and their relation to temperature need further evaluation (Kim and Walker, 2009). Nevertheless, Morales et al. (2011) argued in favor of studying microspheres as pathogen surrogates, considering that the abiotic processes are, as yet, insufficiently understood. A key point is to maintain a conservative approach with regard to size (sections 2.6.1 and 3.4.1), surface charge (section 2.6.5) and hydrophilicity (section 2.3.4 and 2.3.2), as compared to pathogenic *E. coli* strains.

6.4.3 E. COLI AS A REPRESENTATIVE OF PATHOGENIC MICROBES

Another issue concerns the relevance of studying the transport of a harmless *E. coli* strain to predict that of pathogenic *E. coli* strains or other manure-borne pathogens. There are differences in retention rates between various manure-borne bacterial species, such as *E. coli*, *Salmonella* or *Vibrio cholerae*. Haznedaroglu et al. (2009) compared *E. coli* O157:H7 transport with *Salmonella pullorum* SA1685 in columns, considering various ionic strength and cell concentrations, and found that the latter was less adhesive, likely due to its more negative surface charge. In columns Chen and Walker (2012) further reported that retention of the fecal indicator bacterium *Enterococcus faecalis* (Gram-positive) is more sensitive to solution ionic strength and water saturation than *E. coli*. Additionally, Salvucci et al. (2009) reported that the relationship between the ability to form biofilm and the deposition rate that applied to *E. coli* did not apply to the studied *Salmonella* isolates. On the other hand, Pang (2009) found that the attenuation rates of *E. coli*, fecal coliforms, *streptococci*, and *enterococci* were relatively similar in a review of field studies.

Further, *E. coli* behavior has been shown to be heterogeneous within the strain (section 2.6.6). Regarding inactivation, some conservative strains have, however, been identified. Wood et al. (2010) found *E. coli* NAR to be a possible surrogate for the non-pathogenic *E. coli* O157:H7 strain DM3n, considering survival rates after spray irrigation on spinach plants (phylosphere and soil). Further, Mubiru et al. (2000) found that the mortality of the nonpathogenic strain *E. coli* ATCC 27662 was slightly lower than *E. coli* O157:H7 in silt loam. Considering the biofilm formation behavior, Yang et al. (2006) evaluated a wide range of *E. coli* isolates and observed that (motile) *E. coli* O157:H7 formed very little biofilm, while (motile) *E. coli* K-12 formed significant biofilm mass in external (postshedding) environments. This indicates that *E. coli* K-12 might be a conservative strain with regard to growth. Understanding of the variability in long-term survival and growth of various *E. coli* strains in the natural environment, however, remains limited. Regarding transport in porous media conservative representatives of the motile and nonmotile pathogenic *E. coli* strains are as yet to be identified. Considering the transport of 54 strains in saturated sand columns Foppen et al. (2010) reported that no single bacterial factor could statistically significantly explain the retention variations within the *E. coli* strain. Apparently, combinations of the different *E. coli* characteristics determine the transport rates, and these are as yet to be identified.

7. CONCLUDING REMARKS AND RECOMMENDATIONS

Research indicates that the presence of *E. coli* in soil might not be a reliable indicator of the presence pathogenic *E. coli* or other pathogenic fecal bacteria (section 6.4.3). Even so, improved understanding of transport and survival of

pathogenic *E. coli* strains is clearly important to improve the assessment of health risks related to groundwater contamination or wastewater irrigation on ready-to-eat crops. It is, therefore, recommended that future studies better specify the differences in subsurface transport behavior between strains in various contexts in order to identify conservative ones, with regard to motile and nonmotile pathogens. Studies of abiotic colloids with similar surface charge, hydrophily and size as a particular pathogenic strain could, moreover, provide important understanding of the physicochemical attenuation processes. This, however, also requires further research into the differences in transport and retention behavior between such abiotic particles and the *E. coli* they represent, as related to motility and steric interactions due to the presence of surface polymers. The recommendations below are thus primarily related to pathogenic *E. coli* strains or particles that could represent them.

7.1 Articulate the Effects of Heavy Infiltration and Preferential Flow

The most pertinent health risks related to *E. coli* transport in the vadose zone is arguably groundwater vulnerability after flooding events and subsequent transport in macropores. These are typically induced by biological activity and cracks in clayey soil (section 5.3.3). In this case bacterial removal in soil could be practically negligible, and in contrast, it could be nearly complete in the presence of thick biofilms that host predators, such as *protozoa ciliate* (section 6.1.1). Attenuation thus essentially varies with the system characteristics.

Important related findings can, however, appear incoherent. For example, the supply of organic matter could provide nutrients that enable biofilm development and *E. coli* predation (sections 4.3.1 and 4.1); however, a higher presence of organic matter also increases biological activity that induces macropores and fast transport (section 5.3.4), and dissolved humic acid in the transporting solution reduces retention in the matrix as compared to de-ionized water (section 2.6.3). Further, frequent infiltration could increase the saturated hydraulic conductivity (section 5.3.3); however, wetter soil could reduce the risk of cracks and macropore flow in the event of heavy rainfall (sections 5.1.2 and 5.3.3), and frequent dosing of wastewater could enable the formation of thick biofilms that remove *E. coli* (section 6.1.1).

Future research in these topics is thus suggested, especially considering a changing climate with increased frequency of flooding and droughts, which might induce fast contaminant transport. Field studies are, for example, recommended to evaluate in which types of soil the initial moisture content increases or decreases the tendency for preferential flow; the *E. coli* migration velocities considering different rainfall or ponding events in various soils; which temperatures and subsurface environments cause crack

formation; under which circumstances biofilms provide significant *E. coli* attenuation in the vadose zone; how such biofilms are affected by preferential flow paths and sudden infiltration events; as well as whether pathogenic *E. coli* could persist extensively in the natural subsurface.

7.2 Modeling for More Reliable Predictions of Transport and Fate

Predictive modeling is involved with important challenges relating to flow, transport and reactions. Preferential flow could occur in almost any type of soil, and dual-continua flow models might be required to describe heterogeneous and macropore flow in graded and structured soil if the saturation is high (sections 5.1.1, 5.1.2, and 5.2). Such modeling entails important challenges related to parameter identification. Considering transport, the differences in migration velocity and dispersivity between tracers, colloids and motile and nonmotile bacteria have not yet been fully identified (section 6.3.2). Regarding reactions, traditional theories clearly fail to satisfactorily predict the fate of *E. coli* in the unsaturated zone (sections 2.2, 2.3, and 2.4), and in their place, many different models have been proposed. However, these have often only been evaluated in relation to integrated results (section 6.3.3), and it is not clear in which settings they could be applied for reliable predictions (section 6.3.2).

A recommendation is thus that previously derived models, including parameters, would be applied to predict a new set of data. It is suggested that the experimental conditions that applied when the model was derived and tested are described in detail, and, for example, include the pore-water velocity and the water retention curve. This would facilitate inferences relating to model applicability, and comparisons of factors between experiments. Relevant parameter ranges and worst-case transport scenarios could thus be better specified. A suggested starting point is the first-order kinetic removal model, with all attenuation processes lumped in one coefficient, in order to promote simplicity and reduce the risk for over-parameterization. Correlation analyses in this study indicated that the temporal deposition rate in this model ought to vary with the pore-water velocity (sections 6.1.2 and 6.3.2). The characteristics of the relationship, however, need further evaluations. Last, results suggest that depth-dependent deposition, as induced by bioclogging and flocculation, needs to be accounted for in the case of high soil organic matter content and high bacterial concentrations (sections 2.2.2, 2.6.3, 2.6.6 and 3.1.2).

7.3 Imaging to Assess Bacterial Fate Due to Variations in Flow and Nutrient Flux

Findings indicate that the relative importance of each retention mechanism in the matrix varies with the system: attachment to grains in second energy

minima (section 6.2.1), as well as air-water meniscus-solid interface straining (section 6.2.2) are likely to be key retention processes; the relative importance of the latter increases with lower moisture contents and lower solution ionic strength; and further, both mechanisms become more prevalent with slower flow and higher collector surface heterogeneity.

Recent research progress can mainly be attributed to direct visualization of colloidal fate in macroscopically homogeneous sand, and it is recommended that future imaging relate to more heterogeneous media. The relationships between results at various scales are also important areas for future research. Predictive modeling would benefit from better developed theoretical up-scaling approaches relating to unsaturated soil. Imaging at the pore- and meso-scale would allow visualization and quantification of biological and physicochemical processes as related to various types of filter media, nutrient flux patterns and infiltration. This would facilitate analyses of the retention and survival processes in, and near, various types of preferential flow paths during, and between, heavy infiltrations; the effect of the presence of dissolved organic matter on retention versus predation in biofilm as related to time; the underlying reasons for the observed non-exponential deposition profiles; the degree of saturation needed to enable chemotaxis in unsaturated porous media; and the differences in pore-velocity and dispersivity between bacteria, colloids, and tracers.

NOMENCLATURE

Units

L	Length
M	Mass
N	Number of particles
T	Time

Variables and Coefficients

a_p	Particle diameter [L]
C	Concentration of particles in water [NL^{-3}]
C_0	Influent concentration of particles [NL^{-3}]
C_{Aatt}	Concentration of air-phase attached particles [NL^{-3}]
$C_{Aattmax}$	Maximum air-phase concentration of attached particles [NL^{-3}]
C_{im}	Concentration of particles in the immobile region [NL^{-3}]
C_m	Concentration of particles in the mobile region [NL^{-3}]
C_{SAdep}	Concentration of deposited particles [NM^{-1}]
C_{SAdep_i}	Concentration of deposited particles for compartment i [NM^{-1}]
C_{Satt}	Concentration of solid-phase attached particles [NM^{-1}]
C_{str}	Concentration of strained particles [NM^{-1}]
C_{strmax}	Maximum of the strained concentration of particles [NM^{-1}]

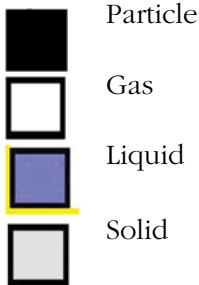
C_{sub}	Concentration of substrate in the fluid [ML^{-3}]
D	Dispersion coefficient [L^2T^{-1}]
D_m	Dispersion coefficient in the mobile phase [L^2T^{-1}]
d_{50}	Median grain diameter [L]
E	Colloid pore exclusion factor [—]
h	Pressure head [L]
k_{Aatt}	Air-phase attachment coefficient [T^{-1}]
k_{gro}	Growth coefficient [T^{-1}]
k_{gromax}	Maximum specific utilization rate [T^{-1}]
K_{half}	Half-velocity coefficient [ML^{-3}]
k_{ina}	Inactivation coefficient in solution [T^{-1}]
$K_r(h)$	Relative hydraulic conductivity [—]
k_{remt}	Total temporal kinetic first-order removal coefficient [T^{-1}]
k_{remx}	Total spatial kinetic first-order removal coefficient [L^{-1}]
K_S	Saturated hydraulic conductivity [LT^{-1}]
k_{SAdep}	Total solid- and air-phase deposition coefficient [T^{-1}]
k_{SArel}	Total release from solid- and air-phase coefficient [T^{-1}]
k_{SArel_i}	Release coefficient for compartment i [T^{-1}]
k_{Satt}	Solid-phase attachment coefficient [T^{-1}]
k_{Sdet}	Solid-phase detachment coefficient [T^{-1}]
k_{Sdep}	Solid-phase deposition (attachment and straining) coefficient [T^{-1}]
K_{Seqatt}	Solid-phase equilibrium deposition coefficient [L^3M^{-1}]
k_{str}	Straining coefficient [T^{-1}]
l	Metabolic time lag [—]
m	van Genuchten parameter [—]
N	Parameter relating colloid retention to pore-velocity [$(LT)^{-0.5}$]
N_2	Parameter relating colloid release to the pore-velocity [$L\kappa T\kappa^{-1}$]
N_C	Number of compartments [—]
q	Darcy velocity [LT^{-1}]
S	Source/sink term [T^{-1}]
s_M	Effective microbial saturation (microbial over void volume) [—]
s_T	Effective total saturation (sum of water and microbial saturations) [—]
s_w	Effective water saturation [—]
t	Time [T]
v	Average interstitial (pore) velocity [LT^{-1}]
x	Vertical distance (positive upward) [L]
Y	Yield coefficient, relates substrate utilization to growth [—]

Greek Letters

α	Sticking efficiency
α_{mim}	Rate of exchange between mobile and immobile liquid regions [T^{-1}]
a_p	Width of particle [L]

β	Parameter quantifying deposition depth-dependency [–]
Γ_{mim}	Particle transfer rate between mobile and immobile regions [$\text{NL}^{-3}\text{T}^{-1}$]
Γ_w	Water transfer rate between matrix and fast flow domain [T^{-1}]
ζ	Zeta
η_0	Single-collector contact efficiency [–]
θ	Volumetric moisture content [–]
θ_A	Volumetric air content [–]
θ_C	Colloid accessible water content [–]
θ_{crit}	Critical moisture content [–]
θ_{crit_i}	Critical moisture content in compartment i [–]
θ_{im}	Fraction of immobile colloids [–]
θ_m	Fraction of mobile colloids [–]
θ_r	Residual volumetric moisture content [–]
θ_s	Saturated volumetric moisture content [–]
κ	Parameter quantifying colloidal release [–]
Λ_{att}	Attachment reaction rate source-sink term
Λ_{fate}	Total particle fate reaction rate source-sink term
Λ_{str}	Straining reaction rate source-sink term
Λ_{sur}	Survival reaction source-sink term in water
λ	Dispersivity [L]
μ	Parameter relating the straining rate to the saturation [–]
ρ_b	Filter media bulk density [ML^{-3}]
ψ	Depth- or time dependent straining function
ω	Exposure or residence time [T]

Figure Colors



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